

PathSmart: Designing a Pathfinder for Naga City People's Mall Using A* Algorithm

A Senior Project Study

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of the Requirements for the Degree of
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CERTIFICATION

This is to certify that this project study entitled "**PathSmart: Designing a Pathfinder for Naga City People's Mall Using A* Algorithm**" written and developed by **Luois Peter P. Alforte, Joseph F. Martin Jr., and Jovy Jay P. Dela Cruz** has been examined, edited, and found to be in accordance with suggestions and recommendations by the faculty-in-charge.

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CERTIFICATION

This is to certify that the project study entitled "**PathSmart: Designing a Pathfinder for Naga City People's Mall Using A* Algorithm**" has been carried out by **Luois Peter P. Alforte, Joseph F. Martin, and Jovy Jay P. Dela Cruz** under my guidance in partial fulfillment of the requirement for Capstone Project during the Academic Year 2024-2025. To the best of any knowledge and belief, this work has been examined, edited, and found to be by the suggestions and recommendations of the project

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DEDICATION

We would like to express our sincere gratitude to our teacher, June Arreb Danilla, for suggesting the idea for our senior project and for continuously supporting and guiding us throughout the development of this study.

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ABSTRACT

This study presents the design and development of PathSmart, an indoor navigation system specifically developed for the Naga City People's Mall (NCPM) in Naga City, Camarines Sur. NCPM's complex three-floor layout spanning 30,000 square meters with over a thousand stalls poses significant navigational challenges for unfamiliar shoppers and tourists. Existing navigation tools such as Google Maps rely on GPS, which is unreliable in indoor spaces, highlighting the need for a dedicated indoor navigation solution. PathSmart is a cross-platform mobile and web-based application developed using React Native and Supabase, offering features such as stall and product search, a shopping list with optimized multi-stop routing, shopping preference settings, and a user-defined starting and destination point. The system's pathfinding functionality is powered by the A* algorithm, adapted to suit the layout and navigation needs of NCPM. A mixed-methods research design was used, conducted interviews and document analysis with MEPO employees and City Hall of Naga City personnel for requirements gathering, and User Acceptance Testing alongside algorithm performance testing for evaluation. The A* The algorithm was evaluated in terms of speed and accuracy.

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CHAPTER 1

INTRODUCTION

As technology moves forward, so does innovation. New inventions and technologies are emerging in every corner of the world. Old systems are replaced with new systems that are easy and efficient. This study is no different. A Navigation System for an indoor marketplace, replacing the old traditional way of navigation—asking people for directions.

This chapter establishes the groundwork for this study, providing a clear understanding of what it is about. It also serves as a preliminary to understanding the next chapters.

Project Context

Ever been lost in a very unfamiliar place? Want to go somewhere, but you do not know which direction to take to get to that desired location? This is the typical problem faced by unfamiliar shoppers and tourists at the Naga City People's Mall (NCPM).

Indoor navigation is a growing field and is in demand due to the need for businesses to enhance the digital wayfinding in complex and large indoor spaces like malls, markets, and many more. According to Ipera [2024], an indoor navigation system can significantly improve customer experience and increase

shopper satisfaction by guiding customers to find their way through their desired location in indoor spaces. The Naga City People's Mall (NCPM) size with its floor area of 30,000 square meters, having three-floor levels, and a complex layout with over a thousand stalls Naga City Tourism [2024], presents navigational challenges for tourists shoppers which justified the need for implementation of an indoor navigation system specifically designed for NCPM, to ensure shoppers can easily navigate the market and go to their desired location.

The study is conducted in the Naga City People's Mall (NCPM) located in General Luna Street, Naga City, Camarines Sur. Naga City is known as the "Heart of Bicol." Naga City attracted over 2.2 million visitors in 2022, according to the City Government of Naga [2023], due to the city's rich culture, natural attractions, and events like the Peñafrancia Festival. NCPM's position is perfectly centrally located where anyone such as tourists and locals can easily find and access, and NCPM also offers a wide range of products—from seafood to souvenir items and handicrafts—making it a go-to market to both locals and tourists Naga City Tourism [2024].

The purpose of this study is to improve the shopping experience of shoppers by addressing the challenges that unfamiliar shoppers face when navigating in NCPM. There are a lot of navigation tools available out there, for example, like Google Maps. Google Maps uses GPS for navigation. Mappedin [2022] explains that although GPS is great for outdoors, in indoor spaces like NCPM GPS signals are often weak or unavailable due to walls, and hence unreliable to use for indoor spaces. So Google Maps is not ideal to use when you

want to navigate in indoor spaces like NCPM. To address this gap, proponents propose PathSmart, it is an indoor navigation system specifically designed for the Naga City People's Mall. Which will help and guide shoppers to navigate the market, making it less overwhelming and difficult for these shoppers. Shoppers will manually select their starting point and destination within the market, and the system will compute the most efficient route using the A* pathfinding algorithm. PathSmart will offer step-by-step guidance, PathSmart aims to improve navigation, reduce time spent wandering, and enhance the overall shopping experience.

Statement of the Problem

The large size, limited signage, and complex layout of Naga City People's Mall are the reasons for the navigation challenges that unfamiliar shoppers and tourists face.

The following problem statements shall be properly addressed:

1. What input requirements are needed to develop the PathSmart system?
2. What modifications are needed to the A* algorithm to work the layout and navigation needs of the NCPM properly?
3. What is the accuracy and speed of the A* algorithm when generating routes in the PathSmart system?

Objective of the Study

The main objective of this study is to address the navigational challenges and enhance the shopping experience of shoppers in Naga City People's Mall by developing a navigation system.

The following objectives should be achieved:

1. To identify the input requirements needed for the development of the PathSmart system.
2. To develop a pathfinder by adapting the A* pathfinding algorithm that suits and works based on the structure and navigation needs of Naga City People's Mall.
3. To test the performance of the A* algorithm when providing routes by evaluating its speed and accuracy.

Purpose and Description

The purpose of this study is to create a solution for the navigational challenges that unfamiliar shoppers and tourists face in Naga City People's Mall. The solution is to develop an application that serves as a guide for shoppers by providing them with a pathfinder and other tools that can help with their shopping.

The application will be named "PathSmart". It is mainly a pathfinding application, but it's not just a pathfinder. The developers kept in mind the user experience, which is why the app is equipped with tools and features designed to make the application easy to use.

The key components and features are discussed in the following:

Search. The application is equipped with a search functionality that has a filtering feature, which makes it easy for users to search for what they are looking for inside the market, whether it's products, stores, or services.

Shopping list. A product listing functionality to help shoppers avoid forgetting items. However, it's not just a simple notepad; using the list, the pathfinder will generate the most efficient route that passes through all item locations.

Pathfinder. The pathfinder is the main functionality of the application. It uses the A* algorithm to calculate the best route to the desired destination of the shopper.

Shopping Preference. A functionality to modify the pathfinder route generation. The application has a selection of tags (e.g., farthest first) to cater to the shopping preferences of the shopper.

User-defined starting point. The system will allow users to manually select their starting point from a selection of pre-defined nodes(starting points). Starting points can also be selected using block and stall numbers.\

Quality Guide. Quality Guide is an additional function. It serves as a helper in determining quality by giving information about the description of good and bad products.

The PathSmart system offers benefits to various stakeholders. The beneficiaries are listed below:

Shoppers and Tourists. The main beneficiary of this

system. The pathfinder that helps in navigating the market, and the quality guide that assists their product quality assessment, will greatly enhance their shopping experience

Stall Owners. The Stall owners will benefit from the PathSmart application, as the system will enhance the visibility of their stall and increase the number of customers. The shoppers will also no longer ask for directions.

MEPO Employee. The MEPO employees will benefit from the PathSmart application since the app will serve as a guide for unfamiliar shoppers when they

navigate inside the market, MEPO employees can now focus more on important market related tasks, and not spend it handling constant queries about directions for unfamiliar shoppers.

Scope and Delimitations

This study focuses on the development of PathSmart, an indoor navigation system designed for the Naga City People's Mall (NCPM) located along General Luna Street, Naga City, Camarines Sur. The system is intended for shoppers and tourists who experience difficulty navigating the mall's complex structure, which consists of three floors, approximately 30,000 square meters of floor area, and over a thousand stalls.

The system includes features such as stall and product search, an A* pathfinding algorithm adapted to the layout of NCPM, shopping list functionality that supports multi-stop route generation, user-defined starting point selection, quality guide and shopping preference settings. These features aim to improve navigation efficiency and enhance the overall shopping experience within the market.

PathSmart is developed as a mobile and web-based application to ensure accessibility across smartphones and computers. These platforms were chosen due to their convenience, usability, and wide availability among users.

However, the study is limited in several aspects. The system does not utilize hardware-based indoor positioning technologies such as Bluetooth Low Energy (BLE) beacons, Wi-Fi fingerprinting, or Ultra-Wideband (UWB) due to cost and

maintenance constraints. Instead, users manually select their starting and destination points within the system. In addition, the study is geographically limited to Naga City People's Mall, and the system is specifically designed based on its unique layout and structure. Therefore, its functionality and performance may not be directly applicable to other market environments without modification.

Limitations of the Study

Outdating Information. The system will definitely require updating if there are any changes in the markets layout, stalls or products. Outdated information may affect the efficiency of the navigation result.

No Real-Time Inventory. The system will not allow live monitoring of product item stock. It will simply tell users about product and stall information, but stall owners will still be responsible for actual availability.

Definition of Terms

Algorithm – A process or set of rules to be followed in calculations or other problem-solving operations, especially by a computer.

NCPM - stands for “Naga City People’s Mall”

Efficiency – The ability to accomplish a task with minimal waste of time or resources.

Geolocation – The identification of a device's real-world geographic position.

Hybrid Navigation System – A system that combines multiple technologies to enhance location tracking accuracy.

Indoor Navigation System – A system designed to help users find their way inside buildings or enclosed areas.

MEPO employee - A stakeholder in NCPM oversees market-related operations.

Market Enterprise and Promotions Office (MEPO) - A division of the Naga City government responsible for managing and promoting the Naga City People's Mall (NCPM) and other public markets.

Navigation – The process of determining and following a route to a destination.

Optimization – The process of making a system as effective or functional as possible.

Pathfinding – A method used in computing to determine the shortest or most efficient route from one point to another.

Quality Guide – A feature that provides a set of standards or criteria used to assess the quality of a product, such as freshness or durability.

Real-time Positioning – The ability to track and update a user's location instantly.

Stakeholders – Individuals or groups with an interest in a particular project, such as shoppers, store owners, and mepo employees.

WiFi Fingerprinting - is a form of location tracking method that uses wifi to track.

BLE - stands for Bluetooth Low Latency. It's a form of location tracking method that uses Bluetooth to track.

AP - stands for access point. This is a network device that sends out signals.

RSSI - stands for Received Signal Strength Indicator. This is a measurement of the distance of a device from an access point or beacon

IPS - IPS stands for Indoor Positioning System is a technology that locates people or objects inside indoor spaces where GPS is unreliable.

REVIEW OF RELATED LITERATURE

This chapter explores various relevant literature, studies, and ideas that contribute to the development of an indoor navigation system, specifically the PathSmart System for Naga City People's Mall (NCPM).

Technical Background

The proponents have gathered related literature that align with the project. The information that these literatures give will be brainstormed by the proponents and will then serve as the basis to develop the project.

There are two essential technical components required for the successful implementation of the PathSmart system for Naga City People's Mall: The algorithm for Pathfinding and user-defined starting-point method. In this study, the system will use the A* algorithm. And a User-Defined Starting Point method wherein users will manually select their starting point and destination, which the system will then use as input for route generation.

In this study, the A* algorithm is selected. According to Patel [1997], A* is a combination of two key approaches: the exhaustive search capability of Dijkstra's algorithm and the heuristic-driven approach of Greedy Best-First Search, that allows the algorithm to find the shortest path while reducing the number of paths being searched. Patel [1997] emphasized that the algorithm stands out compared

to other algorithms (Depth-first search, Dijkstra, Best-first search, etc.) due to its accuracy and efficiency, this is supported by [Kasim et al. 2017; Barnouti et al. 2016; Elgabry 2016]. On the other hand, Foead et al. [2021] highlighted the algorithm's modularity, performance, and effectiveness; making it a reliable and go-to pathfinding algorithm to choose for indoor navigation systems.

Another crucial technical aspect of the PathSmart system is the ability to determine the location or the starting point of the shopper in the market. GPS is a positioning system that provides users their geographic location using a satellite. While GPS is great for providing a user's geographic location, it is not very effective when it comes to providing location in indoor spaces due to satellite signals blocked by walls emphasized by Mappedin [2022]. So Indoor Position System (IPS) is commonly used to pinpoint the location of the user in indoor spaces, and there are a lot of available IPS out there such as Bluetooth Low Energy (BLE) beacons, Wi-Fi fingerprinting, and Ultra-Wideband (UWB), and many more. While these IPS are great, these options are not feasible for the project due to resource constraints. For instance, according to [Oriient 2020; Moran 2017] BLE beacons require a large number of devices for extensive spaces, which will result in a high installation and maintenance cost (e.g. when the beacons need a battery replacement and recalibration). This is also the case for other methods like Wifi-fingerprinting and Ultra-Wideband (UWB), it requires additional hardware, costly due to installation, maintenance, and recalibration [Mapsted Blog 2025; Alarifi et al. 2016]. In this study, proponents will use an alternative method, the User-Defined Starting Point approach which removes the requirement of such

hardware, no setup complexity, and decreases installation and maintenance costs, thus being a more practical solution for indoor navigation in NCPM.

Related Literatures

The Dijkstra algorithm works by selecting the nodes that have the shortest distance from the source, and then assigning the travel distance to those nodes. The assigned travel distance to each node updates whenever a shorter route is discovered. This process repeats until the shortest path is found. However, according to IRJMETs [2024], the Dijkstra algorithm is inefficient when it comes to large-scale graphs as it visits from the shortest to the longest distance; if the actual shortest path has the longest distance to the initial node, the algorithm would have to check all initial nodes first before finding the right path. The study also highlights that modern applications use optimized pathfinding algorithms or heuristic-based algorithms. That does not mean that Dijkstra is obsolete; it is still suitable for applications that prioritize accuracy over efficiency.

Ipera [2024] This article explores the use of indoor navigation in retail spaces. According to the article, an indoor navigation system enhances both customer experience and business efficiency. The shoppers were provided with interactive maps and real-time directions, making it easier to locate products or departments, which ultimately led to shoppers' increased satisfaction and longer store visits. Additionally, the article also makes use of the data from foot traffic patterns to identify popular areas and shopper behaviors. Retailers can offer

personalized offers based on the data gathered. The indoor navigation helped both shoppers and retailers.

McGee [2025] On March 12, 2025, Scopely, a mobile game maker owned by the Saudi Royal Investment Fund, bought a video games unit of Niantic. Scopely acquired the game “Pokémon Go” which is a game that tracks the location of the player in real-time; the game revolves around that feature. The acquisition of this game sparked discussions about data privacy. According to this article, since Pokémon Go collects location data, it could be used incorrectly. This article provides insight into data privacy with sensitive information like location data.

Patel [1997] His article “Introduction A*” provides an overview of the A* pathfinding algorithm and highlights the differences between pathfinding algorithms. The article emphasizes more on the A* pathfinding algorithm, explaining how the A* algorithm is the best choice for pathfinding due to its characteristics. It integrates both Dijkstra’s exhaustive search and the heuristic-driven approach of the Greedy Best-First Search, which makes the A* algorithm a good pathfinding algorithm to choose from due to its accuracy and efficiency in pathfinding.

Foead et al. [2021] in their research paper "A Systematic Literature Review of A* Pathfinding," study the A* algorithm by reviewing past works, looking at its strengths, weaknesses, challenges, characteristics, and future possibilities. In the study, A* pathfinding algorithm is generally a good choice due to its accuracy, modularity, performance, and efficiency, and the number one choice algorithm if

memory storage is not a major issue. It mixes Dijkstra's careful search with fast guesses (heuristics), making it 40% more efficient than other methods. The strength of the heuristic employed significantly determines the performance of A* — higher-quality heuristics = faster, more reliable results.

Barnouti et al. [2016] In their study entitled "Pathfinding in Strategy Games and Maze Solving Using A* Search Algorithm," they designed an image-based navigation system that transforms game maps or mazes into grid forms and uses the A* search algorithm to calculate optimal routes between an end-user-specified source and destination. Tested on 100 map and maze images, the system successfully found the shortest path in more than 85 % of cases, demonstrating a high level of accuracy and reliability for grid-based indoor navigation tasks.

Javaid [2013] Discusses an in-depth walkthrough of Dijkstra's Algorithm, demonstrating its operation in finding the shortest distance from a single source node to all other nodes in a graph. The research outlines how Dijkstra's algorithm updates the path costs iteratively and efficiently by choosing the node of minimum cost at each iteration. Javaid shows its real-world application in a six-node network topology, illustrating its efficiency in routing within networks and topology optimization. The results concluded with precise path reconstruction, which enhances Dijkstra's credibility in the computation of optimal routing paths.

Rachmawati and Gustin [2020], in their study *"Analysis of Dijkstra's Algorithm and A Algorithm in Shortest Path Problem"*, compare the performance of *Dijkstra's Algorithm and A* in solving shortest-path problems. They note that

although Dijkstra's algorithm produces the optimal path, it achieves this by scanning all the potential nodes, thereby being less efficient, especially with large graphs. Compared to that, A* cuts computation time dramatically using heuristics to focus the search effort. While the optimality of the path for A* relies upon admissibility of its heuristic (i.e., not more than true cost), the study shows A* always performed better than Dijkstra's algorithm while computing single-pair shortest paths (SPSP). A* is particularly appropriate for use with applications where navigation needs to happen rapidly, point-to-point.

Kasim et al. [2017] The study introduces an Android-based indoor navigation system developed to help new students at the University Tun Hussein Onn Malaysia (UTHM) locate classrooms within the Faculty of Computer Science and Information Technology. The application implements the A* (A-Star) algorithm, a pathfinding method that calculates the shortest route from a selected starting point to a destination on a 2D floor plan. The system guides users by visualizing the path on the map, offering a user-friendly and time-saving alternative to conventional navigation apps, which are often ineffective indoors. According to the study, the A* algorithm improves efficiency by integrating a heuristic function that estimates the distance to the goal, making it more suitable than traditional algorithms for indoor environments. This approach emphasizes accuracy and speed, which are critical for real-time navigation in complex structures like university buildings or markets.

Yan [2023] In his paper titled "Research on the A Star Algorithm for Finding Shortest Path", Yan examined the A* algorithm principles and compared it to the

Dijkstra algorithm, which he found out that A* is more efficient for determining the shortest path in grid environments, focusing on path planning. Yan discovered that heuristic search greatly limits the number of nodes searched by favoring those most likely to reach the goal, thus optimizing computational efficiency. They emphasize that an appropriately designed heuristic function can reduce search costs and determine an optimal path efficiently.

Chumbley et al. [2020] In their article titled “Bellman-Ford Algorithm”. They provided a detailed overview of the Bellman-Ford algorithm. It is a graph search algorithm designed to find the shortest paths from a given source vertex to all other vertices within a graph. Unlike Dijkstra, Bellman-Ford is capable of working on graphs with negative weights for edges, which makes it more versatile. It can also find negative cycles. Bellman-Ford is slower than Dijkstra, though.

Elgabry [2016] In his article entitled "Path Finding Algorithms", Elgabry discusses how different pathfinding algorithms—BFS, DFS (recursive and iterative), Dijkstra, Greedy, and A*—are utilized to traverse trees and compute the shortest path between a start node and a goal node. He states that uninformed or blind search algorithms, such as breadth-first search and depth-first search, tend to be slower compared to informed search algorithms like A* and Greedy Best-First Search. This is because uninformed algorithms do not know anything about the goal node and have to search through every possible path, hence, they are less efficient than informed algorithms that apply heuristic approaches to direct the search.

Elkari et al. [2024] conducted a comparative study to evaluate the performance of Depth-First Search (DFS), Breadth-First Search (BFS), and A* search algorithms in maze navigation. The research utilized Python and the Pmaze library to simulate path planning in a maze environment, focusing on algorithmic efficiency in terms of path cost and computational complexity. The study found that A* is better at computing optimal paths efficiently, with the help of the heuristic to reduce node exploration. BFS guarantees optimal paths but has higher computational complexity as a result of exhaustive node exploration and is best used when there are lots of resources available. DFS tends to create nonoptimal paths with less memory requirement but greater time complexity.

Codecademy [2023] In their article "Greedy Best-First Search", they discuss the Greedy Best-First Search (GBFS) algorithm, an informed search method applied to pathfinding in graphs. The article describes GBFS's operation, which depends only on a heuristic function to select nodes nearest to the goal, and addresses its efficiency, pros and cons. GBFS provides quick exploration by expanding nodes with the minimum heuristic value, with low memory requirements and easy implementation. But it generates non-optimal paths, disregards edge weights, and is incomplete, with a chance to miss solutions or get trapped in local minima. Its performance relies on the accuracy of heuristics.

Iyanda [2023] Comparative analysis of Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms, two basic graph traversal methods employed in pathfinding, network analysis, and puzzle-solving. The research discusses their working principles, applications, time and space complexities, and appropriateness

for various problem domains. The study discusses BFS and DFS in the context of graph traversal techniques, contrasting the way they explore graphs. BFS ensures a minimal path in unweighted graphs through level-by-level neighbor exploration, being ideal in maze solving and network broadcasting, but needing lots of memory as a result of the FIFO queue. DFS, more conserving in terms of memory, digs branches more deeply before attempting to backtrack, being ideal in cycle detection as well as games and puzzles, but prone to giving suboptimal paths.

Moran [2017], in this article, entitled “4 reasons why Mapsted Technology doesn’t require beacons for indoor positioning” explained the cons of the BLE beacons. Moran states that “each beacon typically provides a signal that only covers about 150 square feet”, so it requires a lot of beacons especially for extensive indoor spaces like malls, hence making it costly. That means that if the indoor space has its floor area of 30,000 square meters then it would require about 200 BLE beacons. The article also states that the BLE beacons maintenance are costly and time consuming, BLE beacons only last about six months.

Oriente [2020], in this article, entitled “Comparing Indoor Positioning Technologies: Which one is right for you?”, talks about various IPS technologies such as BLE beacons, WIFI fingerprinting, UWB, etc and its pros and cons. The article stated that the required number of beacons is determined by the size of the building and layout of signal obstructions (furniture, people, etc.), the similar idea stated in Moran [2017]. The article also said that the beacons are usually battery-operated, this means that the installation and maintenance costs of the beacons are high and should be taken into account.

Mappedin [2022], in this article, entitled “Blue Dot Navigation for Stadiums” talks about Google Maps using GPS as a tool to provide navigation and geographic location to users outdoors. In the article, they explain that Google Maps uses a Global Positioning System which uses satellites to pinpoint a location geographically. Furthermore, they explain that although Google Maps is great for outdoor navigation, it falls short and limited when it comes to providing wayfinding in indoor spaces due to GPS signals being blocked by walls resulting in GPS not reliable to use as ISP for indoor spaces.

Naga City Tourism [2024]. In this article, entitled “Naga City People’s Mall”. It mentioned there that Naga City People’s Mall houses over thousands of stalls, offers a wide range of products, the total floor area of 30,000 square meters and has three floor levels.

City Government of Naga [2023]. In this article, entitled “Naga welcomes 2.2 million tourists in ‘22”. In the article it talks about the total number of tourist visits in Naga City both domestic and foreign in 2022. It stated in the article that they have recorded 2,126,765 million domestic tourists while the other 57,522 came from foreign countries in the span of a year, recorded every quarter (e.g. Jan-March, April-June, etc.).

Alarifi et al. [2016]. In this study, entitled “Ultra Wideband Indoor Positioning Techniques: Analysis and Recent advances” conducted a detailed comparative analysis of UWB compared to other IPS methods. In the study, they emphasized that although UWB IPS is highly accurate (centimeter-level accuracy) but it is

expensive compared to other IPS methods and has complex setup processes, for example, it stated in the study that “The design and implementation of antennas for UWB systems can be more challenging”.

Mapsted Blog [2025]. In this article, entitled “What is Fingerprint Positioning—Advantages and Disadvantages and How Does it Work?” They talked about the WIFI fingerprinting, its pros and cons, and how the IPS works. Although we could use a pre-existing wifi infrastructure, it demands a frequent calibration of reference points whenever environmental changes occur, especially for hybrid environments where change is constant. It also demands labor-intensive surveys, which means that before the Wi-Fi fingerprinting can work, someone has to walk around while carrying the device physically to the whole area and measure Wi-Fi signal strength at many points to understand the surroundings, the information and the location coordinates are then now is stored in a database.

Related Studies

Renaudin et al. [2019] An event called “The International Conference on Indoor Positioning and Indoor Navigation (IPIN) 2018” was held to test the capabilities of indoor positioning technologies. The evaluation took place in a shopping mall. The results were based on the accuracy and effectiveness of the technology. The results showed that there is no single technology that can provide flawless position tracking because of environmental factors. However, combining technologies to create a hybrid navigation system will make a more accurate positioning system.

Zhang & Zhang [2023] The A-star algorithm is an improved Dijkstra algorithm, it uses a heuristic function to find the optimal path. The study discusses different heuristic approaches, the Euclidean distance, and the Manhattan distance. Euclidean distance measures the distance in a straight line from source to destination. Manhattan distance measures the number of horizontal and vertical moves. The study stresses choosing the best heuristic approach. A well-chosen heuristic will significantly improve the efficiency of the A-star pathfinding algorithm. According to the study, the A-star with a well-chosen heuristic is better than the Dijkstra algorithm.

Alija, A. S [2015] The study compares Dijkstra and A-star algorithms. Using Visual C++ 2008, both algorithms were tested on 8×8, 10×10, 22×22, and 33×33 grid environments. Both algorithms resulted in the same shortest path. A-star is significantly better than Dijkstra, according to the tests. A-star averages 466ms across all tests, while Dijkstra's Algorithm averaged 1102ms. On the 33x33 grid, A-star found the shortest path at 1123ms, while Dijkstra took 2781ms. The result shows that A-star is better in larger environments. The study implies that A-star is faster due to its heuristic function, which reduces unnecessary node explorations.

Rupasinghe [2023] The study conducted an analysis of different pathfinding algorithms, placing them in different environments, the study tested the efficiency, accuracy, and adaptability of the algorithms. The study examined over 150 research papers related to pathfinding. The study highlights that there are performance trade-offs between pathfinding algorithms when tested in changing terrains and real-time navigation. The findings suggest that while the A* star

algorithm provides a balance between speed and optimality, other algorithms can still outperform it if it's placed in the right environment.

Nash et al. [2007] This study acknowledges the limitations of common pathfinding algorithms that can only map vertically and horizontally. The authors introduced Theta*, their pathfinding algorithm that is designed to compute any-angle paths on grid-based maps. There are 2 variants of Theta*, Basic Theta* and Angle-Propagation Theta*. Basic Theta* is simple and fast, but doesn't always show the true shortest path. Angle-Propagation Theta*, on the other hand, has angle propagation, which allows it to determine paths more efficiently while maintaining flexibility in path selection, but it may produce slightly longer paths. This study is significant because it proves that any angle paths are possible.

Hammami et al. [2020] The study is about how consumers are affected by the changes in the appearance of fruits and vegetables. The authors surveyed 366 US consumers to determine how these consumers assess the freshness of products. According to the study, a noticeable change is the discoloration, such as browning in apples, dark spots on bananas, and yellowing in leafy greens. This study also includes the reason for this discoloration. This study provides useful information that can be used for the development of the quality guide feature in this project. Some people might not be aware of these visual indicators of spoilage, but with the quality guide, they will be aware of them.

De Hooge et al. [2017] The study explores the preference of consumers when it comes to products that have slight imperfections. The authors surveyed

4,214 consumers across Denmark, Germany, Norway, and the Netherlands. They also observed the demographics of the consumers. According to the study, consumers would reject items with slight imperfections such as brown spots, blemishes, odd shapes, and “off” smell. However, in supermarkets, the rejection is lower, it is tolerable for the consumers as long as the smell and taste are acceptable. Per the demographics, the study states that the younger consumers are more tolerant when it comes to slightly imperfect products. This study is significant in understanding the preferences of consumers in different age groups.

Mubarak et al. [2019] The study is about the integration of technology in businesses. The authors of this study gathered data from 250 business owners in Pakistan. They focused on 4 technologies: big data, Cyber-Physical Systems(CPS), IoT(Internet of Things), and Interoperability. Through questionnaires and surveys, they found that businesses with lots of data, smart machines, and connected systems did better, with an increase of 20%-31% in speed and profit. However, the Internet of Things is excluded from this.

Wielgos et al. [2021] The study is about how skills in using digital technology impact firms and customer performance. The author wants to know if being good at using digital tech helps businesses make more money. The data gathering method used was interviews across industries. They did 2 studies. Study 1 assessed 356 firms across different industries. According to the study, being good with technology beats just “knowing the market” or “being customer-focused”. Study 2 focused on how customers feel, they surveyed 203 firms, collecting data from customers and managers. According to the study, customers like it when

businesses integrate digital technology since it makes it easier for them, making them happy and stick around longer.

Rahman & Alam [2018] The article provides an overview of algorithms employed in indoor navigation, such as Dijkstra, A*, and D*. It discusses the strengths and weaknesses of each algorithm in the indoor environment and how hybrid solutions or sensor fusion can enhance accuracy. The article is a general guide to pathfinding system developers in small environments.

Cheng & Wang [2022] The research examines what indoor navigation systems in shopping malls are expected to provide by users. It defies the premise that the shortest route is the best, presenting evidence that users find routes based on comfort, familiarity, or convenience more desirable. The research indicates that indoor navigation systems consider user preferences such as avoiding congested areas or traversing certain stores over shortest-path algorithms.

Deliyska et al. [2021] The article provides indoor navigation system construction with an emphasis on the use of popular navigation apps. It points out the need for personalization of navigation solutions for particular environments, e.g., markets, taking into account organizational and design considerations.

Khairy et al. [2017] The study introduces an indoor navigation app that uses the A* algorithm to calculate the shortest path for users. Deployed on Android phones, the system offers real-time navigation directions, showing the use of pathfinding algorithms in indoor settings such as markets.

Related Systems

Techbreeze [2025] The article discusses how google maps navigation works. The article talks about the algorithms that Google Maps uses. According to the article, Google uses a combination of Dijkstra and A-star algorithms; the Dijkstra algorithm ensures the correct shortest path, while A-star improves the efficiency by using a heuristic function. Additionally, the article discusses other features of Google Maps, such as real-time traffic data and machine learning models that analyze the weather, time of day, and road infrastructure.

Mappedin [2024] Mappedin is an indoor mapping platform that provides digital interactive maps accessible through web and mobile applications. The article explores places where indoor mapping can be adopted to improve navigation and user experience, such as malls, hospitals, airports, and office buildings.

Living Map [2025] Their article “Case Study: Canary Wharf” discusses the successful implementation of a case study in Canary Wharf, a renowned shopping destination in London. In the case study, Living Map showcases how digital wayfinding has enhanced visitors’ experiences in Canary Wharf. The system provides interactive maps accessible via kiosks and mobile devices (accessible through QR codes), allowing users to easily search for shops, restaurants, and amenities using their mobile phones.

Mapsted [2024] Mapsted's article describes a cutting-edge indoor navigation solution that is aimed at navigating users through complex indoor and outdoor

environments, including malls or campuses. The solution provides advanced real-time indoor navigation and mapping, delivering precise turn-by-turn guidance. It incorporates horizontal and vertical positioning, along with multi-building navigation capabilities, enabling seamless navigation across complex indoor-outdoor venues.

Synthesis

The conclusive summary of relevant literature, studies, and existing systems provides insights that the proposed project is not a duplicate and is a continuation/improvement of a previous work.

The related literature shows different pathfinding algorithms to determine the best and reliable pathfinding algorithm to be used for the development of PathSmart. Pathfinding algorithms listed are A*, Bellman-Ford, Dijkstra, DFS, BFS and many more. In this literature it shows that A* stand out from the rest listed algorithms, and also generally a good algorithm to choose from due to its characteristics such as accuracy and efficiency, since its combination of power of Dijkstra algorithm exhaustive search and the heuristic-driven of Greedy Best-First search, resulting in a good accuracy and efficiency. These characteristics are also supported by [Kasim et al. 2017; Foead et al. 2021; Yan 2023; Elkari et al. 2024; Elgabry 2016]. For this reason, proponents used A* algorithm to develop PathSmart.

Similarly, findings from related studies supports how A* the pathfinding algorithm outperformed other pathfinding algorithms, and is generally a good

choice when it comes to providing an optimal path due to the help of a heuristic function. In the listed studies, it emphasizes that A*'s advantage lies from the use of heuristic functions, which guide the search and lessen the unnecessary node exploration, resulting in an optimal path [Alija, A. S 2015; Zhang & Zhang 2023;].

The related system demonstrates that the indoor navigation system has been widely used in big and complex environments such as airports, universities, malls, markets and many more to improve its wayfinding, and guide customers find their way through these environments. For example, [Mapsted 2024; Living Map 2025; Mappedin 2024], showcase how an indoor navigation system can be used to complex and big environments to improve its wayfinding, resulting in better customer experience. This supports the relevance of the development of PathSmart for NCPM, the PathSmart aims to improve NCPM's customer experience.

Additionally, in the literatures, it shows that most of the indoor solutions or systems use an indoor positioning system (IPS) that requires additional hardware, have complex setup processes, and infrastructure [Oriient 2023; Moran 2019; Alarifi et al. 2016; Mapsted Blog 2025] just to determine the location of the user inside indoor spaces. None of them uses a lightweight and cost-effective solution. The PathSmart system addresses this gap by using the A* pathfinding algorithm with a user-defined starting point method. User-Defined starting point is the alternative method for the PathSmart system for finding the location of the shopper inside the market, this approach eliminates the need for costly or additional

hardware, installation, maintenance, and recalibration, offering a practical, low-cost navigation indoor positioning system (IPS) solution.

DESIGN AND METHODOLOGY

This chapter provides a comprehensive overview of the research and development methodology, including the requirements analysis and specification, system design, development methodology, and testing and evaluation procedures.

Research Methodology

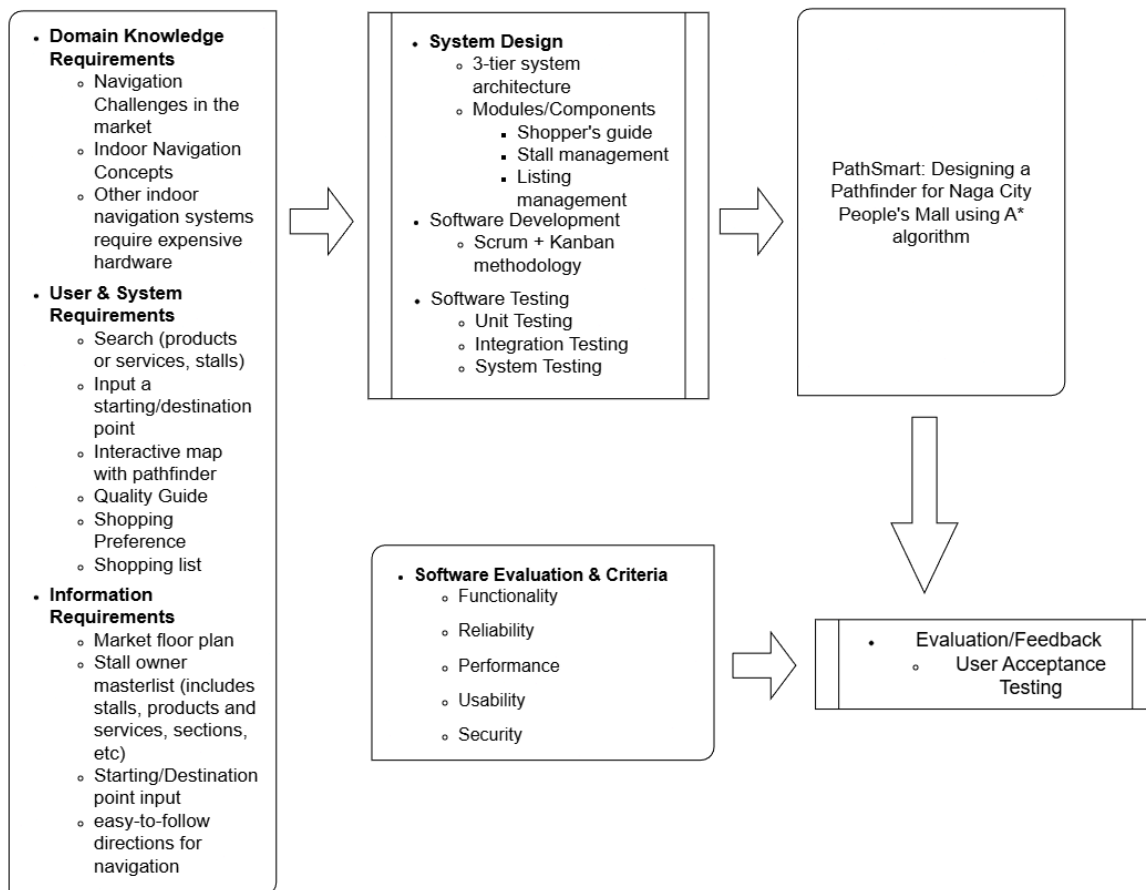


Figure 1: Conceptual Framework of Exploratory Research for the Development of the PathSmart system

Figure 1 illustrates how exploratory research will guide the development of the PathSmart system through a qualitative and quantitative research approach. The framework follows an Input–Process–Output (IPO) model with a feedback loop. The interviews and document analysis are used to derive user, system, and input requirements, and also understand pain points and problems faced by the stakeholder. This information helps us to derive the system architecture and identify the necessary modules for the system, and also determine the software development methodology suitable for the development of the project and the testing plan (unit, integration, and system). The system is evaluated by using defined criteria such as functionality, reliability, security, and so on. These findings from evaluations reflected back into the requirements and design for the purpose of refinement and feedback from its real users until they are satisfied and acceptance is met before releasing it, and going live.

Data Collection Methods

The following data collection methods were employed in this project study:

- Interviews. An interview is a direct communication with someone or group of people. In this study, proponents have conducted interviews with the stakeholders in Naga City People's Mall (NCPM). In this method, proponents have prepared interview questions that we are going to ask to

stakeholders, and then went to the MEPO office to talk , and have a direct, and a one-on-one interview with the stakeholders for the purpose of gaining an understanding of their needs, the problems or pain points they are currently facing, get the documents and other important details.

- Document Analysis. Document Analysis is a qualitative data collecting method that involves reviewing and analyzing documents (digital or physical) to get a better understanding, context or insights about the data. In this study, proponents went to the stakeholders, and requested the necessary documents, physical and digital to gain understanding and insights, and also help proponents gather the data needed to develop PathSmart. Physical and digital documents such as floor plan of the market, masterlist that includes the stalls, the sections in the market, numbers of stalls, and products and services being offered in the market are gathered.

Data Analysis Techniques

The data collected was analyzed using the following techniques:

- **Content Analysis.** Content analysis is an analysis technique for reviewing and interpreting documents whether digital or physical for understanding. In this study, content analysis was used on the documents gathered from the stakeholders of the market such as the masterlist of stall owners that includes important data such as list of stalls and their owners, number of stalls, market's floor plan, and products and services available. This technique helps proponents better understand, review, and interpret the data that was collected from the stakeholders.
- **Thematic Analysis.** Analysis technique where qualitative data is analyzed by finding repeating ideas or meaning and patterns from people through methods like interviews. In this study, proponents used thematic analysis to the responses from interviews with stakeholders. This technique helps proponents to gain understanding of ideas, problems or pain points, and needs from interview sessions from stakeholders.

Software Development Methodology

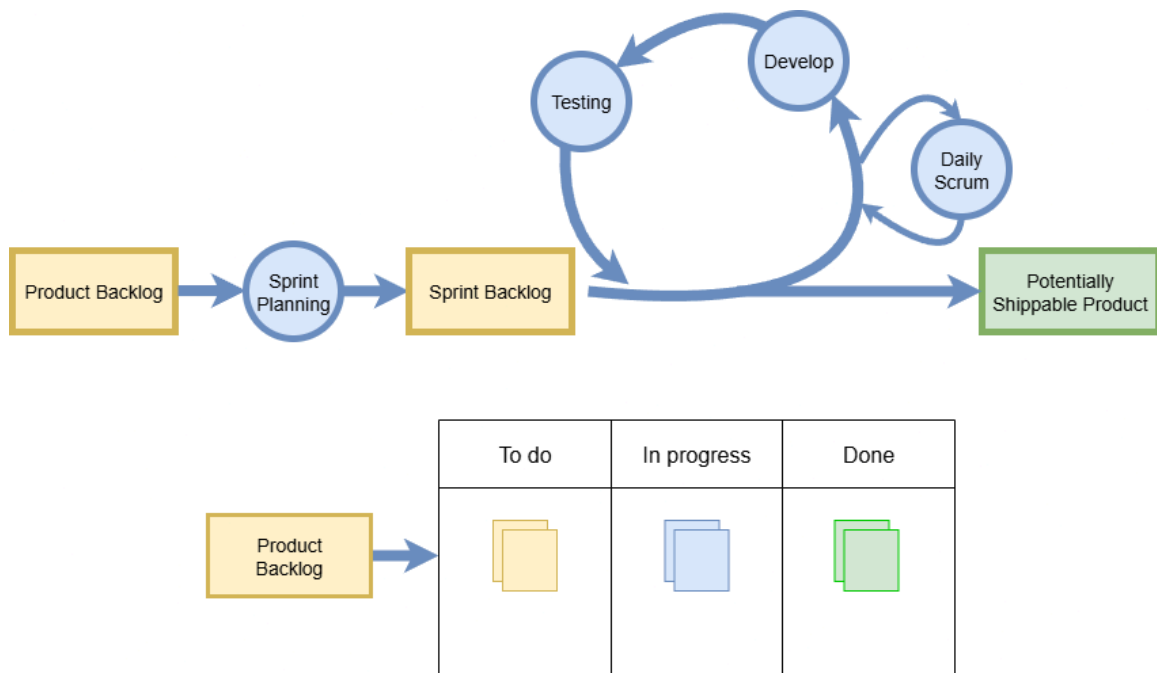


Figure 2: Scrum + Kanban

In this study, proponents used the Scrum + Kanban as a software development methodology; it guided the planning, development, and testing of the PathSmart indoor navigation system. This approach helped the proponents implement and evaluate the A* algorithm in accordance with the objectives of the study. Scrum supported the step-by-step development of system features and functionalities, while Kanban provided clear visualization of tasks and progress, Martins, J. [2025].

Product Backlog. This phase contains the list of all the necessary features and functionalities that the system must have for the project to be completed, Indeed Editorial Team. [2021]. In this study, the Product Backlog phase guided the

proponents in identifying all system features and functionalities that the system needed to develop the PathSmart system. e.g. Indoor mapping. This is where we also defined the roles such as the Scrum Master and development members.

Milestone: Completion of the product backlog with all listed system features and functionalities by the proponents, assigned team roles is identified.

Output: A complete list of all features and functionalities for the project, and assigned team roles are identified.

Deliverable: The complete Product Backlog that contains the features and functionalities for the project and team roles are also known after this phase.

Sprint Planning. The phase where proponents discuss and create their plan for the next sprint, Indeed Editorial Team. [2021]. Sprint planning guided proponents plan for the next sprint, which involves picking the next features or functionalities from the product backlog, and then move them to the sprint backlog which will determine the tasks needed that the proponents need to work on for this upcoming sprint. Tasks or features such as indoor mapping, implementing the A* algorithm, and so on.

Milestone: Sprint plan Complete, product backlogs and sprint backlog updated.

Output: A list of prioritized tasks for the sprint.

Deliverable: The finalized Sprint Backlog, which determined the goals and tasks for the upcoming sprint.

Spring Backlog. In this study, Sprint Backlog guided proponents by clearly defining the list of tasks that was selected from the product backlog which the proponents will work on during the sprint, Indeed Editorial Team. [2021]. These tasks included implementing the A* algorithm, integrating indoor navigation data, and preparing the A* for testing.

Milestone: Completion of sprint backlog containing the tasks selected for the upcoming sprint.

Output: A list of selected tasks, features or user stories assigned to each proponent.

Deliverable: The Sprint Backlog.

Implementation (develop, test). This phase is where proponents will implement the plan for the sprint, Indeed Editorial Team. [2021]. This phase guided proponents in developing and testing the PathSmart system. During this phase, the A* algorithm was executed on the NCPM indoor layout. Route accuracy was verified by checking if the generated paths were optimal, while efficiency was measured by observing the time taken to generate routes

Milestone: Completion of the sprint's development and testing

Output: Developed and tested system features based on the sprint backlog.

Deliverable: A shippable product for stakeholders.

Requirement Analysis & Specifications

This section illustrates the proposed system processes and defines the functional and non-functional requirements that the PathSmart system must meet to satisfy the user's needs.

Process Flow Models

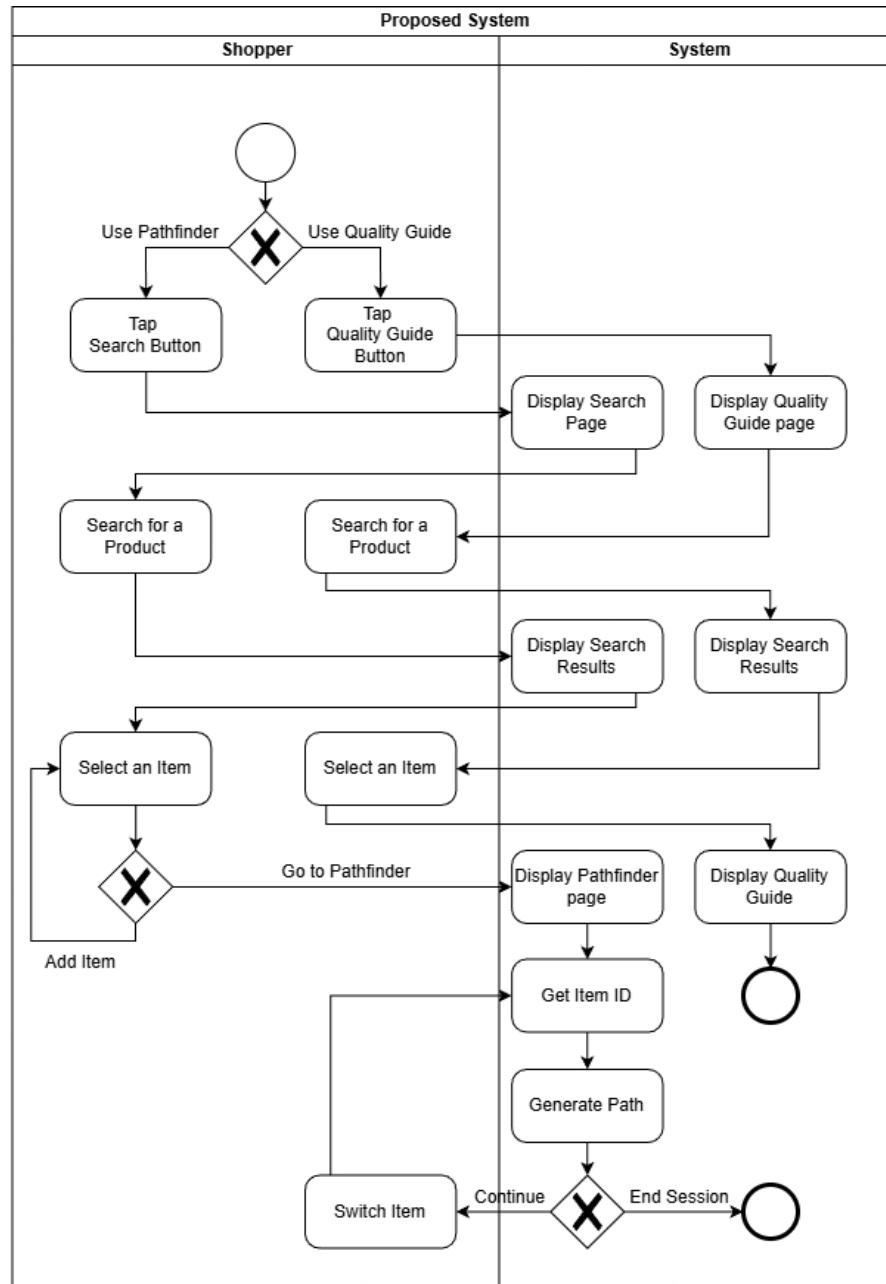


Figure 3: Process Flow Diagram of Proposed [Shopper Guide] (BPMN2.0)

The figure above shows the BPMN2.0 for the Shopper Guide which consists of core components, the pathfinder and the quality guide. The diagram shows the procedure that the shopper will go through while using the shopper guide.

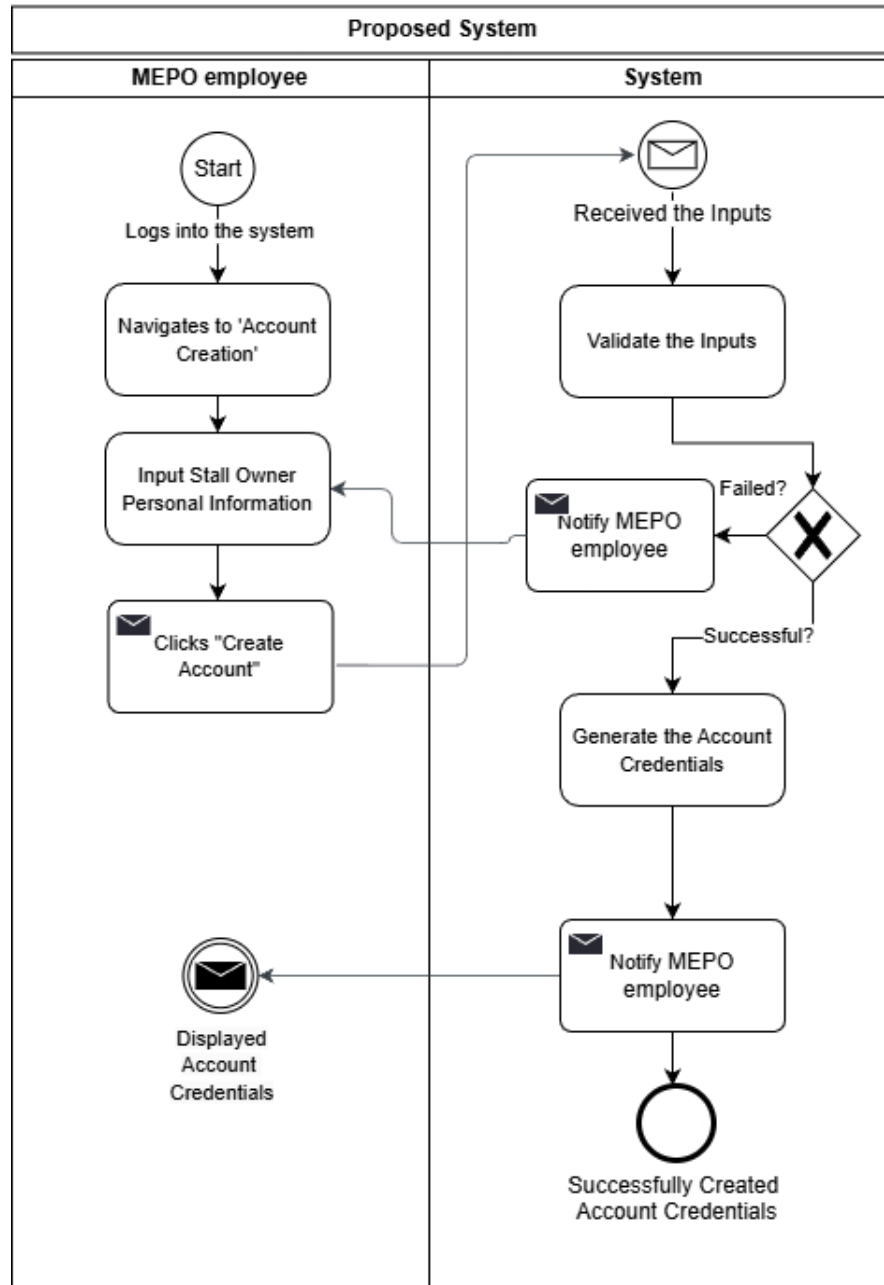
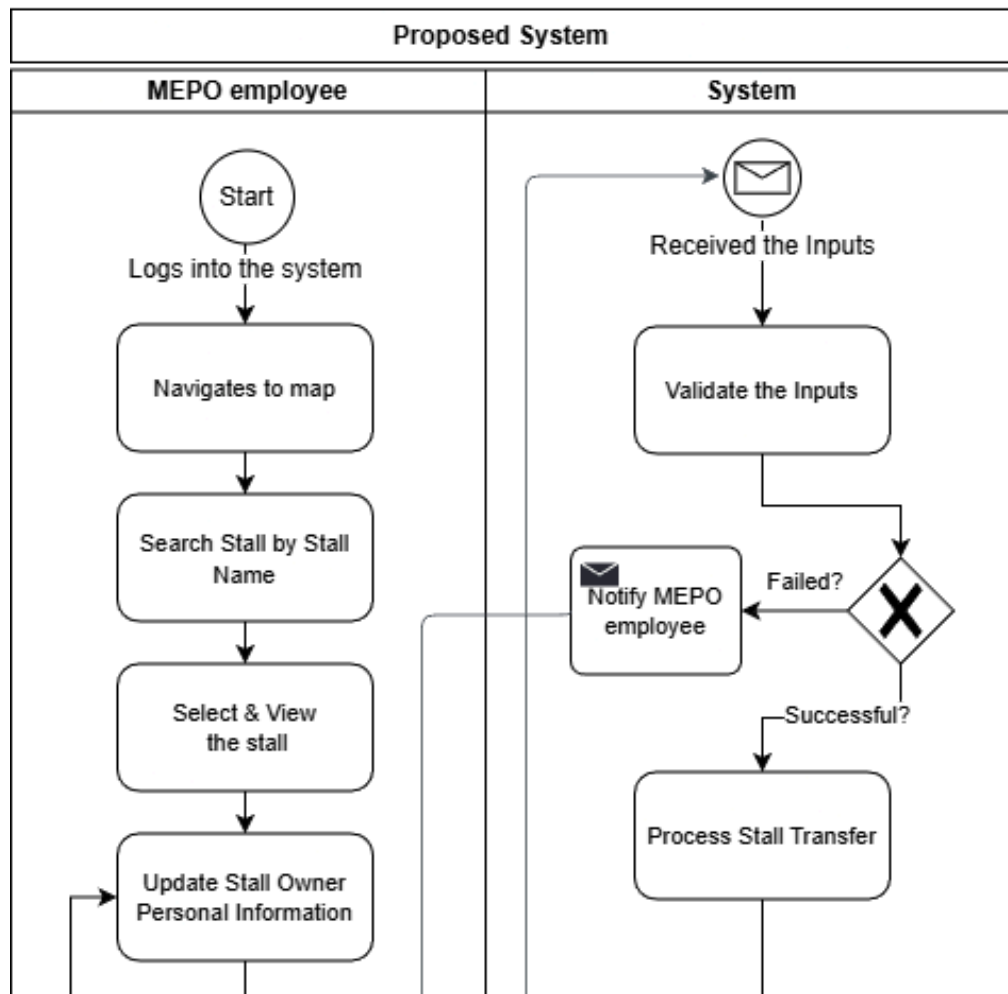


Figure 4: Process Flow Diagram of Proposed [Account Creation] (BPMN2.0)

The figure above shows the BPMN2.0 for Account creation. In this diagram, it visually shows the steps that MEPO will go through when creating an account for stall owners in the system.



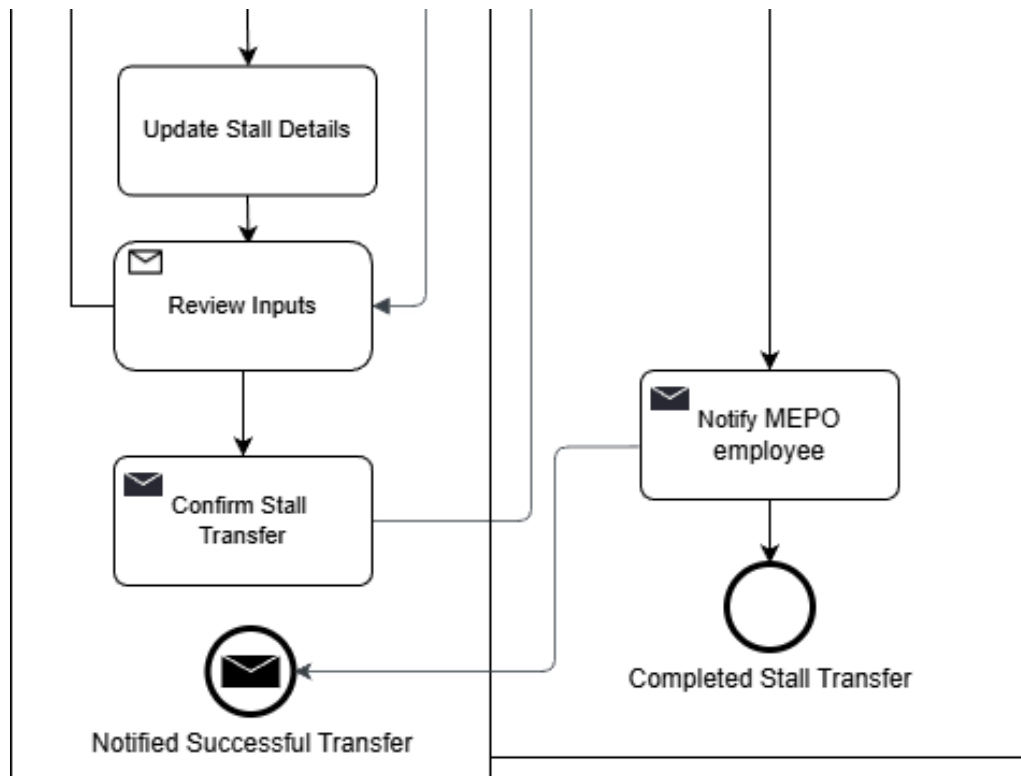


Figure 5: Process Flow Diagram of Proposed [Transfer Stall Ownership] (BPMN2.0)

The figure above shows the BPMN2.0 for Transfer Stall Ownership or simply updating the stall details. In this diagram, it visually shows the steps that MEPO will go through the process of updating the stall details.

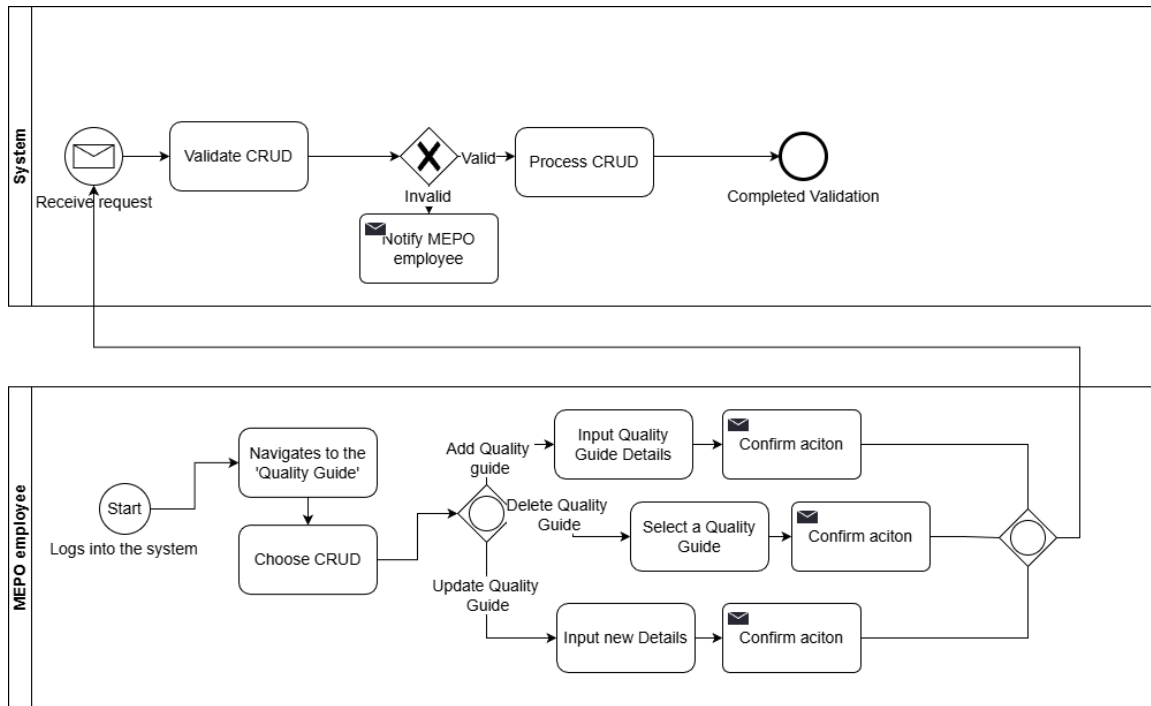


Figure 6: Process Flow Diagram of Proposed [Manage Quality Guide] (BPMN2.0)

The figure above shows the BPMN2.0 for Managing quality guide. In this diagram, it visually shows the steps that MEPO will go through in the process of managing quality guides, which includes adding, deleting, and updating the quality guide entry.

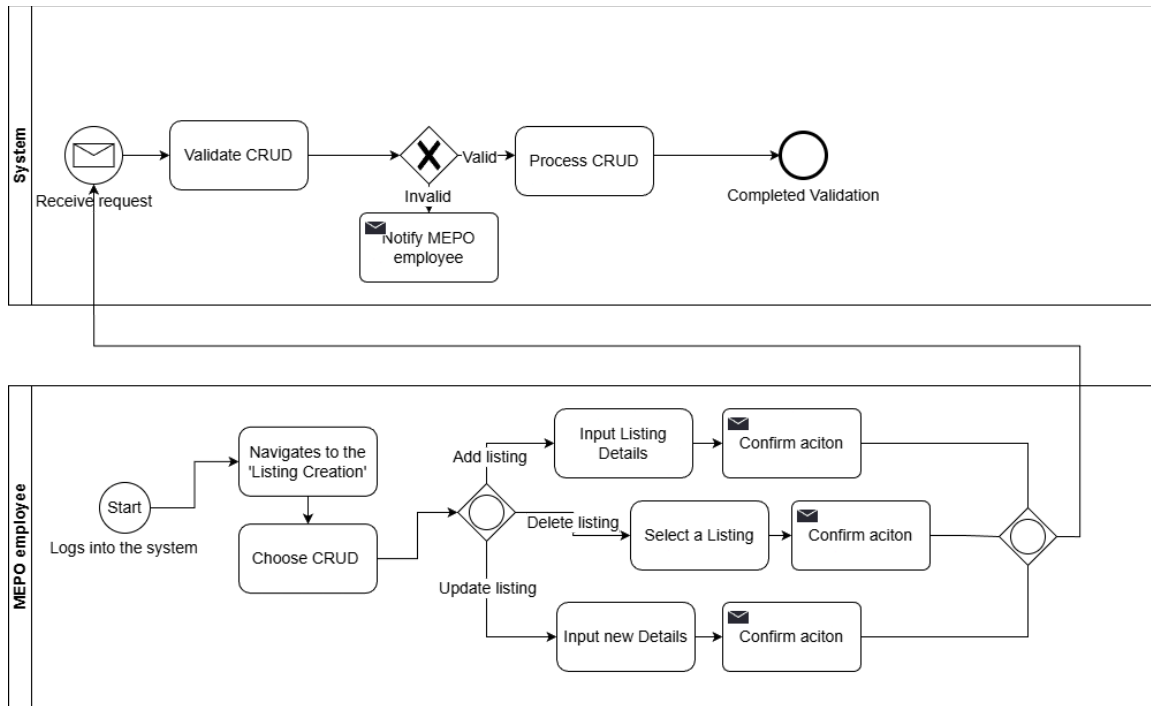


Figure 7: Process Flow Diagram of Current [Listing Creation] (BPMN2.0)

The figure above shows the BPMN2.0 for Listing creation. In this diagram, it visually shows the steps that MEPO will go through in the process of creating the listings. The data entry for the listing management module that allows stall owners to select the products and services they are offering in their stall.

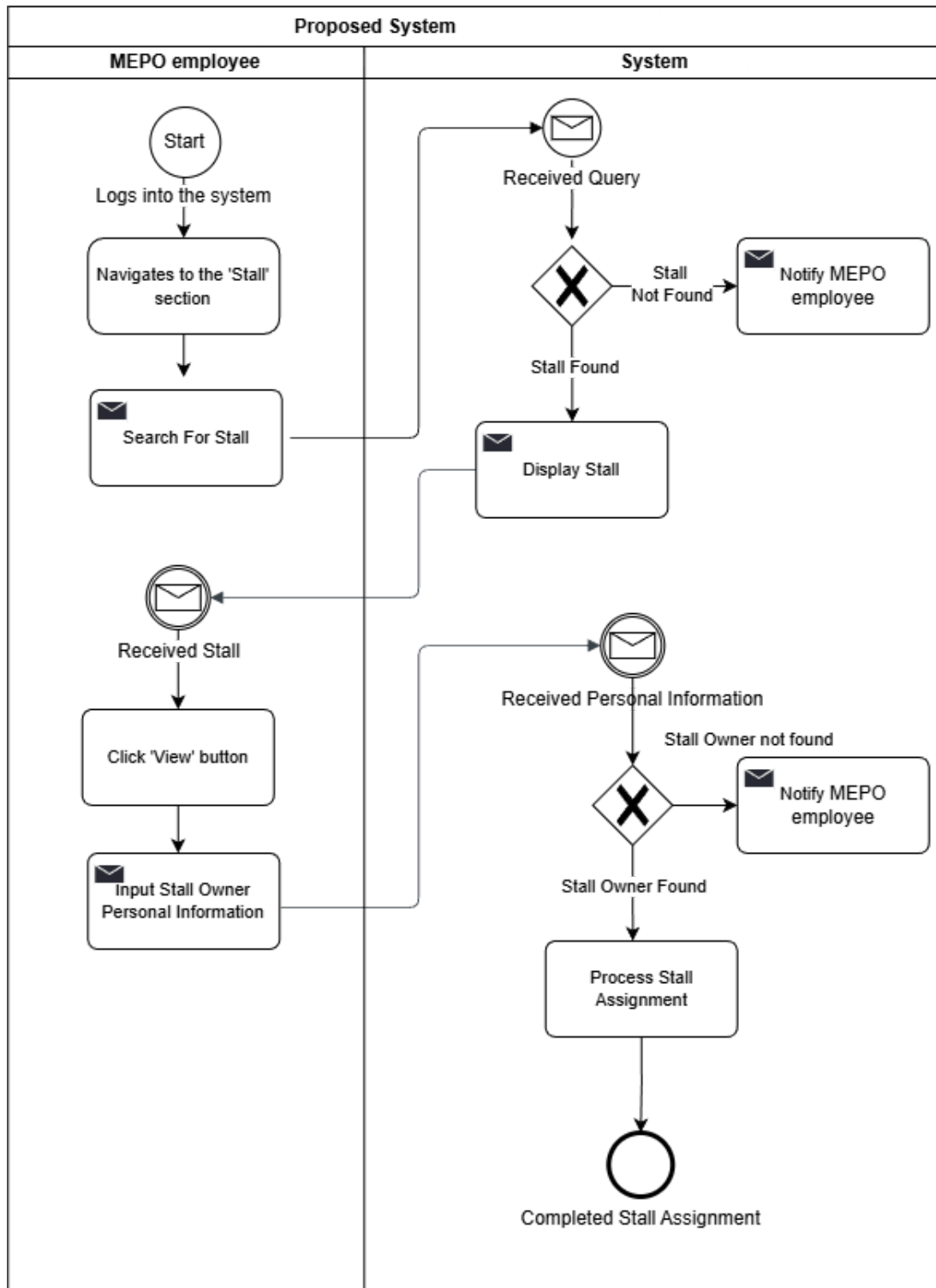


Figure 8: Process Flow Diagram of Proposed [Assign Stall] (BPMN2.0)

The figure above shows the BPMN2.0 for Assigning of stall. In this diagram, it visually shows the steps that MEPO will go through the process of assigning a specific stall to the stall owner.

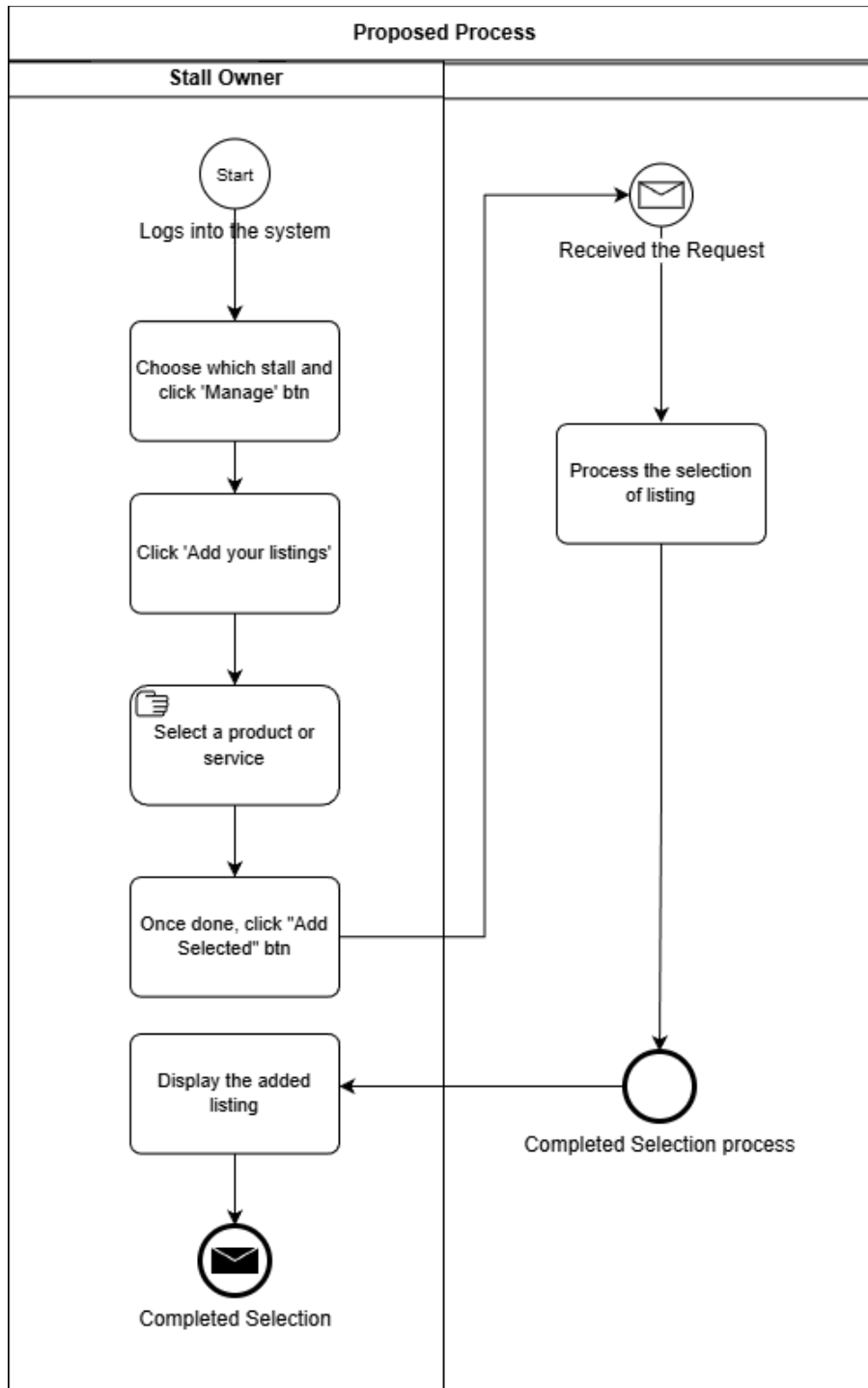


Figure 9: Process Flow Diagram of Proposed [Select/Add Listing] (BPMN2.0)

The figure above shows the BPMN2.0 for Select/Add of Listing. In this diagram, it visually shows the steps that Stall owners will go through the process of selecting/adding listings for their stall.

Functional Requirements

Shopper Guide

1. Performance

- 1.1. The application shall be compatible with both Android and iOS devices.
- 1.2. The application shall support offline mode.
- 1.3. The application shall support multiple users at the same time.

2. Search

- 2.1. The application shall allow users to search for products, stores, or services.
- 2.2. The application shall allow users to filter search results based on category(product, stores, services).
- 2.3. The application shall allow searches in Bicol, Tagalog, and English languages.

3. Pathfinder

3.1. The application shall track the user's location in real-time within the market.

3.2. The application shall provide an interactive map for navigation within the market.

3.3. The application shall provide the shortest route to the target location.

3.4. The application shall have a rerouting feature.

3.5. The application shall highlight stores that offer the searched product.

4. Quality Guide

4.1. Users shall be able to see the suggested retail prices of products.

4.2. The quality guide shall show pictures of good and bad product quality.

4.3. The quality guide shall show descriptions of good and bad product quality.

4.4. The quality guide shall allow users to report misleading or incorrect information.

4.5. The quality guide shall be accessible from both the search result and a dedicated guide section.

Stall Management

1. Account creation

1.1. The system shall allow the MEPO employee to create stall owner accounts.

1.2. The system shall generate a unique username and a password for stall owners to log in and access the system.

2. Login auth

1.3. The system shall allow MEPO employees to log in using a username and password.

4. CRUD Quality Guide

1.4. The system shall allow the MEPO employee to perform CRUD operations on quality guide entries.

1.5. The system shall allow the MEPO employee to search for quality guide entries.

5. Listings Creation

1.6. The system shall allow the MEPO employee to navigate through pages in the listings creation

1.7. The system shall allow the MEPO employee to perform CRUD operations on products and services that will serve as a data entry for the listing management module.

1.8. The system shall allow the MEPO employee to search for specific products or services within the listings.

7. Manage Stall

1.9. The system shall allow the MEPO employee to assign specific stalls to individual stall owners.

1.10. The system shall allow MEPO employees to Update stall details.

Listing Management

1. Login and Authentication

1.1. The system shall allow stall owners to log in using a username and password.

2. Dashboard and Stall Access

2.1. The system should get and show all the stalls that the authenticated Stall Owner owns.

3. View and Manage Listings

3.1. The system must let the Stall Owner see all of the listings for the stall they chose.

3.2. The system should show the following information for each listing: name of the product or service, category, price, and availability.

3.3. The system should let the Stall Owner change a listing, such as the price and availability.

4. Add New Listings

4.1. The Stall Owner will be able to search and filter products by category using the system.

Non-functional Requirements

Shopper's Guide

- Performance:
 - The application shall load within 1 second after the loading animation.
 - The pathfinder shall show the generated path within 2 seconds.
 - The application shall switch between levels in the pathfinder within 3 seconds.
 - The search shall show results within 3 seconds after entering a search query.

- The quality guide shall load within 3 seconds.
- Security:
 - The application shall not store shoppers' location data.
 - The application shall not share shoppers' location data.
- Reliability:
 - The quality guide shall provide factual information.
 - The application shall regularly update the SRP of products.
 - The pathfinder shall give accurate and complete routes.
- Usability:
 - The system shall have a simple user interface that users with minimal smartphone literacy understand.
 - The application features shall be accessible within 3 steps.
 - The application shall not require manual typing aside from search.
- Availability:
 - The application shall be available 24/7.
- Scalability
 - The application shall support concurrent use by at least 500 users without crashing.

- The application shall be able to handle the growing number of products and quality guides without performance degradation.

Stall Management

- Performance:
 - The system shall respond to user actions (e.g., login, search, and form submissions) within 2 seconds under normal load.
- Reliability:
 - The system shall perform its intended functions correctly at least 99.9% of the time
- Scalability:
 - It shall be scalable to support additional products/services listings and quality guide entries.
- Availability:
 - The system shall be available at least 99.9% of the time.
 - The system shall notify users in advance of any planned maintenance.
- Usability:

- The MEPO employee interface shall be designed to be user-friendly and intuitive.
- The system shall display clear error messages with guidance when invalid inputs or errors occur.
- Security:
 - All sensitive data (e.g., passwords) shall be stored using one-way encryption (e.g., bcrypt).
 - Only MEPO employees shall have access to sensitive operations such as stall transfers and quality guide management.

Listing management

- Performance
 - The system must load the dashboard and listings in 2 seconds or less during normal load conditions.
 - The system must update, add, or delete a listing and reflect those changes in real time without requiring the user to manually refresh the page.
 - The search and filtering of the Add Listings screen or the View Listings screen must return results in 2 seconds or less.

- Reliability
 - The system will accurately retrieve and display listings for the appropriate stall owner 99.9% of the time.
 - The system will ensure that listing records remain consistent even in the case of system or network outages.
 - All CRUD (create, read, update, and delete) operations on listings need to be validated to prevent data loss or corruption
- Scalability
 - The system must be able to accommodate an increasing number of stalls, stall owners, and product/service listings without experiencing performance problems.
- Availability
 - With the exception of planned maintenance, the Listing Management module must be accessible at least 99.9% of the time.
 - Users will receive advance notice from the system in the event of scheduled maintenance or outages.
 - A fallback message indicating temporary unavailability will be displayed in the event of system outages.
- Usability

- For stall owners with little technical expertise, the interface should be responsive, easy to use, and intuitive.
- Clear feedback messages for successful operations ("Listing added successfully") and error conditions ("Failed to load listings. Please retry.") must be provided by the system.
- For easy navigation, the design must adhere to consistent color schemes, font sizes, and button placements.
- Security
 - Bcrypt or comparable one-way encryption algorithms must be used to encrypt all sensitive data, including session tokens and passwords.
 - To avoid SQL injection, cross-site scripting (XSS), and other vulnerabilities, the system must validate every input field.

Use Cases

Use Case	#1	Looking for a Product/Stall
Description	The procedure of using the pathfinder to find a product/stall	
Actor(s)	Shopper, System	
Pre-Condition	• The Product/Stall exists in Naga City People's Mall	
Ordinary Sequence	Step	Action
	1	The Shopper uses the search function
	2	The System shows a list of results matching the shopper's search
	3	The shopper selects one or more items then taps on search button
	4	The System generates a path to the selected item
Post-Condition	• The system generates a path that leads to the Product/Stall	

Use Case	#2	Switching between Items in the Shopping List
Description	The procedure of switching products/stalls when using the pathfinder.	
Actor(s)	Shopper, System	
Pre-Condition	• The Product/Stall exists in Naga City People's Mall • The User is already done selecting items • There must be more than one items in the Shopping list	
Ordinary Sequence	Step	
	1	The System generates a path to the closest item on the list
	2	The Shopper taps on "Item Found" button
	3	The System marks the item found and proceeds to the next item on the list
	4	The System regenerates a new path to the newly selected item

Post-Condition	The system regenerates a path that leads to the newly selected Item
-----------------------	---

Use Case	#3	Using the Quality Guide Tool
Description	The procedure of using the quality guide tool	
Actor(s)	Shopper, System	
Pre-Condition	The Product has a Quality Guide Information in the System Database.	
Ordinary Sequence	Step	
	1	The Shopper taps on the Quality Guide Tool button
	2	The Shopper searches for a product in the Quality Guide
	3	The System shows a list of results matching the shopper's search
	4	The Shopper selects a product from the list of results

	5	The system fetches Quality Guide information from the Database
	6	The system displays the Quality Guide for the product
Post-Condition	The system shows a detailed guide to assess the selected product	

Use Case	#4	Create Stall Owner Account
Description	A functionality that allows MEPO employee creates accounts for stall owners	
Actor(s)	MEPO Employee, System	
Pre-Condition	The MEPO employee is logged into the system.	
Ordinary Sequence	Step	Action
	1	The MEPO employee logs into the system.
	2	The MEPO employee navigates to the Account Creation.

	3	The MEPO employee will input stall owner personal information such as first name, middle name, and last name.
	4	The MEPO employee reviews and clicks "Create Account".
	5	The system validates the details inputted: i. The first and last name should not be empty.
	6	The system must generate a unique username and password.
	7	The system successfully creates the account and displays the account credentials.
Post-Condition	• A new stall owner account is successfully created with account credentials. • A new stall owner account is created in the system.	

Use Case	#5	Transfer Stall Ownership
Description	A functionality for updating stall details and stall owner personal information	
Actor(s)	MEPO Employee, System	
Pre-Condition	• The MEPO employee is logged into the system. • The stall should be assigned to a stall owner first before the MEPO employee can transfer or update the stall.	
Ordinary Sequence	Step	Action
	1	The MEPO employee logs into the system
	2	Navigates the Map.
	3	The MEPO employee will search for a specific stall by stall name
	4	The MEPO employee selects and views the stall.

	5	The MEPO employee updates the stall details and stall owner's personal information.
	6	The MEPO employee reviews and confirms the stall transfer.
	7	The system validates the input: i. The stall details and owner information must be complete and valid.
	8	The system processes the stall transfer and notifies the MEPO employee of success.
Post-Condition	• The stall owner's personal information and stall details are updated.	

Use Case	#6	Manage Quality Guide
Description	A functionality for managing quality guides in the system.	
Actor(s)	MEPO Employee, System	
Pre-Condition	• The MEPO employee is logged into the system. • The product should be available in the system before the MEPO employee performs CRUD operations.	
Ordinary Sequence	Step	Action
	1	The MEPO employee logs into the system
	2	The MEPO employee accesses the Quality Guide.
	3	The MEPO employee chooses a CRUD operation i. Add a quality guide ii. Update a quality guide iii. Or remove the quality guide.

	4	The MEPO employee confirms the action.
	5	The system validates the input: i. The fields must be complete and valid.
	6	The system successfully performed the CRUD operation.
Post-Condition	• The Quality Guide has been successfully added, updated, or deleted.	

Use Case	#7	Listing Creation
Description	A functionality that serves as a data entry for listing management module.	
Actor(s)	MEPO Employee, System	
Pre-Condition	The MEPO employee is logged into the system.	
Ordinary Sequence	Step	Action
	1	The MEPO employee logs into the system
	2	The MEPO employee accesses the Listing Creation.
	3	The MEPO employee chooses a CRUD operation: i. Add a new listing ii. Update a listing iii. Delete a listing
	4	The MEPO employee confirms the action.

	5	The system validates the input. i. The product or service listing must not be duplicated. ii. The fields must be complete and valid.
Post-Condition		• The system successfully performed the CRUD operation.

Use Case	#8	Assign Stall
Description	A functionality that allows MEPO employee to assign specific stall to stall owners	
Actor(s)	MEPO Employee, System	

Pre-Condition	• A stall owner must have an account in the system before MEPO employee can assign the stall • The MEPO employee must be logged into the system.	
Ordinary Sequence	Step	Action
	1	The MEPO employee logs into the system
	2	The MEPO employee accesses the map
	3	The MEPO employee will search for a specific stall by its stall name
	4	The MEPO employee selects and views the stall.
	5	The MEPO employee inputs the stall owner's name, and confirms the actions
	6	Assign the stall to the corresponding stall owner if a name match is found.
Post-Condition	• The MEPO employee successfully assigned the stall to the stall owner.	

	i. The stall owner can now manage their stall and select their listings. ii. The stall is now eligible for ownership transfer. iii. The stall status has been updated from "Unassigned" to "Assigned."
--	--

Use Case	#9	Add/Select Listing
Description	A functionality that allows stall owners to select products or services that they are offering in their stall.	
Actor(s)	Stall Owner, System	
Pre-Condition	• The Stall Owner is logged in into the system	
Ordinary Sequence	Step	
	1	Stall Owner logs in into the system.

	2	The Stall Owner chooses a stall to manage and taps "Manage".
	3	On the Manage Business screen, the Stall Owner can: • Tap "View Listings" to see current listings for the selected stall. • Tap "Add your listings" to navigate to Add Listings.
	4	View Listings screen shows each listing with name, category, optional price, and availability. The Stall Owner can: • Edit a listing: change price (numeric) and availability (Available/Not Available); System updates the listing. • Remove a listing: System deletes the listing
	5	Add Listings screen loads catalog. The Stall Owner can:

		• Select one or many items not already in the stall listing (duplicates are prevented).
Post-Condition		• The Stall Owner can view and manage the listings for their stall(s).

System Design

This section contains illustrations and diagrams that provide a detailed description of the architecture, components or modules, data, models, and interfaces for the PathSmart system.

System Architecture

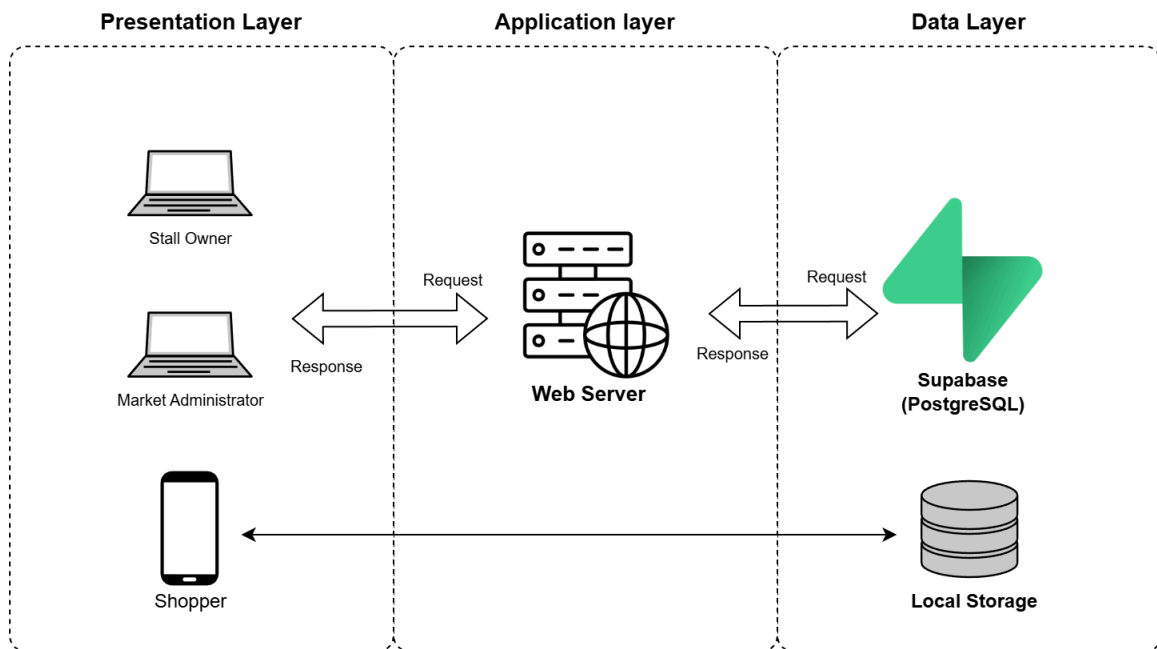


Figure 10: Architecture Design of the PathSmart System

In this study, PathSmart uses a system architecture called 3-tier. This architecture consists of three layers such as the presentation layer (built with Expo React), application layer, and data layer (Supabase). The presentation layer is what the user sees and interacts with directly. The application layer is a bridge that

allows the client and data layer to communicate to each other. The data layer manages the data storage, retrieving data, and so on.

Data Flow Models

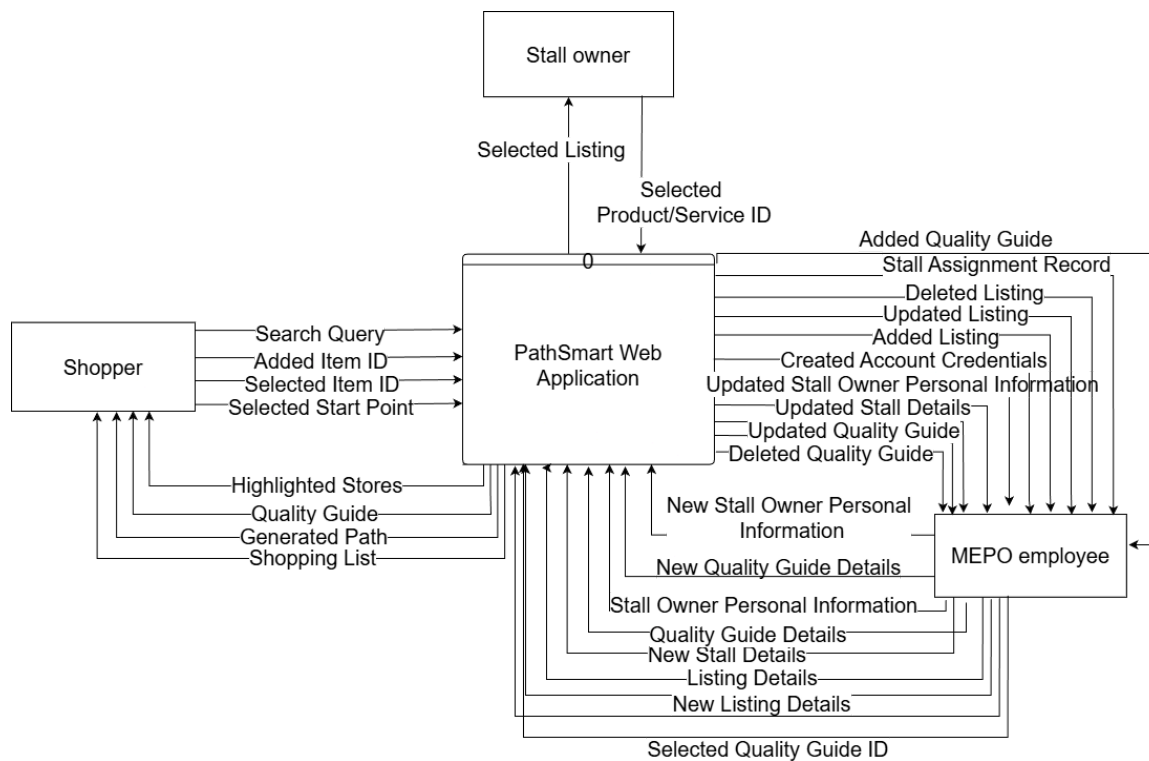


Figure 11: Context Diagram of the PathSmart System

(Gane & Sarson)

The figure above represents the context diagram that gives proponents a high-level overview of how data flows through the PathSmart system and external entities. In this context diagram, we have 3 external entities such as the Shopper, Stall owner, and MEPO employee. This context diagram shows what each external entity sends data to the system and what they receive from the system.

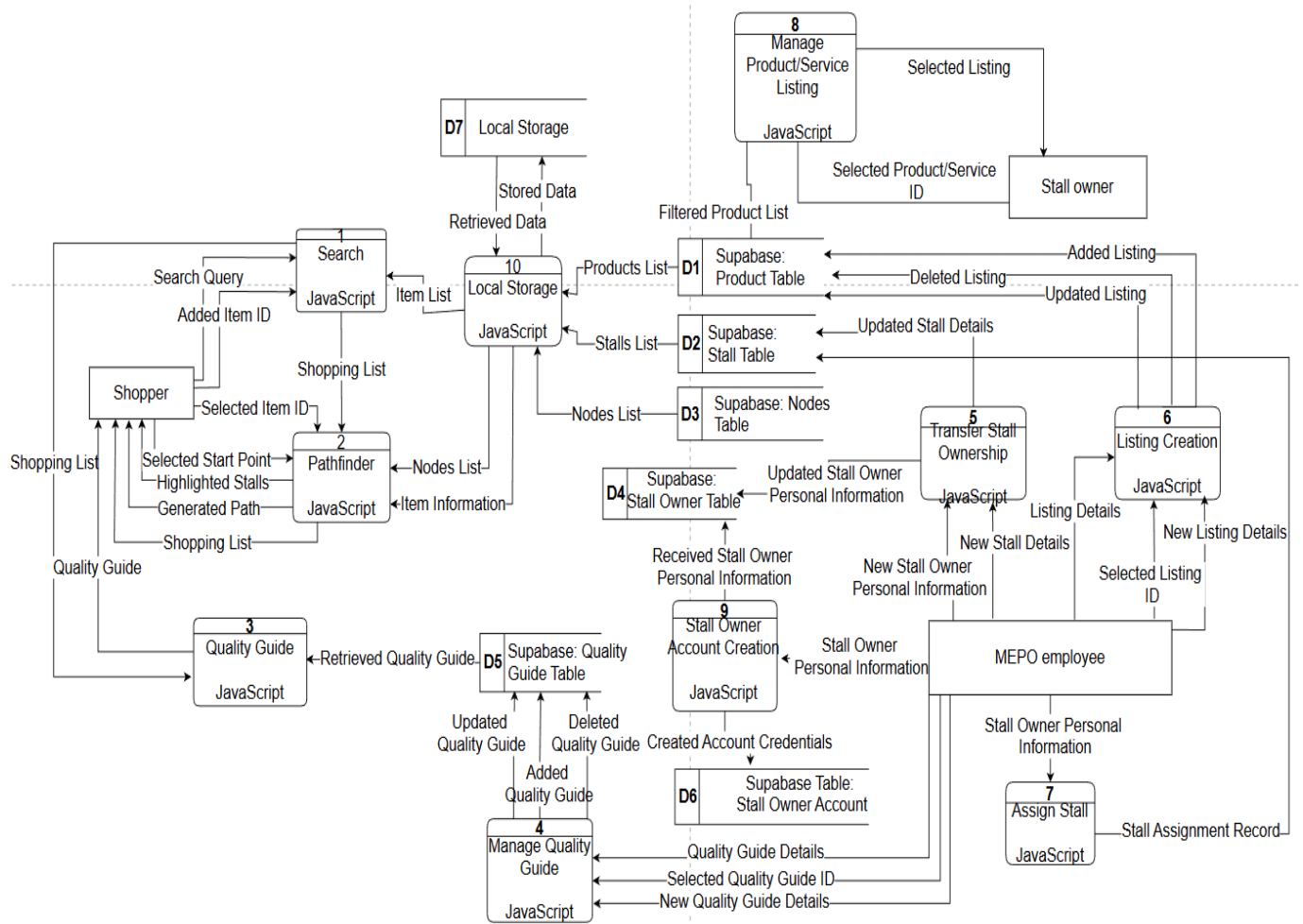


Figure 12: Level 0 DFD of the PathSmart System (Gane & Sarson)

The figure above represents the level 0 DFD for the PathSmart system. In this level 0 DFD it shows a detailed overview of the data flows in the system, what are the processes (10 processes) and data stores (7 data stores) present, where the data comes from, where it goes, and where it's stored.

Shopper's Guide

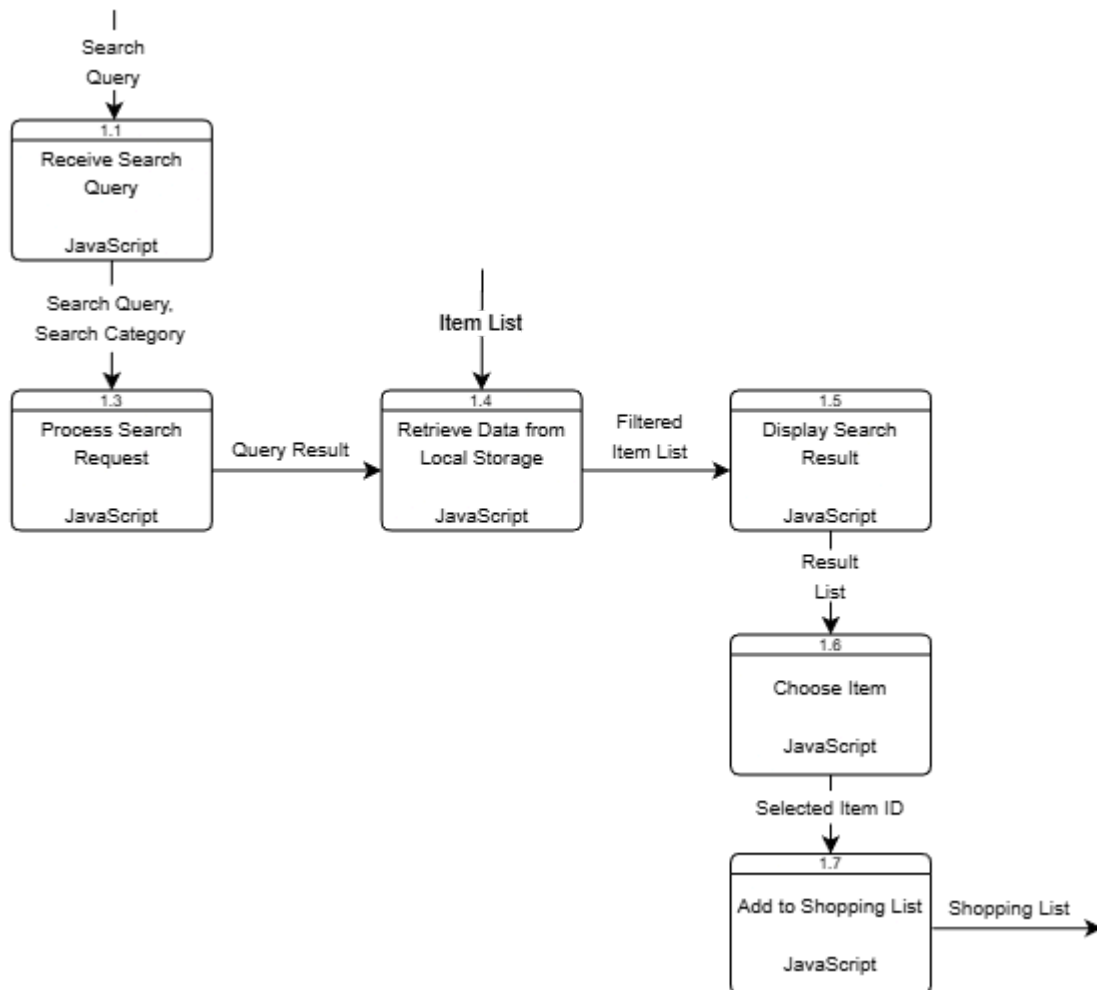


Figure 13: Level 1 - Process 1 Child DFD (Gane & Sarson)

This figure represents level 1 DFD for process 1 (Item Search Function). The diagram shows how the system handles the retrieval and processing of data right before the pathfinder or the quality guide. Process 1.1, the system receives the search query of the user. Process 1.3, the system fetches data from the local storage of the user's device. Process 1.4, the system lists the results. Process 1.5,

the system awaits user input for the selected item. Process 1.6, the item id is listed on a shopping list.

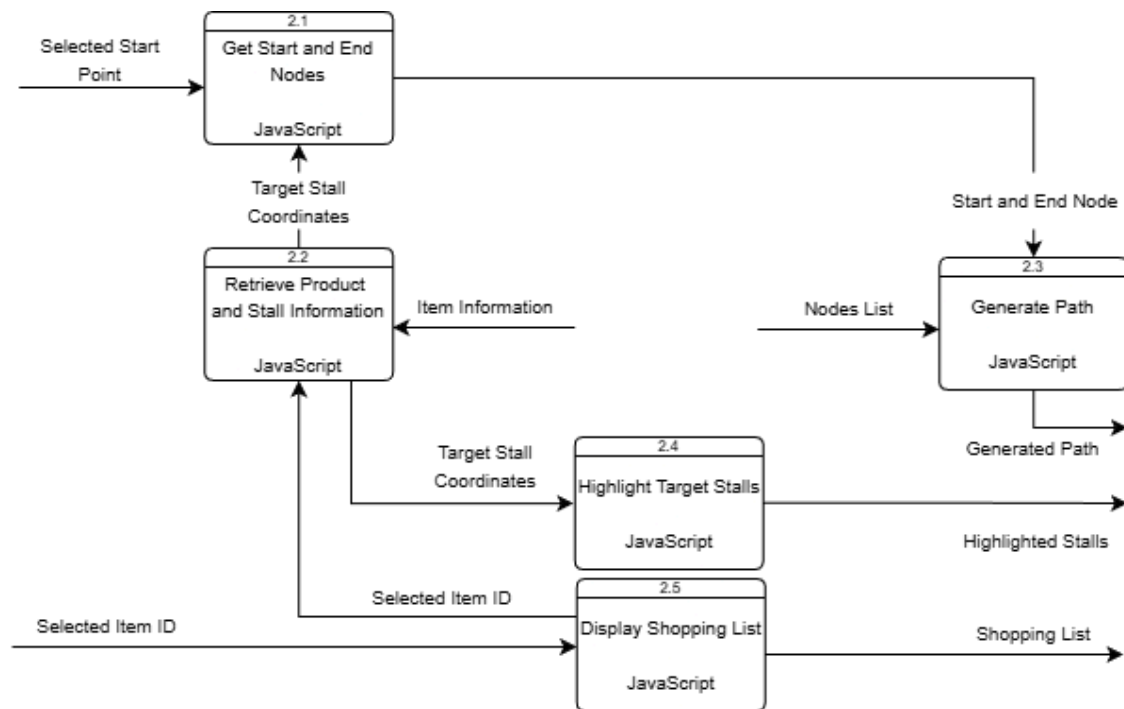


Figure 14: Level 1 - Process 2 Child DFD (Gane & Sarson)

This figure represents level 1 DFD for process 2 (Pathfinder). The figure shows the process of how the system calculates and outputs the results. Process 2.1, fetches the start node from the user input and end node from Process 2.2 for the Process 2.3 to calculate the path. Process 2.2, determines the end nodes using the listed id(s) in the shopping list.

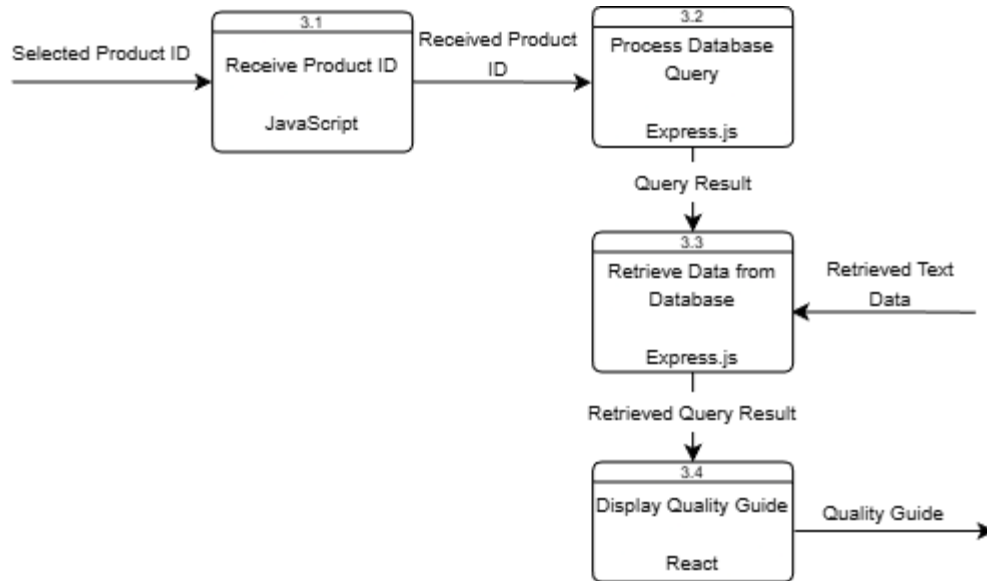


Figure 15: Level 1 - Process 3 Child DFD (Gane & Sarson)

This figure represents level 1 DFD for process 3 (Quality Guide). Process 3.1, the system receives the product id. Process 2.2, the system fetches the quality guide data from the database. Process 3.4, the system displays the quality guide information.

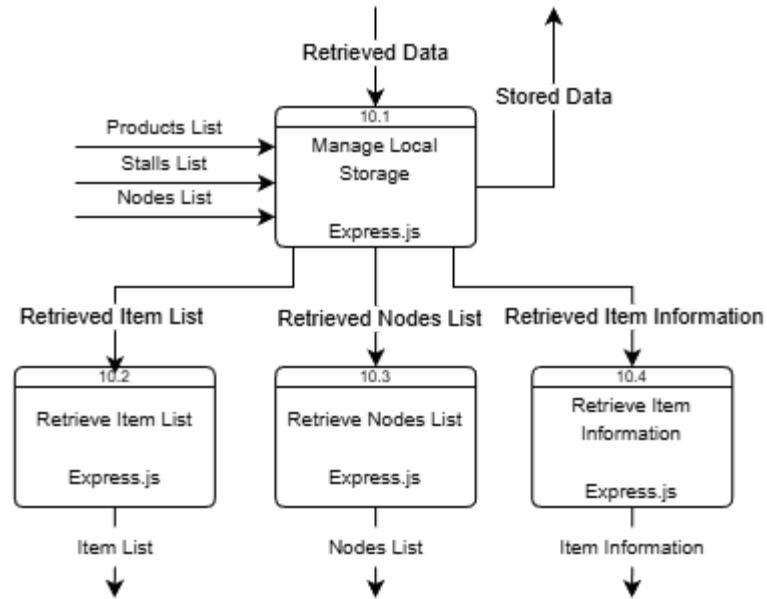


Figure 16: Level 1 - Process 10 Child DFD (Gane & Sarson)

This figure represents level 1 DFD for process 10 (Local Storage). Process 10.1, the local storage stores the data from the database into the device of the user if they permit. The local storage is used to be able to access the pathfinder even in an environment without internet.

Stall Management

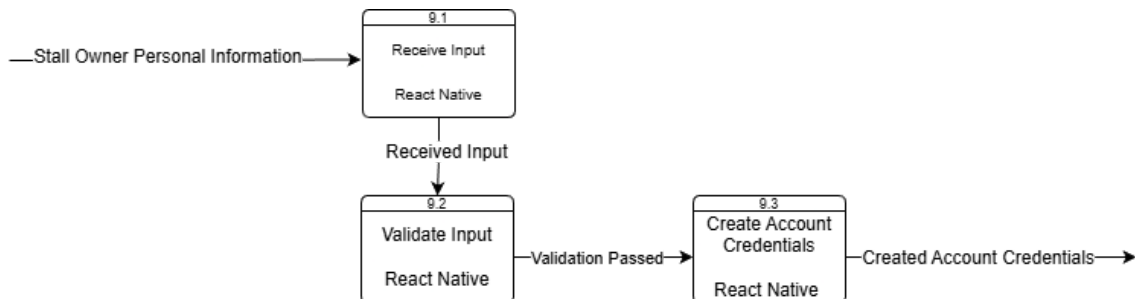


Figure 17: Level 1 - Process 9 Child DFD (Gane & Sarson)

This figure represents level 1 DFD for process 9 (account creation for stall owners). Process 9.1 receives the input which is the stall owner's personal information, and input goes to another process 9.2 that validates it, if the input is valid, process 9.3 processes the account creation, and provides it with the user.

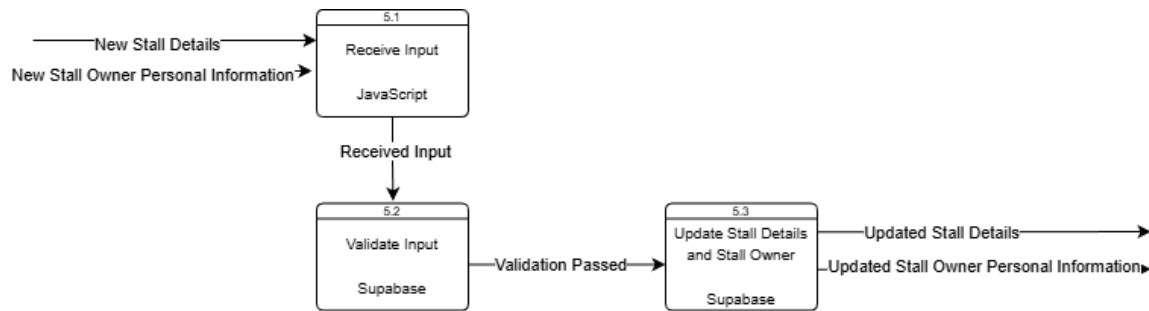


Figure 18: Level 1 - Process 5 Child DFD (Gane & Sarson)

This figure represents level 1 DFD for process 5 (updating of stalls). Process 5.1 receives the input such as the new stall owner's personal information and stall details, and input goes to another process 5.2 that validates it, if the input is valid, process 5.3 processes the updation, and reflects the changes to the user.

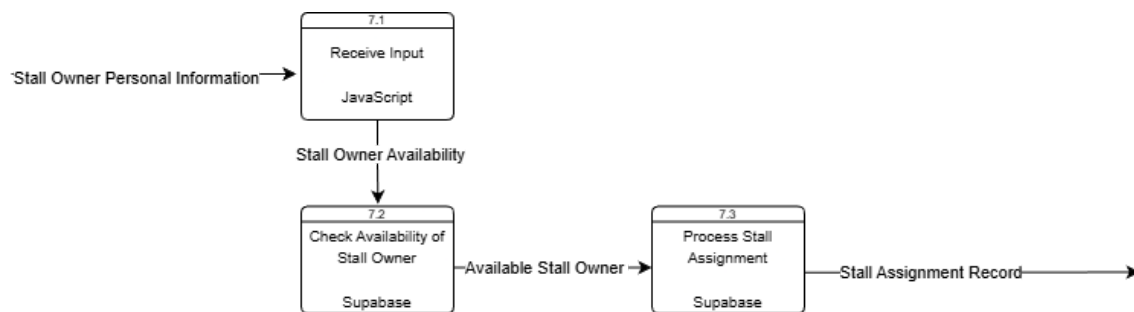


Figure 19: Level 1 - Process 7 Child DFD (Gane & Sarson)

This figure represents level 1 DFD for process 7 (assigning of stalls). Process 7.1 receives the input which is the stall owner's personal information, input goes to another process 7.2 that validates it, it checks whether the stall owner is available, if the stall owner is available, process 7.3 processes the stall assignment and assigns the stall to the specific stall owner.

Data Dictionary

- Shopper = Shopper ID + Device ID
- Stall Owner = Full Name + Stall Owner ID + [Phone Number] + Stall Owner Account ID
 - Full Name = Lastname + Firstname + [Middlename]
 - Stall Owner Account ID = Username + Password
- Stall = Stall ID + Stall Owner ID + Stall Name + Stall Address + Category + Stall Section
 - Stall Address = Block Number + Stall Number
 - Category = ("Vegetables", "Fishes", "Dairy")
 - Stall Section = ("Wet Section", "Pavenas Section", "Squatters Section")
- Product and Service = PnS ID + Name + PnS Img + PnS Category + Type +Alternate Name

- Alternate Name = Bicol Name, Tagalog Name
 - Type = ("Product", "Service")
- Quality Guide = Quality Guide ID + PnS ID + Details
- Nodes = Node ID + Node Category + Node Location
 - Node Category = ("Stall", "Path")
 - Node Location = Latitude + Longitude
- Stall Owner Account = Stall Owner Account ID + Username + Password
- MEPO Employee Account = MEPO Employee ID + Username + Password
+ Name + Role
- Listing = Listing ID + Stall ID + PnS ID + isAvailable + price

Data Models

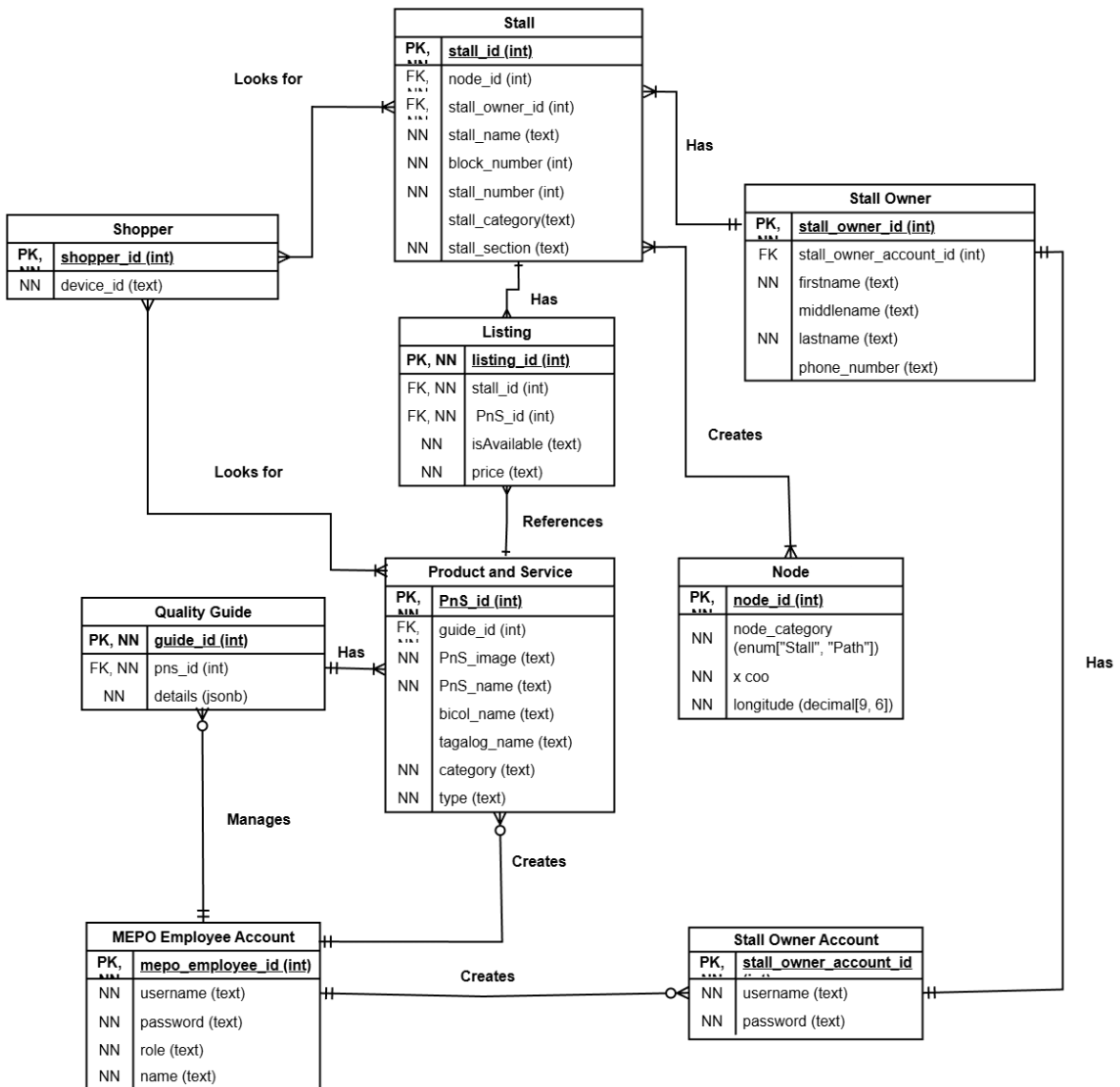


Figure 20: Entity-Relationship Data Model for the PathSmart system (Crow's Foot Notation)

The figure above represents the ERD model for the Pathsmart system. In this ERD model, It shows the important components that will allow proponents to create the relational database in Supabase. In this ERD model, components

present are entities(e.g. stall, stall owners), their attributes (e.g. category, type, pns_image), relationships (e.g. creates, manages, has) to each other, and the data type (e.g. jsonb, int, text).

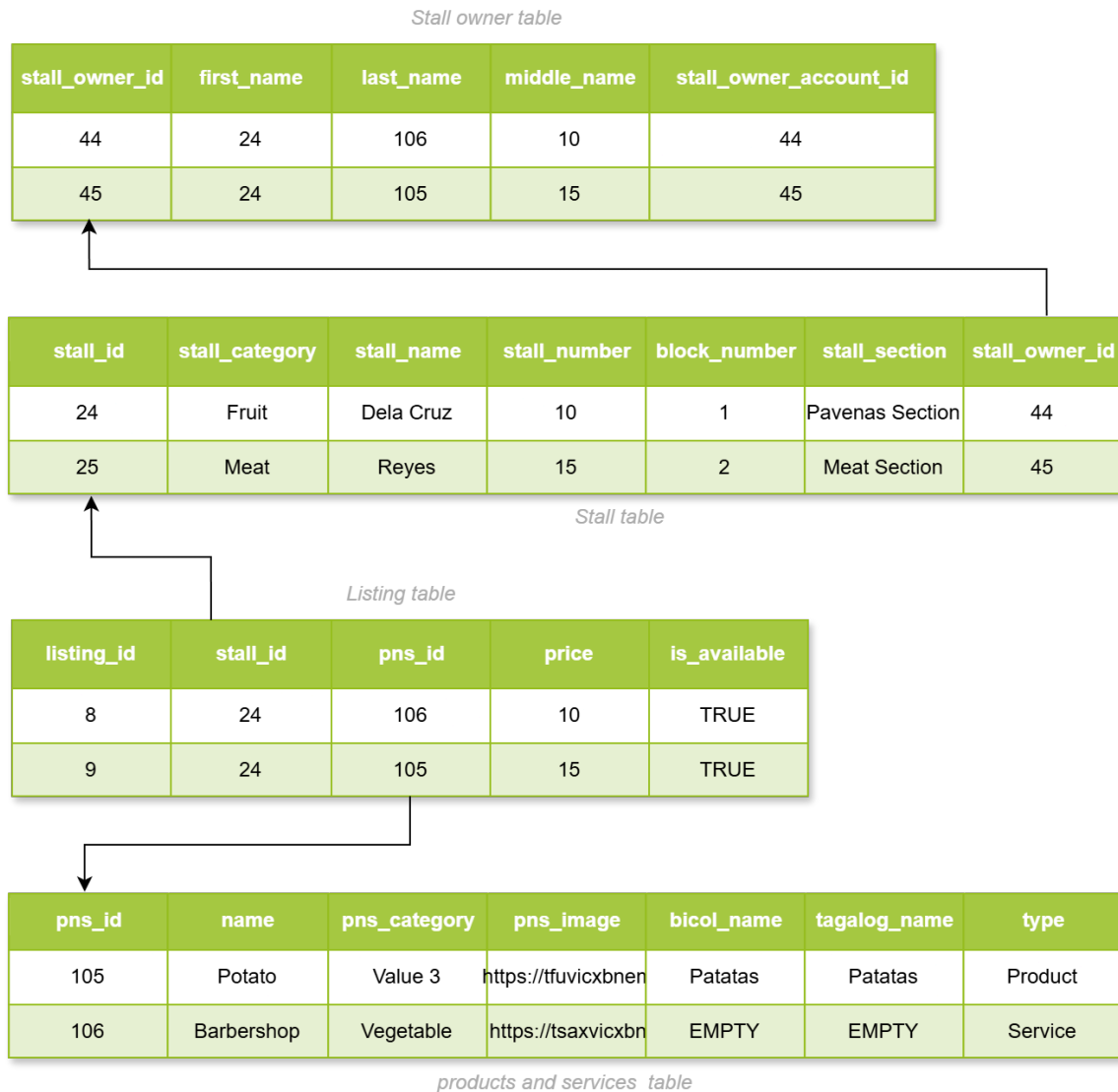


Figure 21: Relational Data Model for the PathSmart System (Crow's Foot)

The figure above represents the relational data model for the Pathsmart system. It shows how these entities such as Listing, products and services, stall, and stall owner are structured and connected to each other. This figure defines and shows the available products and services, the list of stall owners and in which stall they're assigned, and what they offer in that stall or called listing.

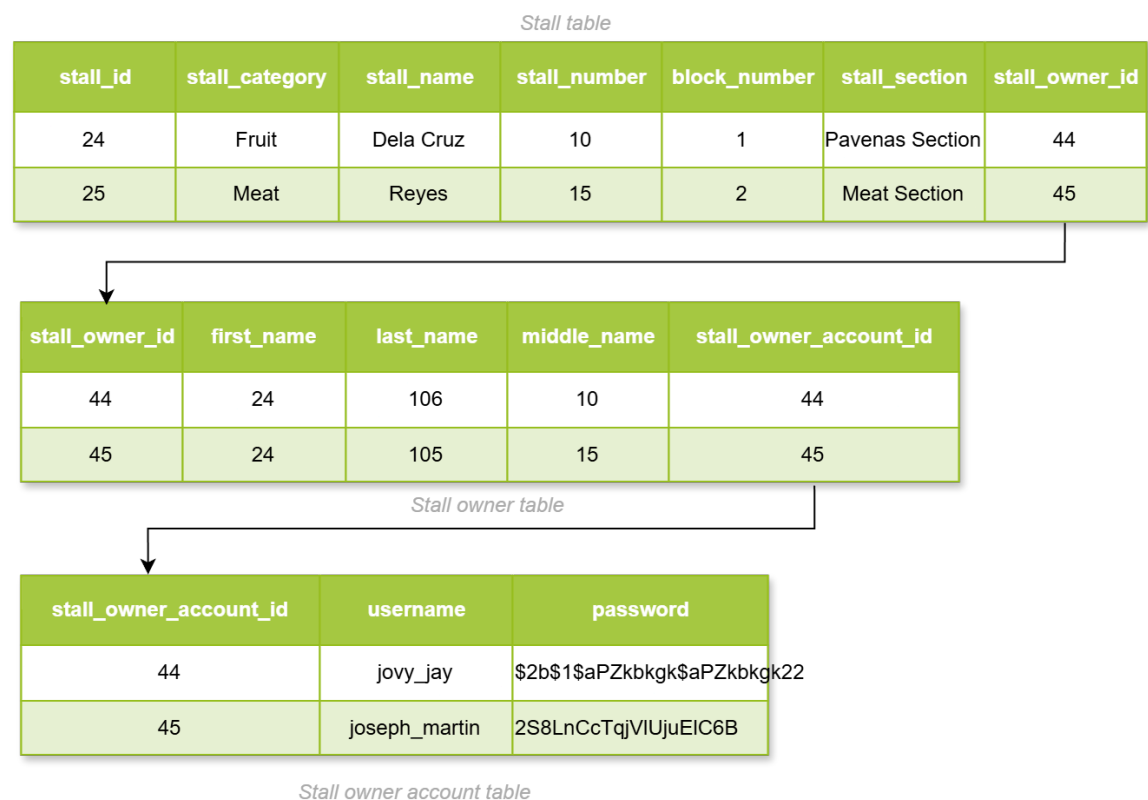


Figure 22: Relational Data Model for the PathSmart System (Crow's Foot)

The figure above represents the relational data model for the Pathsmart system. It shows how these entities such as stall, stall owner account, and stall

owner are structured and connected to each other. This figure defines and shows the list of stalls and who owns that stall, and also the accounts for each stall owner.

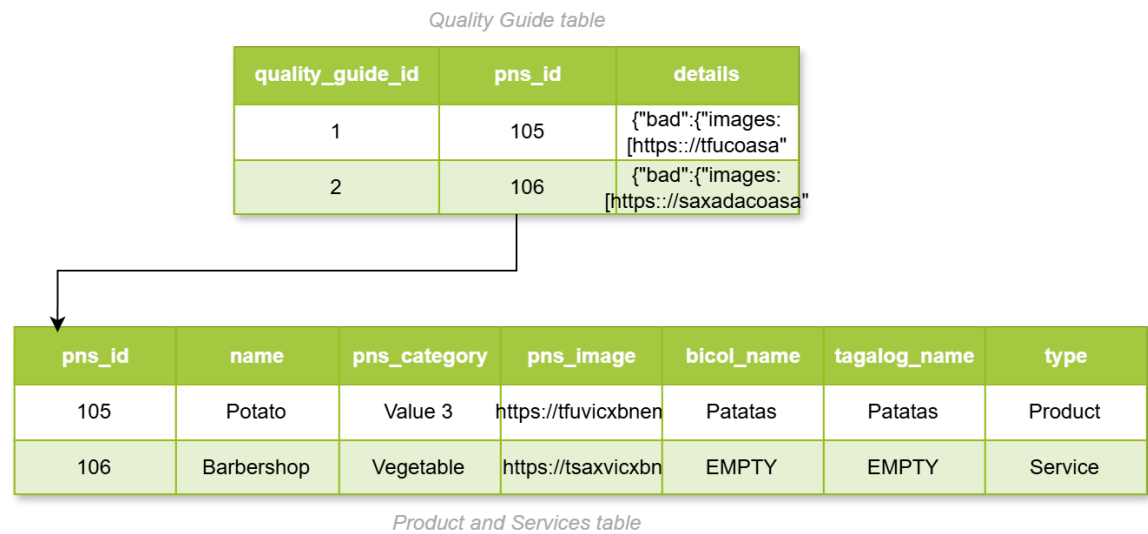


Figure 23: Relational Data Model for the PathSmart System (Crow's Foot)

The figure above represents the relational data model for the Pathsmart system. It shows how these entities such as quality guide and products and services are structured and connected to each other. This figure defines and shows that the product can have a quality guide.

System Models

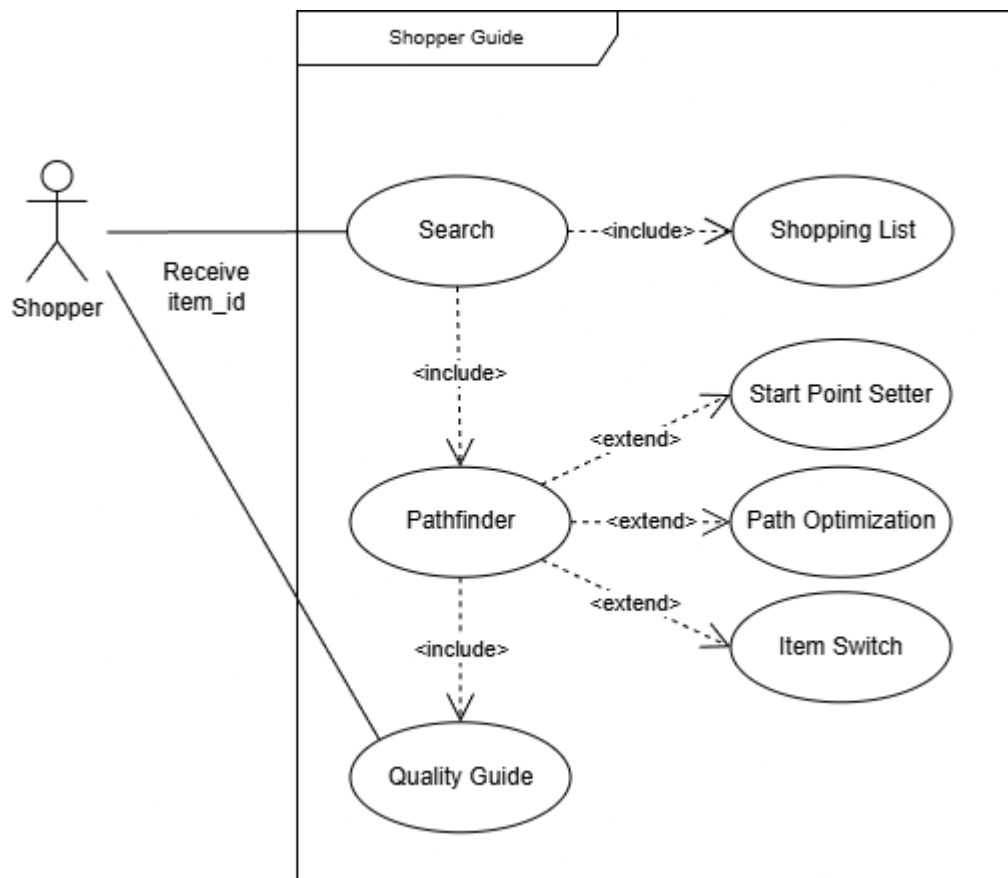


Figure 24: Use Case Diagram of Shopper Guide module

(Unified Modeling Language)

The figure presents a use case diagram showing how users interact with the shopper guide module. Selected ids are always placed on a list, whether it be single or multiple items. Upon executing the pathfinder, there are tools to help that come with it, the start point setter, path optimizer, and item switcher. The quality guide can be accessed manually, but it also shows up together with the pathfinder.

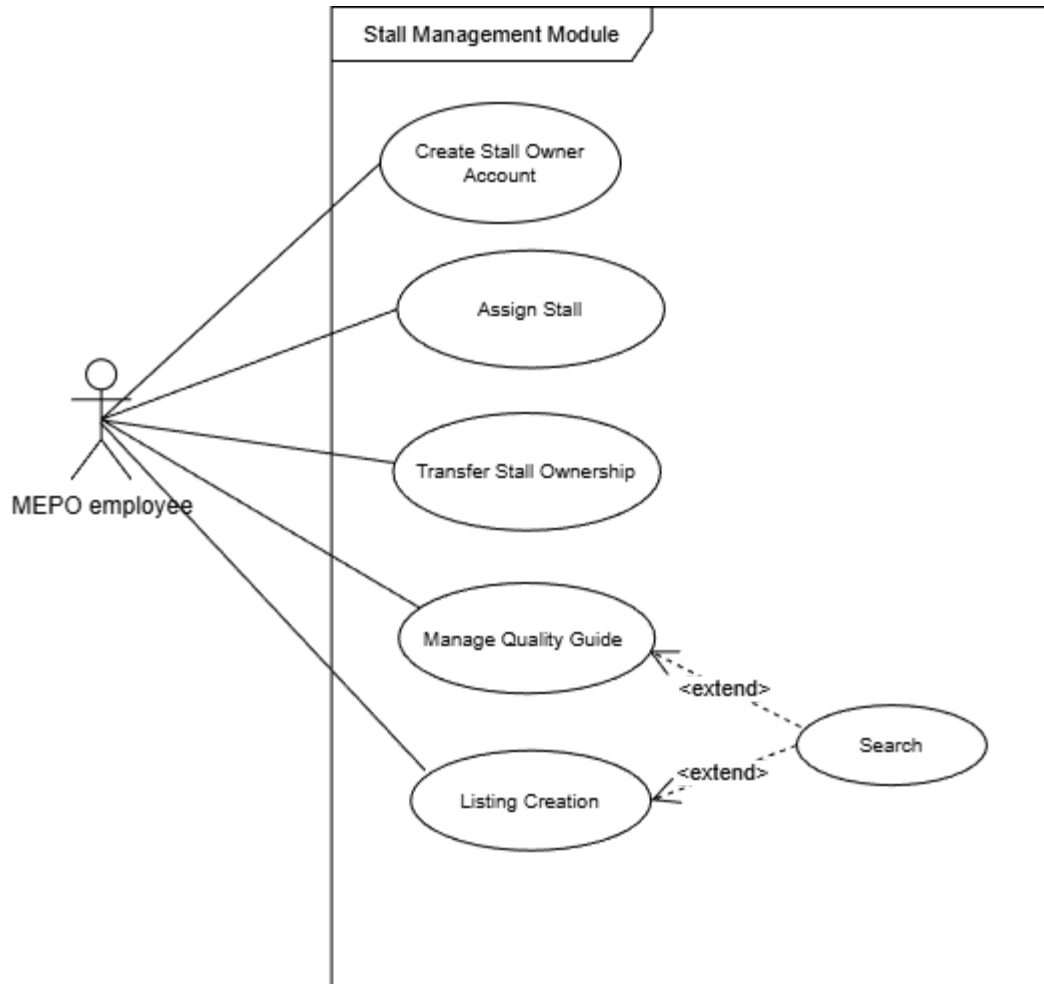


Figure 25: Use Case Diagram of Stall Management module

(Unified Modeling Language)

The figure presents a use case diagram showing how the MEPO employee interacts with the stall management module. The figure shows the tasks that MEPO employees can perform, including creating accounts for stall owners, assigning stalls to specific stall owners, updating stalls details, listing creation that serve as the data entry for listing management module, and so on.

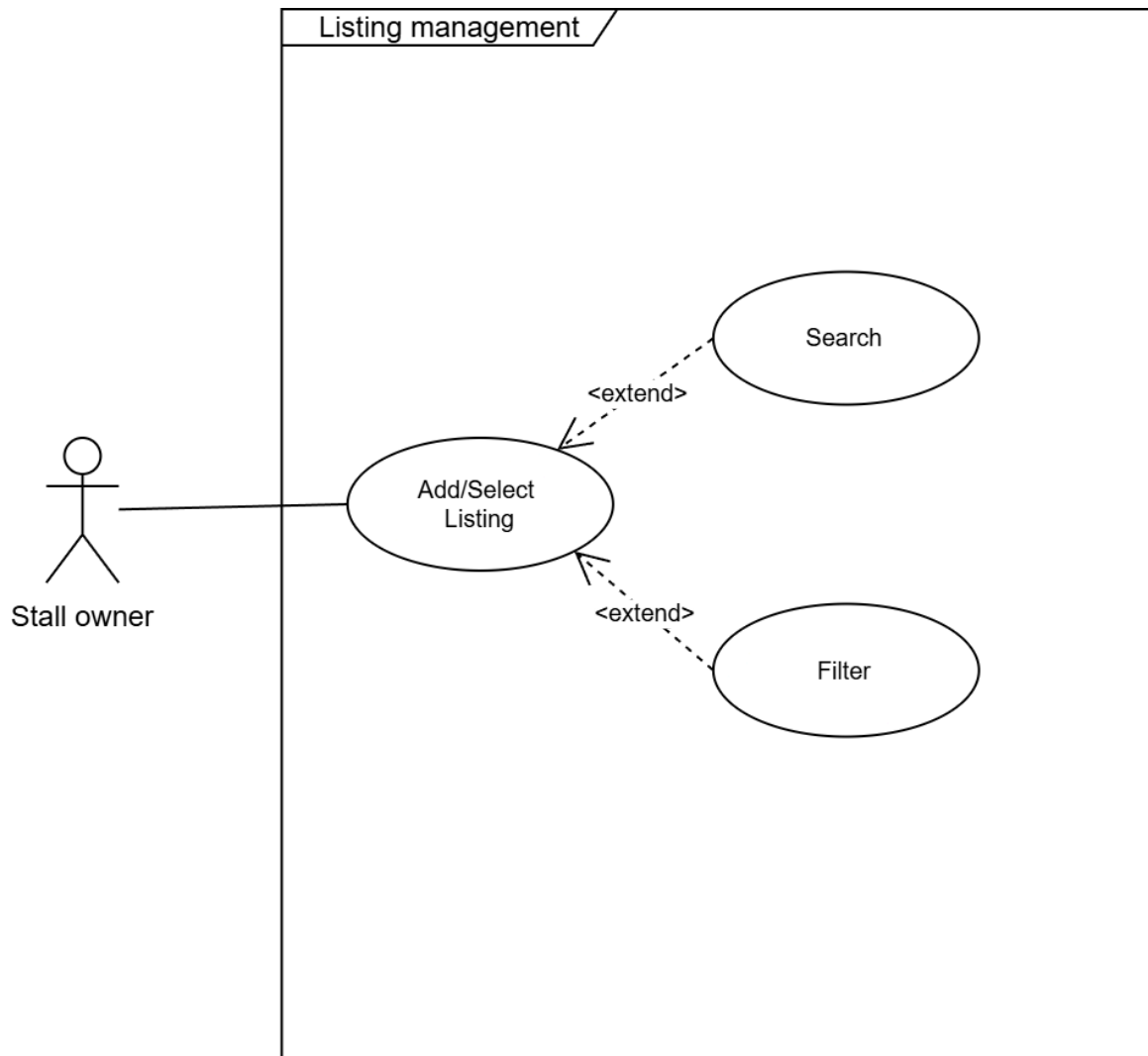


Figure 26: Use Case Diagram of Listing Management module

(Unified Modeling Language)

The figure presents a use case diagram showing how the Stall owner interacts with the listing management module. The figure shows the task that Stall owners can perform, which is adding or selecting their listing which serves as their

offerings in their stall. Stall owners can search products or services and also filter categories (e.g. vegetable, meat, poultry).

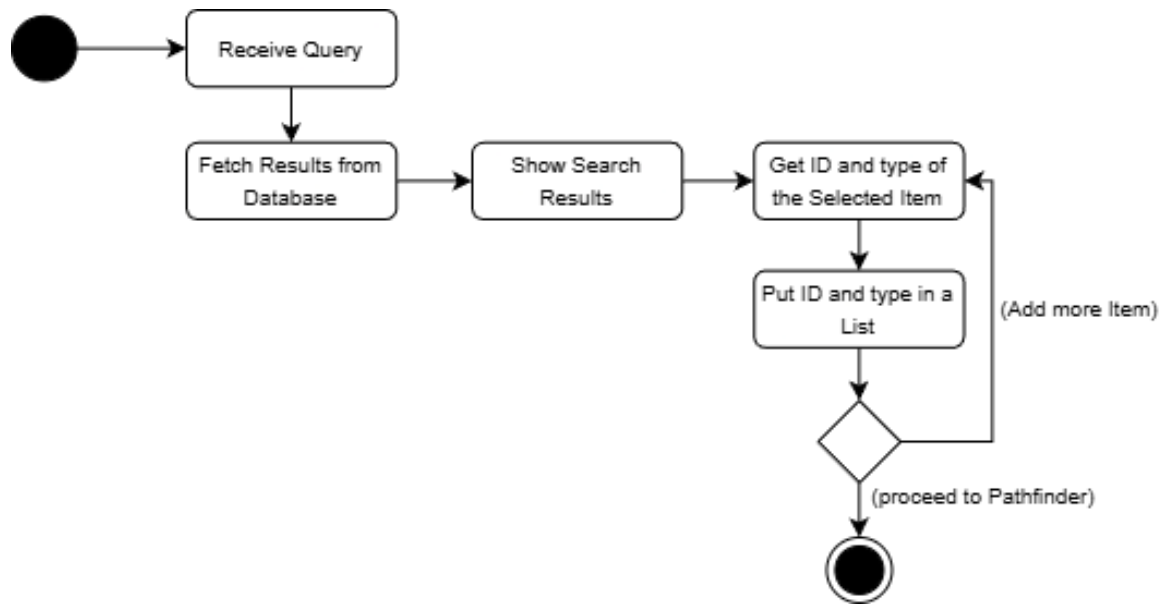


Figure 27: Activity Diagram of Shopper Guide module (Search Function)

The figure above is the activity diagram for the search function. It shows how the system handles search requests. It uses the local storage of the device as a database. Instead of searching one by one, the system creates a list of items selected by the user and stores the ID of each item.

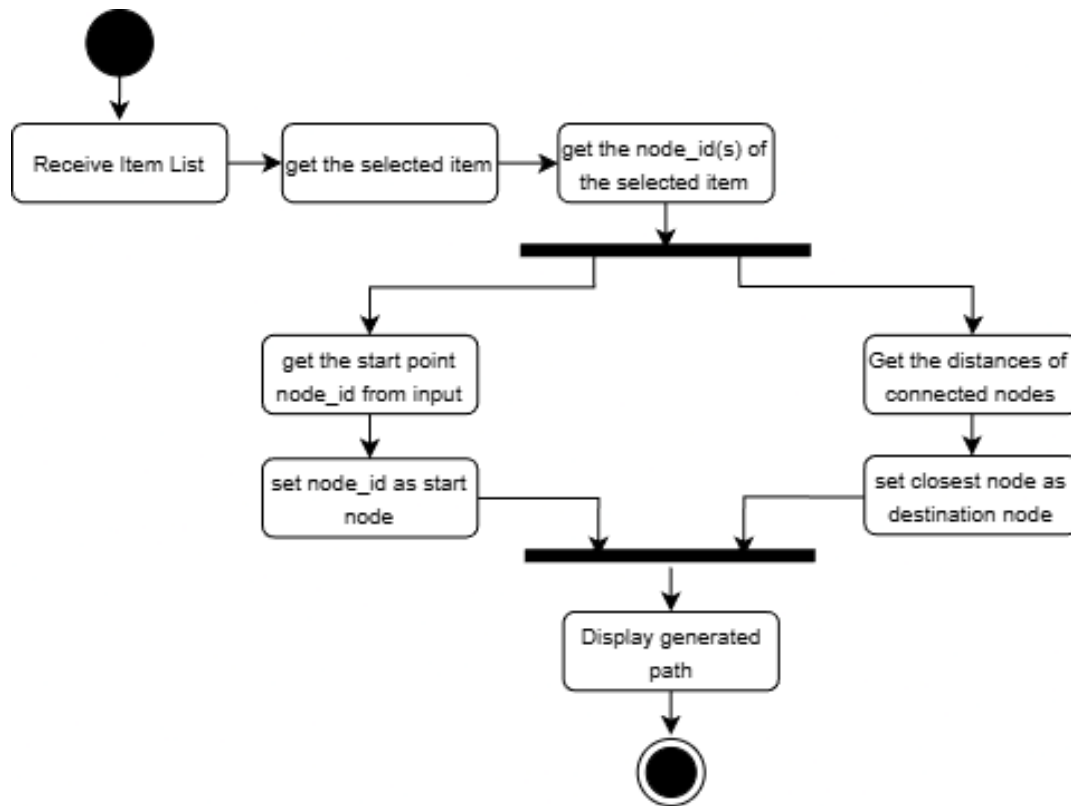


Figure 28: Activity Diagram of Shopper Guide module (Pathfinder)

The figure above is the activity diagram for the Pathfinder. The pathfinder uses the shopping list items from the Search Function to get the destination nodes. The start nodes are from the input of the user. The pathfinder then uses graph data to create the optimal path.

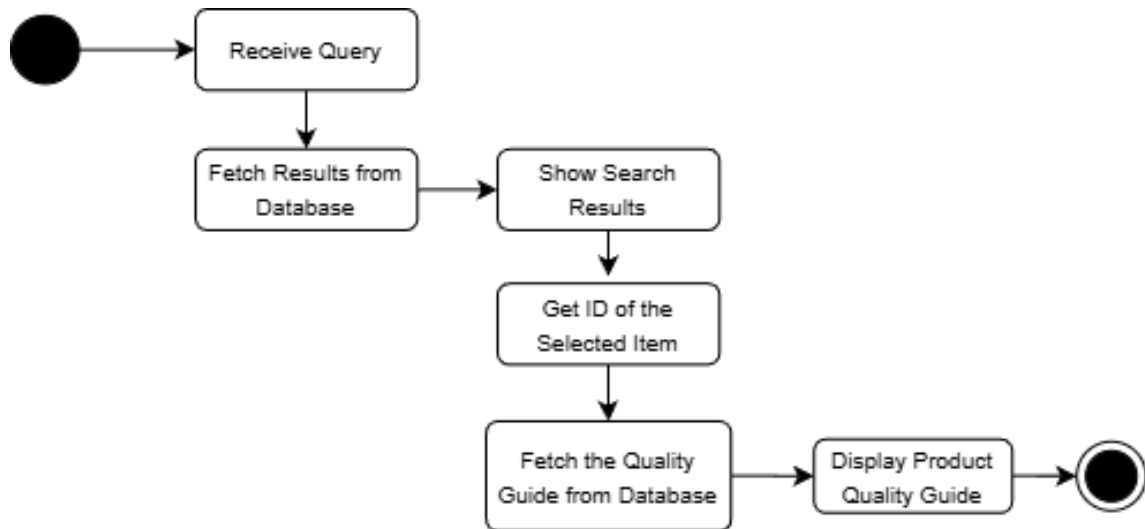


Figure 29: Activity Diagram of Shopper module (Quality Guide)

The figure above is the activity diagram for the Quality Guide. The system fetches from the database for the quality guide information, then the system displays the quality information of the selected product.

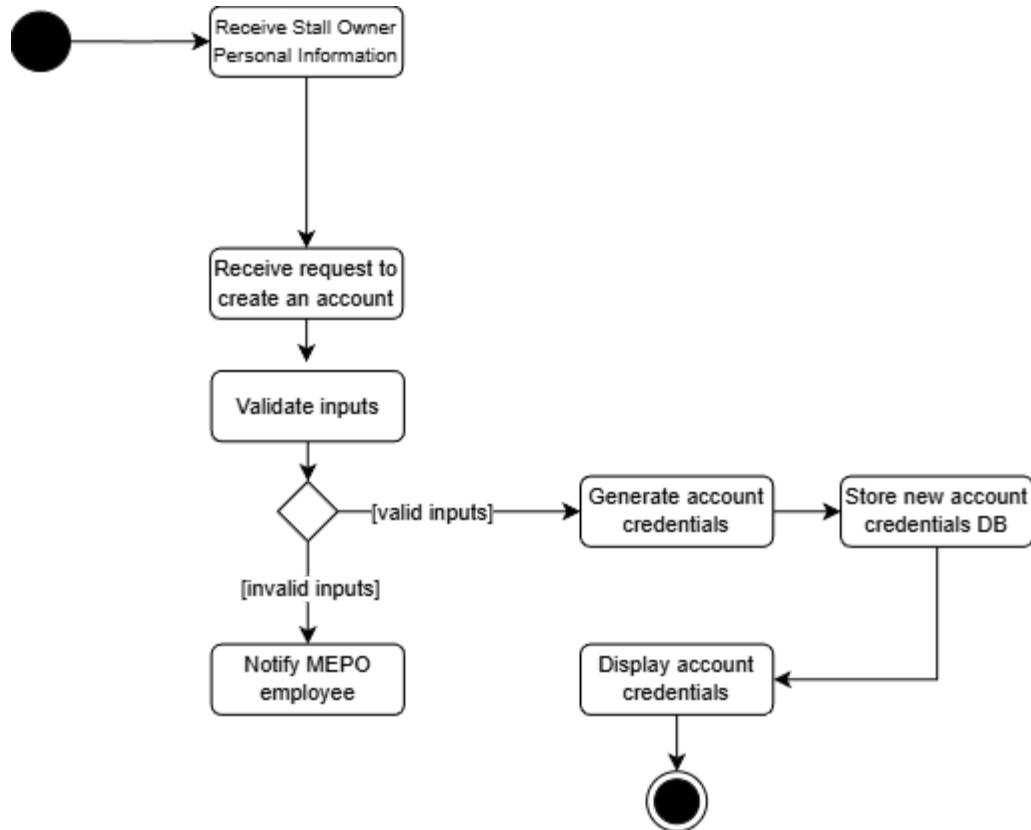


Figure 30: Activity Diagram of Stall Management module (Account Creation)

The figure above represents the activity diagram for account creation. In this diagram, the system receives the stall owner's personal information (firstname, lastname, and middlename). The MEPO employee will now confirm the account creation, and the inputs will be validated first, if the inputs are valid, and there is nothing that went wrong, then process the account creation, store the created credentials to the Supabase DB, and then finally display the account credentials to the MEPO employee. If the inputs are invalid, and something went wrong when creating the account, then notify the MEPO employee.

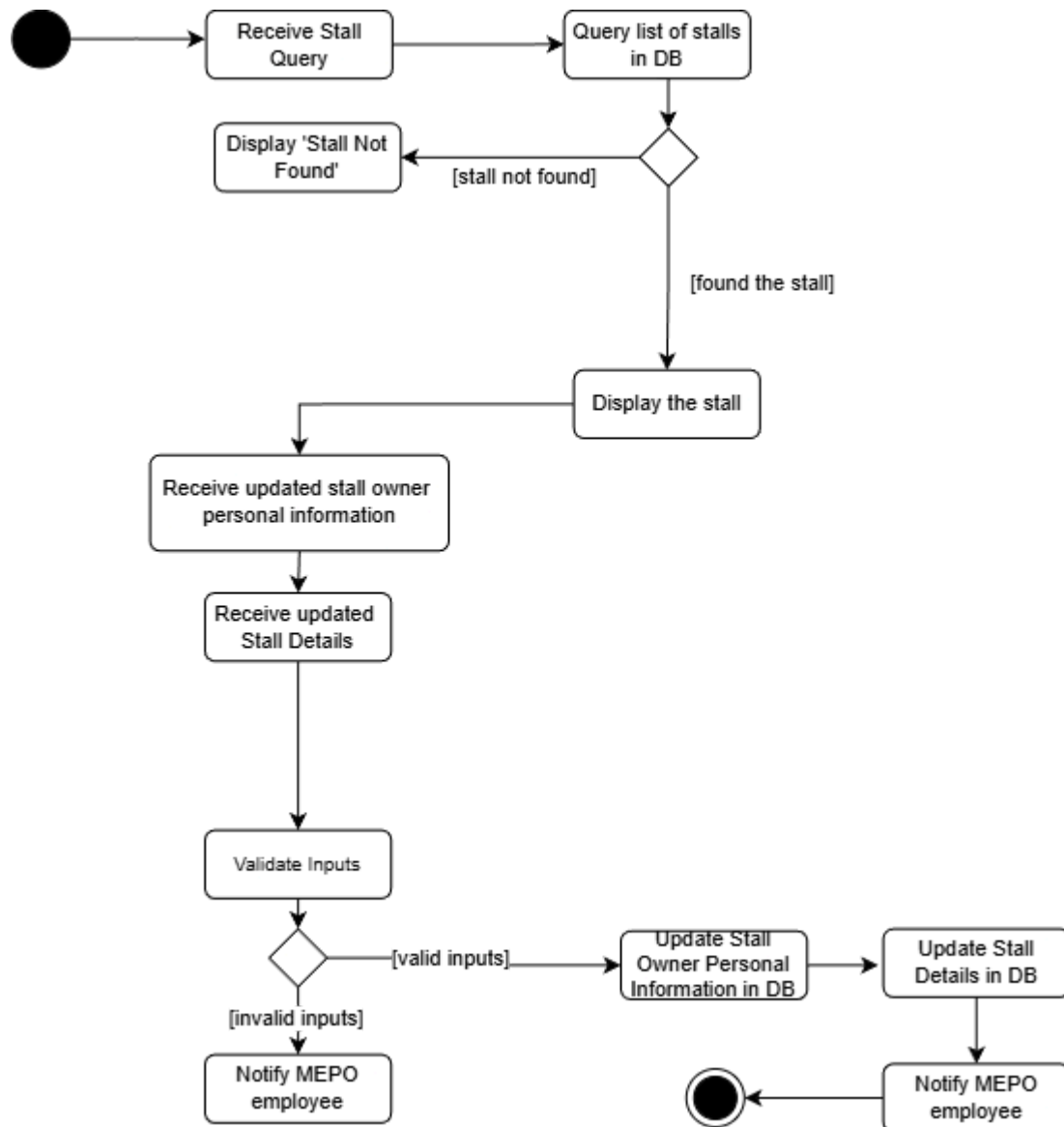


Figure 31: Activity Diagram of Stall Management module (Transfer Stall Ownership)

The figure above represents the activity diagram for transfer stall ownership (or for updating of stalls). In this diagram, the system will receive a stall search

query from the MEPO employee and it will fetch a list of stalls from the DB, if the stall is not found, then notify the MEPO employee, if not display the stall. MEPO employee will either update the stall owner's personal information or stall details, the inputs will be validated first, if valid then process the updation of stall, if not then notify the MEPO employee

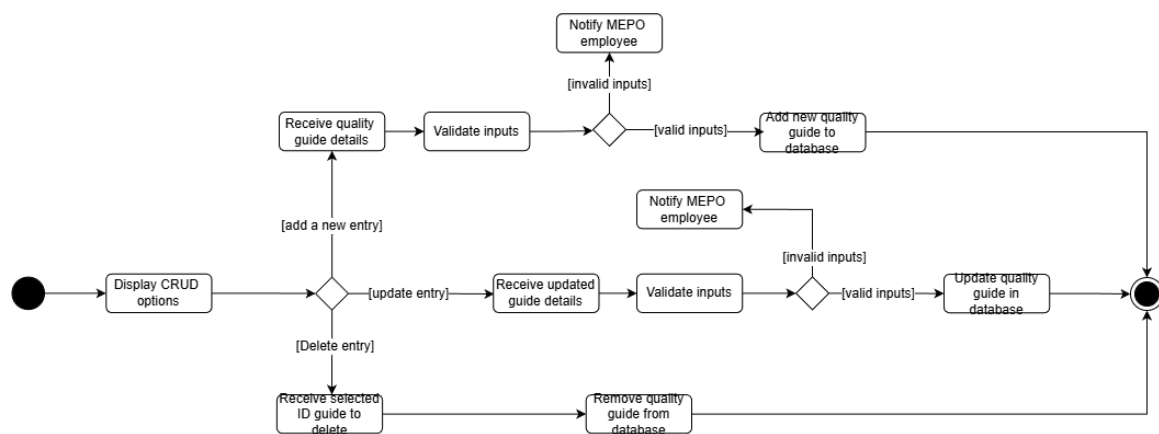


Figure 32: Activity Diagram of Stall Management module (Manage Quality Guide)

The figure above represents the activity diagram for managing the quality guide. In this diagram, it shows the flow of actions that the MEPO employee can perform such as adding, updating, and deleting quality guides, followed by receiving the inputs based on the action, validating the inputs, and either notifying the MEPO employee or saving the changes to the DB.

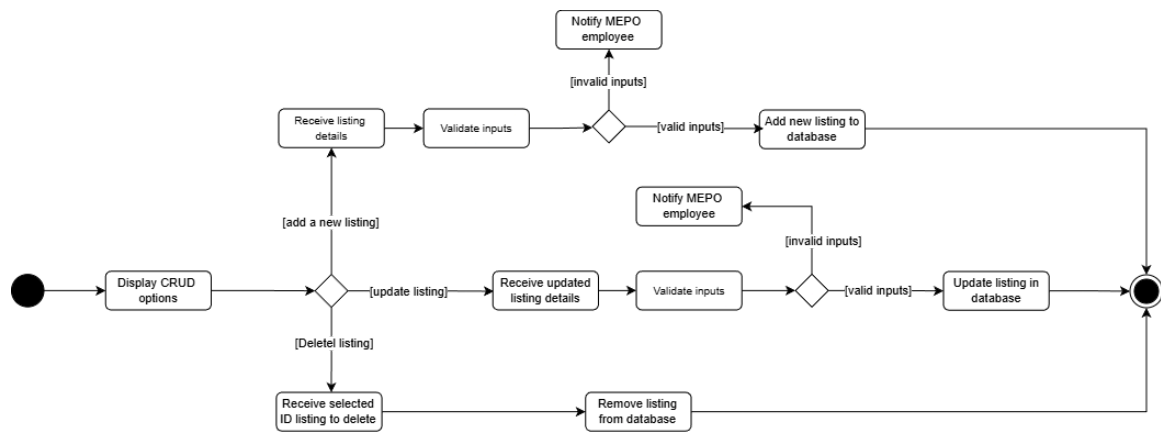


Figure 33: Activity Diagram of Stall Management module (Listing Creation)

The figure above represents the activity diagram for listing creation. In this diagram, it shows the flow of actions that the MEPO employee can perform such as adding, updating, and deleting listings, followed by receiving the inputs based on the action, validating the inputs, and either notifying the MEPO employee or saving the changes to the DB.

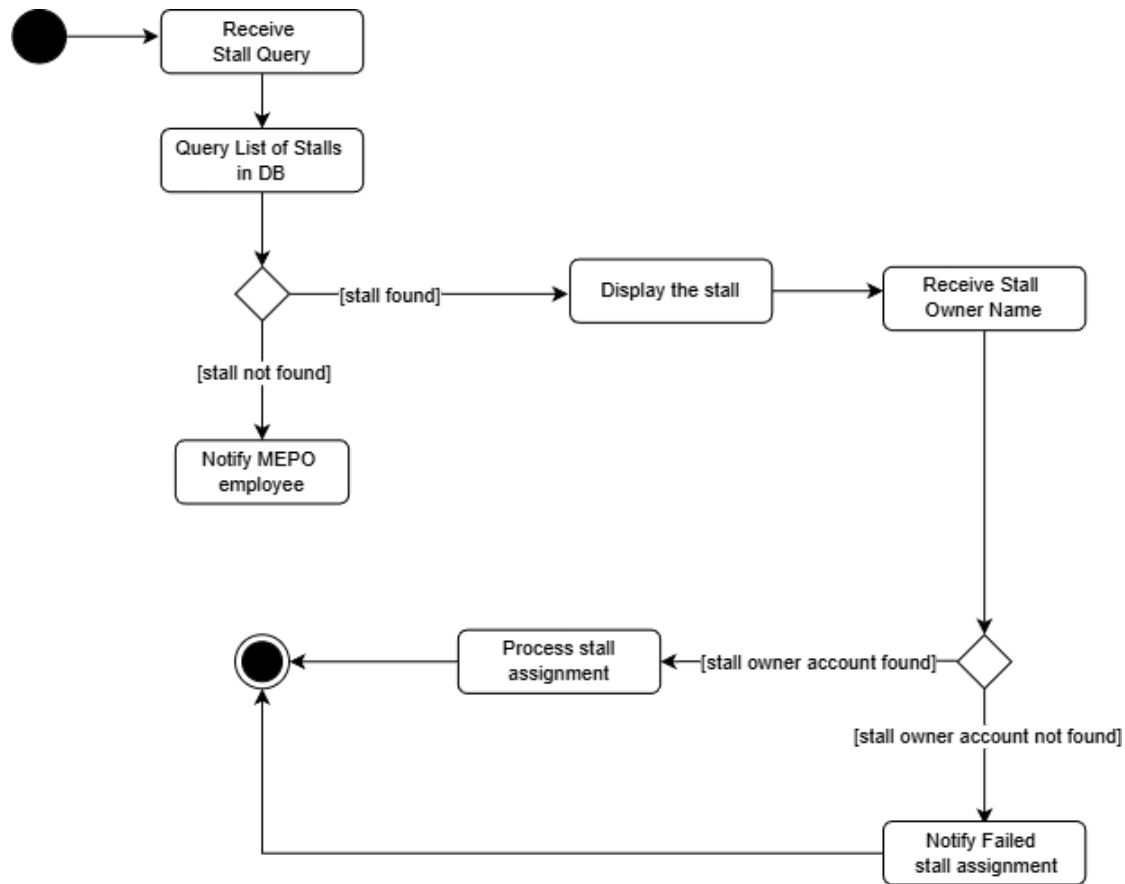


Figure 34: Activity Diagram of Stall Management module (Assign Stall)

The figure above represents the activity diagram for assigning a stall. In this diagram, the system will receive a search stall query from the MEPO employee, and the system will try to find the stall. If the stall is not found, just notify the MEPO employee, and if not, then display the stall. The system will receive the stall owner name, if the stall owner account or name is found which is entered by the MEPO employee, then process the stall assignment, and assign the stall owner to that stall, if not then notify that the assignment is failed.

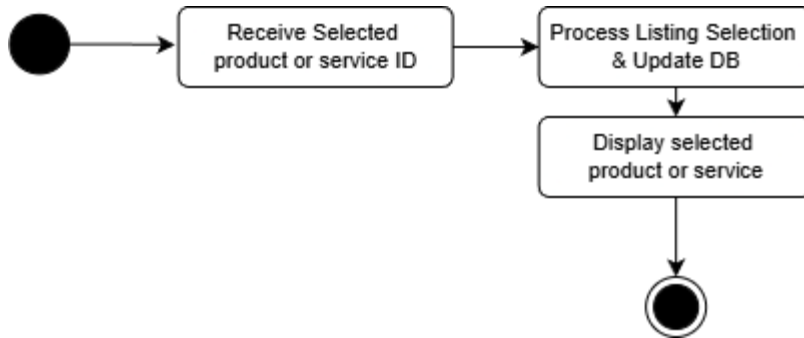


Figure 35: Activity Diagram of Listing Management module (Add/Select Listing)

The figure above represents the activity diagram for Manage Listing that involves CRUD such as adding/selecting, updating, and removing. In this diagram, the MEPO employee will choose either add, update, or remove a product or service.

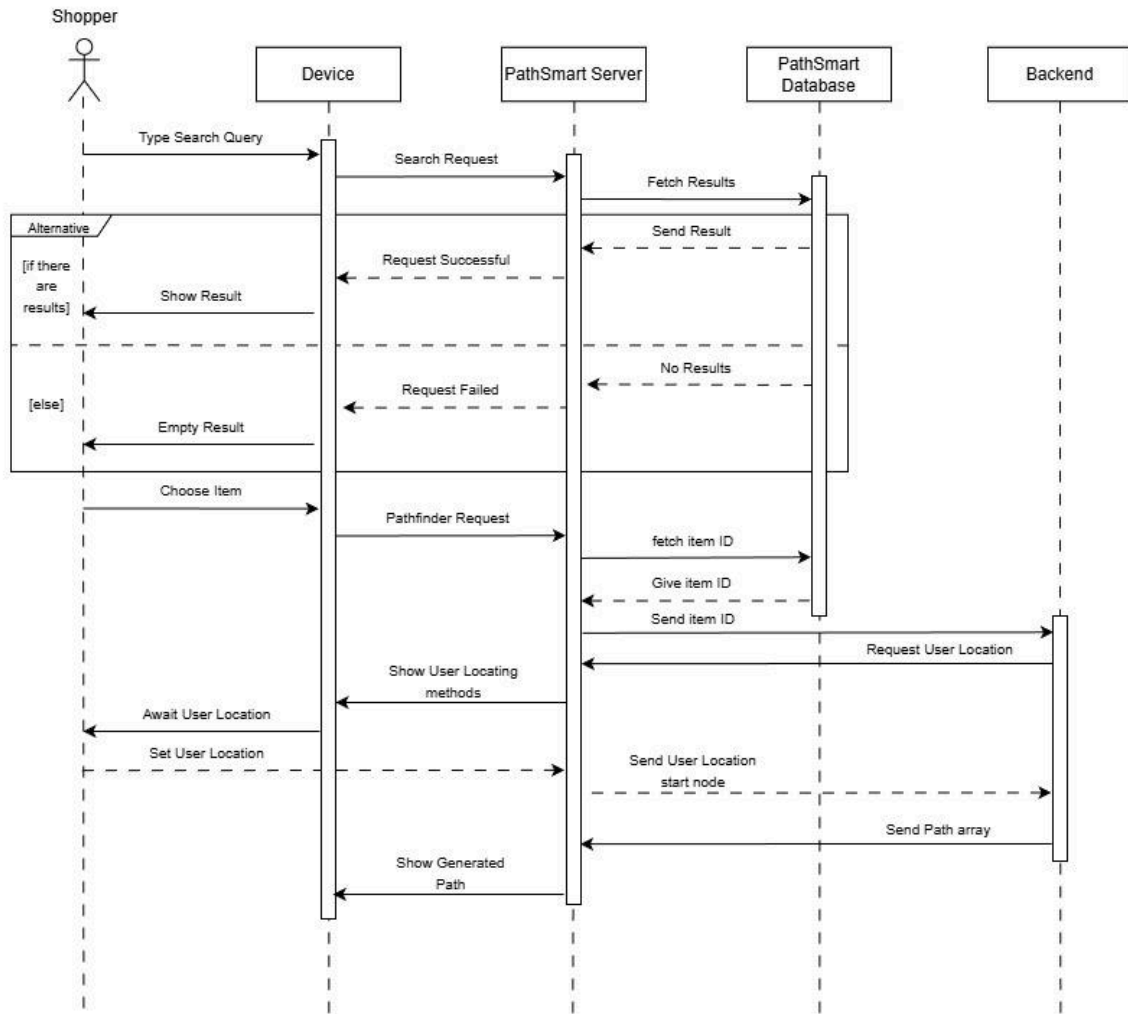


Figure 36: Sequence Diagram of Shopper Guide Module (Pathfinder)

The figure above represents the sequence diagram for Pathfinder.

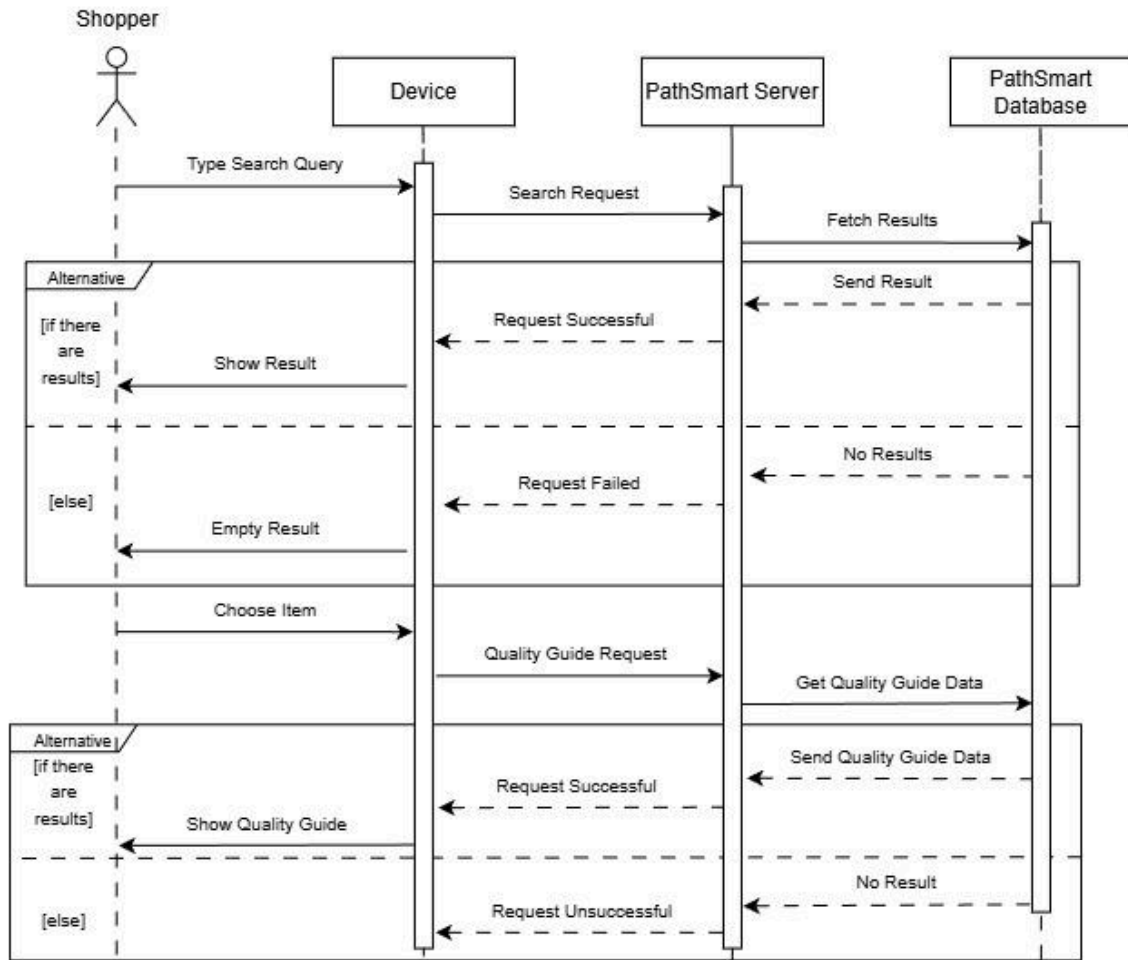


Figure 37: Sequence Diagram of Shopper Guide Module (Quality Guide)

The figure above represents the sequence diagram for Quality Guide.

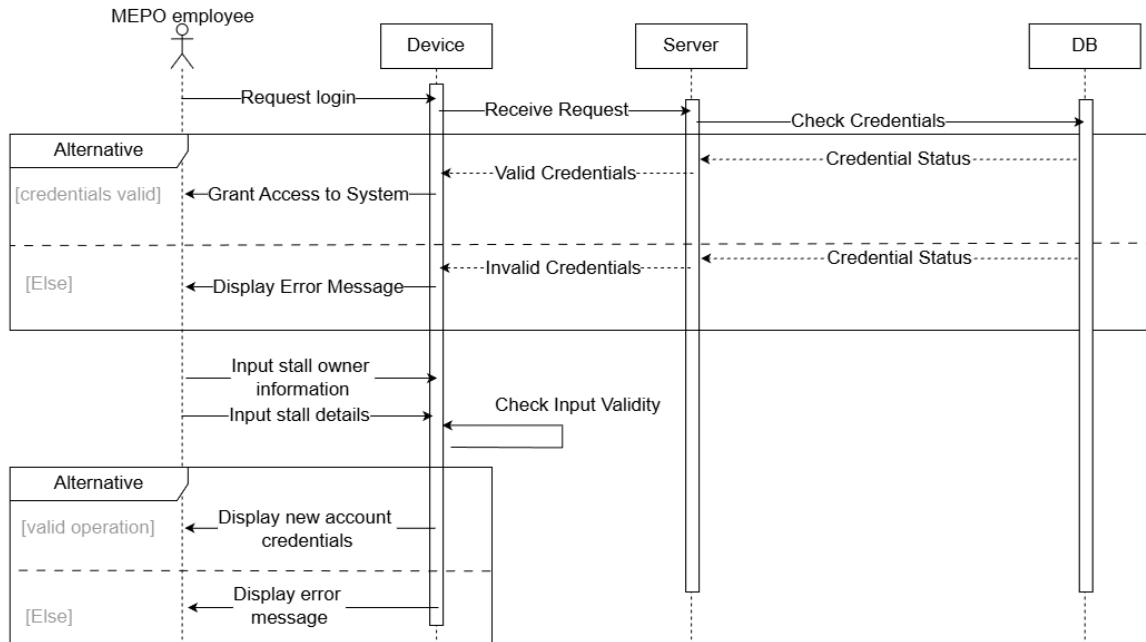


Figure 38: Sequence Diagram of Stall Management Module (Account creation)

The figure above represents the sequence diagram for account creation. In this diagram, MEPO employees first need to login first, Supabase is used to verify the identity of the MEPO employee. Once the MEPO employee is logged in, the MEPO employee will input the stall owner's personal information, front-end validates the inputs, if valid then display the account credentials and store it in the DB, otherwise display error message.

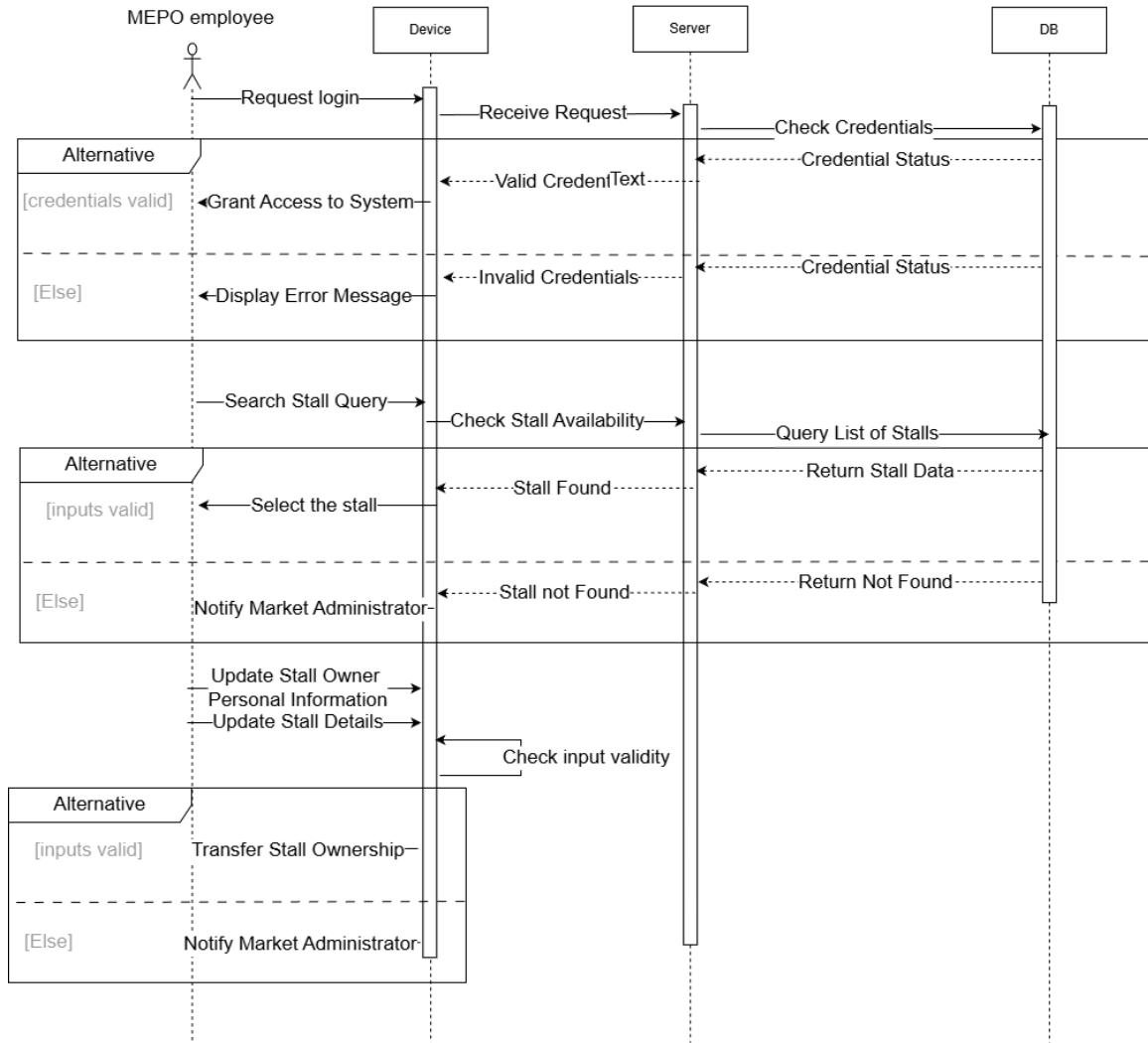


Figure 39: Sequence Diagram of Stall Management Module (Transfer Stall Ownership)

The figure above represents the sequence diagram for transferring stall ownership or simply updating the stall details. In this diagram, MEPO employee first need to login, then search the availability of the stall, and if the stall is found, MEPO employee can update stall details. If the inputs are valid and nothing went wrong, update the stall details.

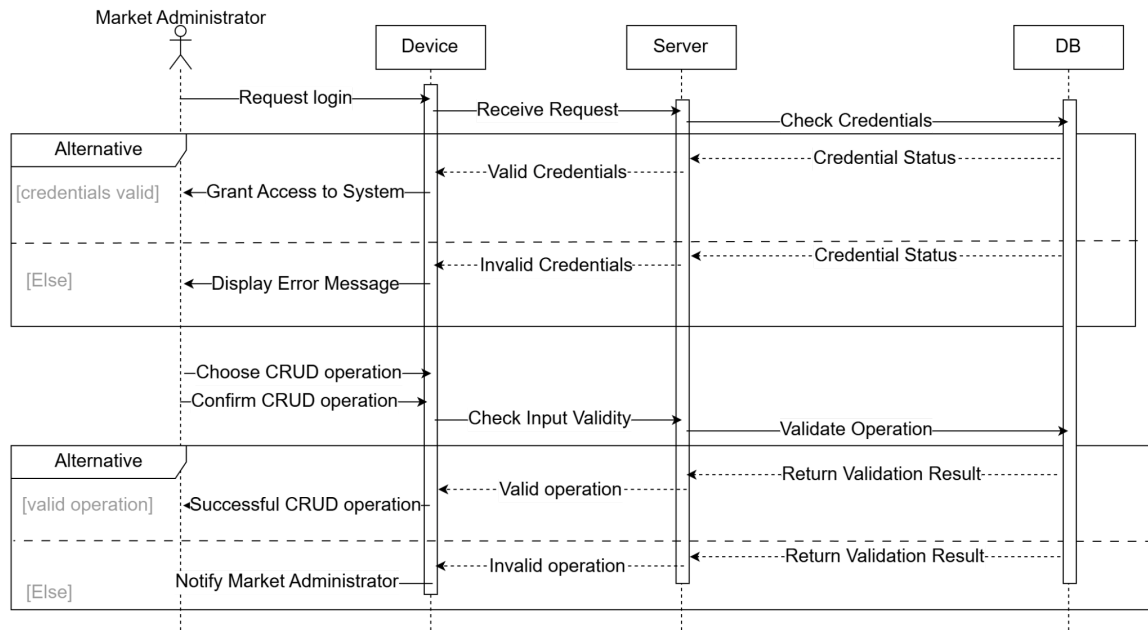


Figure 40: Sequence Diagram of Stall Management Module (Manage Quality Guide)

The figure above represents the sequence diagram for managing quality guides. In this diagram, MEPO employees first need to login in the system, then choose an action to perform (CRUD), validate the action first, if the action is valid and nothing went wrong, the action or CRUD is successful otherwise notify MEPO employee.

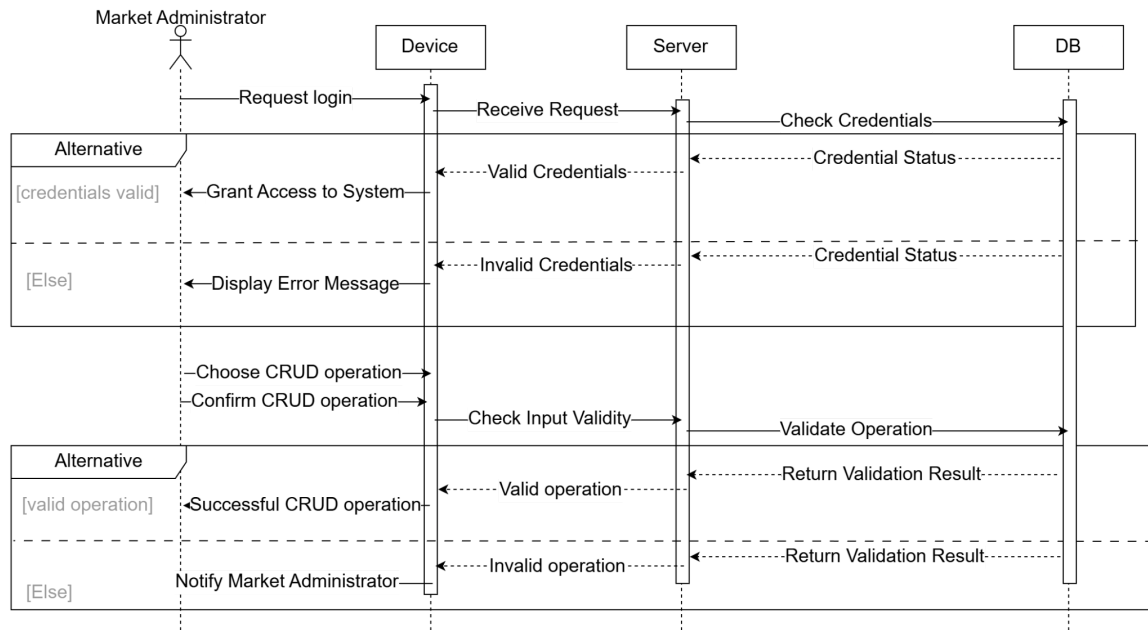


Figure 41: Sequence Diagram of Stall Management Module (Listing Creation)

The figure above represents the sequence diagram for listing creation. In this diagram, MEPO employees first need to login in the system, then choose an action to perform (CRUD), validate the action first, if the action is valid and nothing went wrong, the action or CRUD is successful otherwise notify MEPO employee.

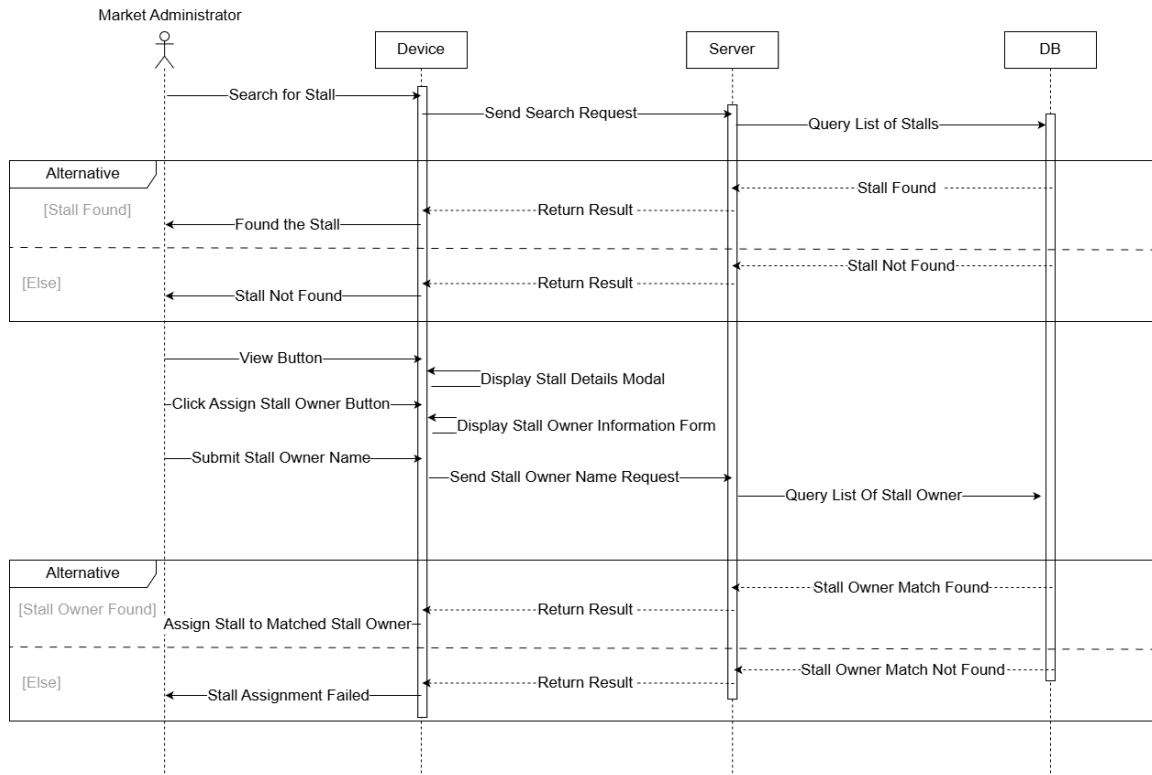


Figure 42: Sequence Diagram of Stall Management Module (Assign Stall)

The figure above represents the sequence diagram for assigning a stall. In this diagram, MEPO employees first need to login in the system, then start to look for the stall. If the stall is found, the system will display it, otherwise notify the MEPO employee. The MEPO employee will now start the stall assignment by searching for the stall owner which will be assigned to the stall they picked, if the stall owner account is found in the system (which is created by the MEPO employee), process the stall assignment, if account is not found, notify MEPO employee.

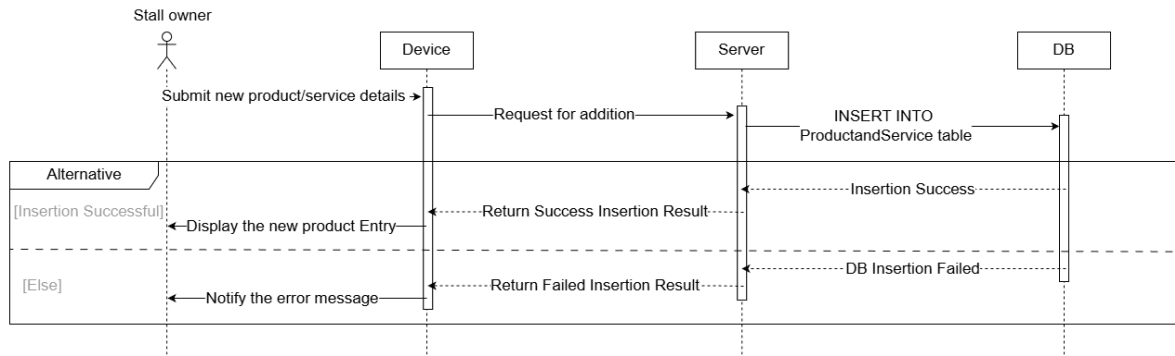


Figure 43: Sequence Diagram of Store Management Module (Add/Select Listing)

The figure above represents the sequence diagram for managing a listing for stall owners. It consists of selecting/adding, updating, and deleting the selected listing.

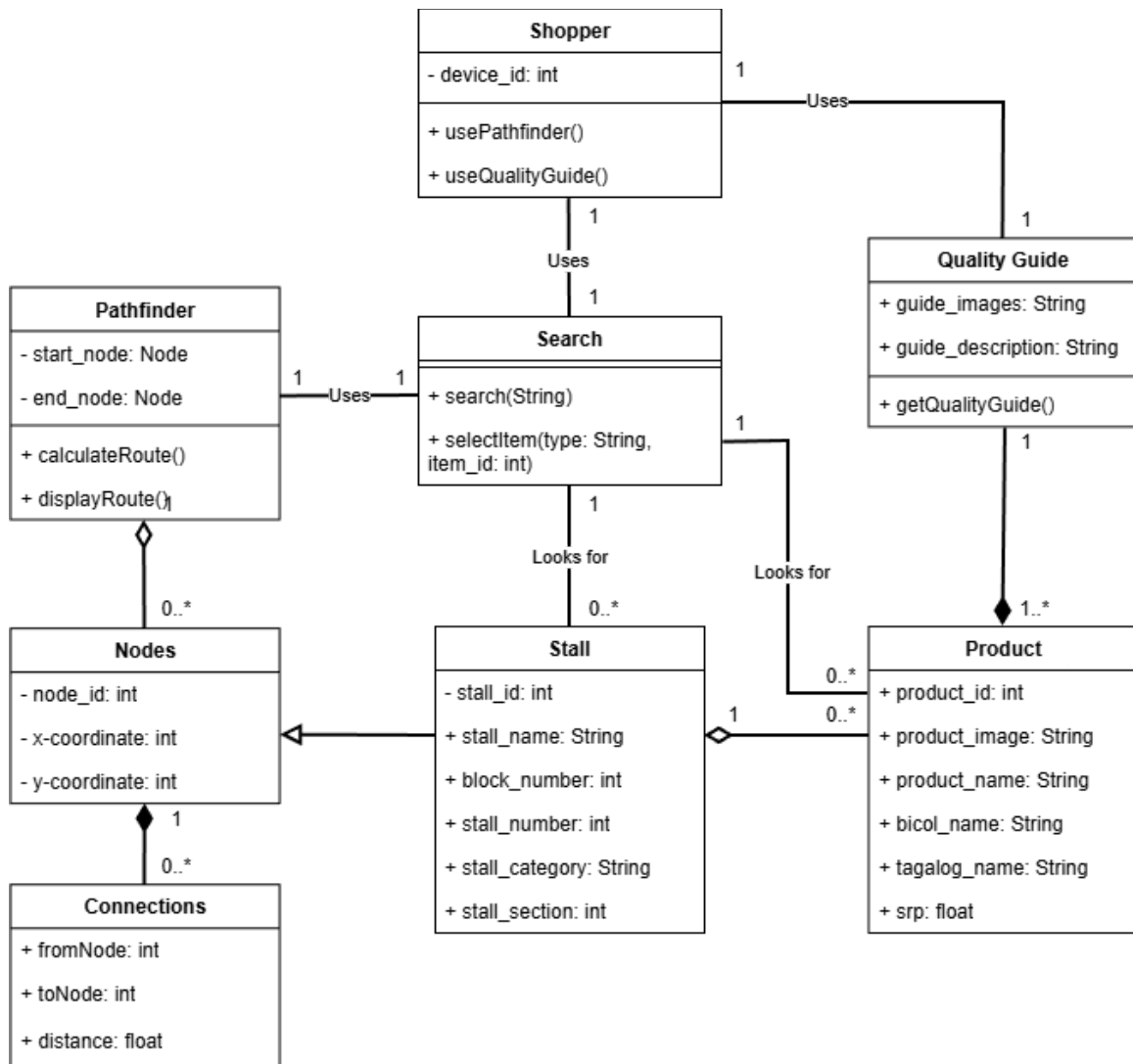


Figure 44: Class Diagram of Shopper Guide Module (Unified Modeling Language)

The figure above represents the class diagram for the shopper guide module. The figure shows the relationship between the entities of the shopper guide module.

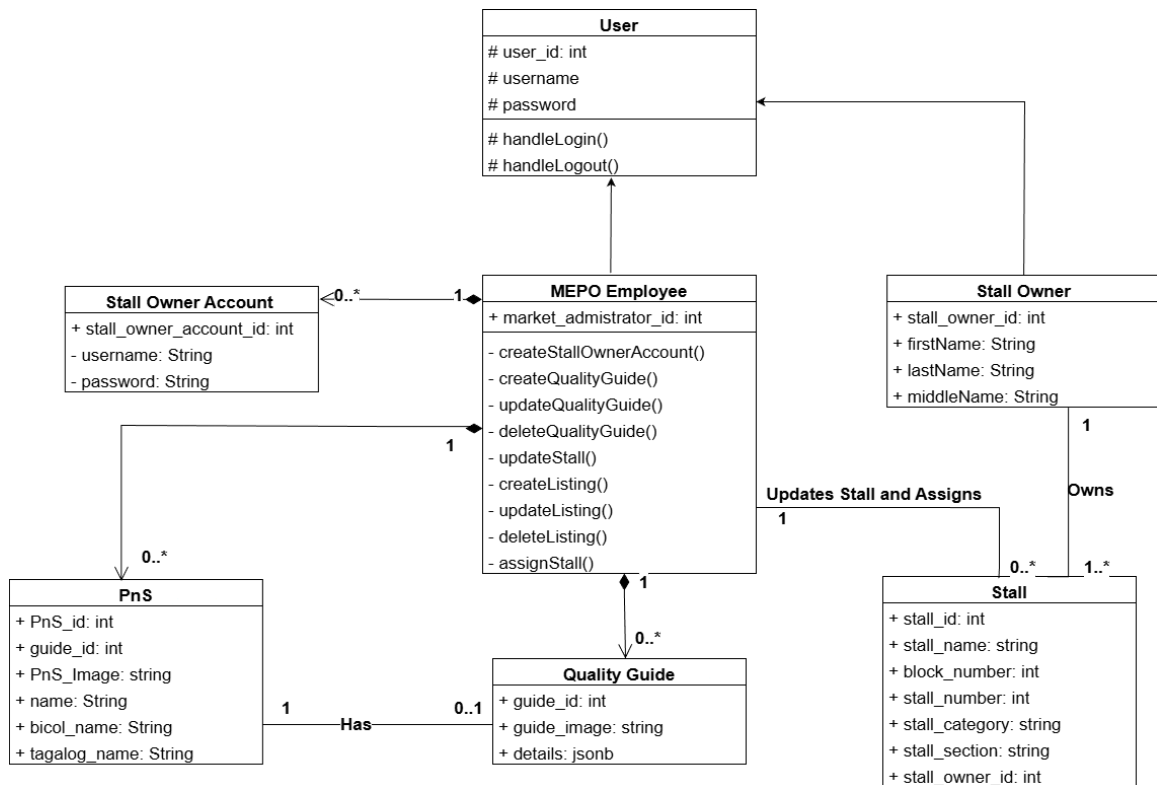


Figure 45: Class Diagram of Stall Management Module (Unified Modeling Language)

The figure above represents the class diagram for the stall management module, showing the structure of the system entities and their relationship to each other. It shows the main classes such as the User, MEPO employee, Stall Owner, Stall, and so on—along with their attributes, operations, and relationships or connection.

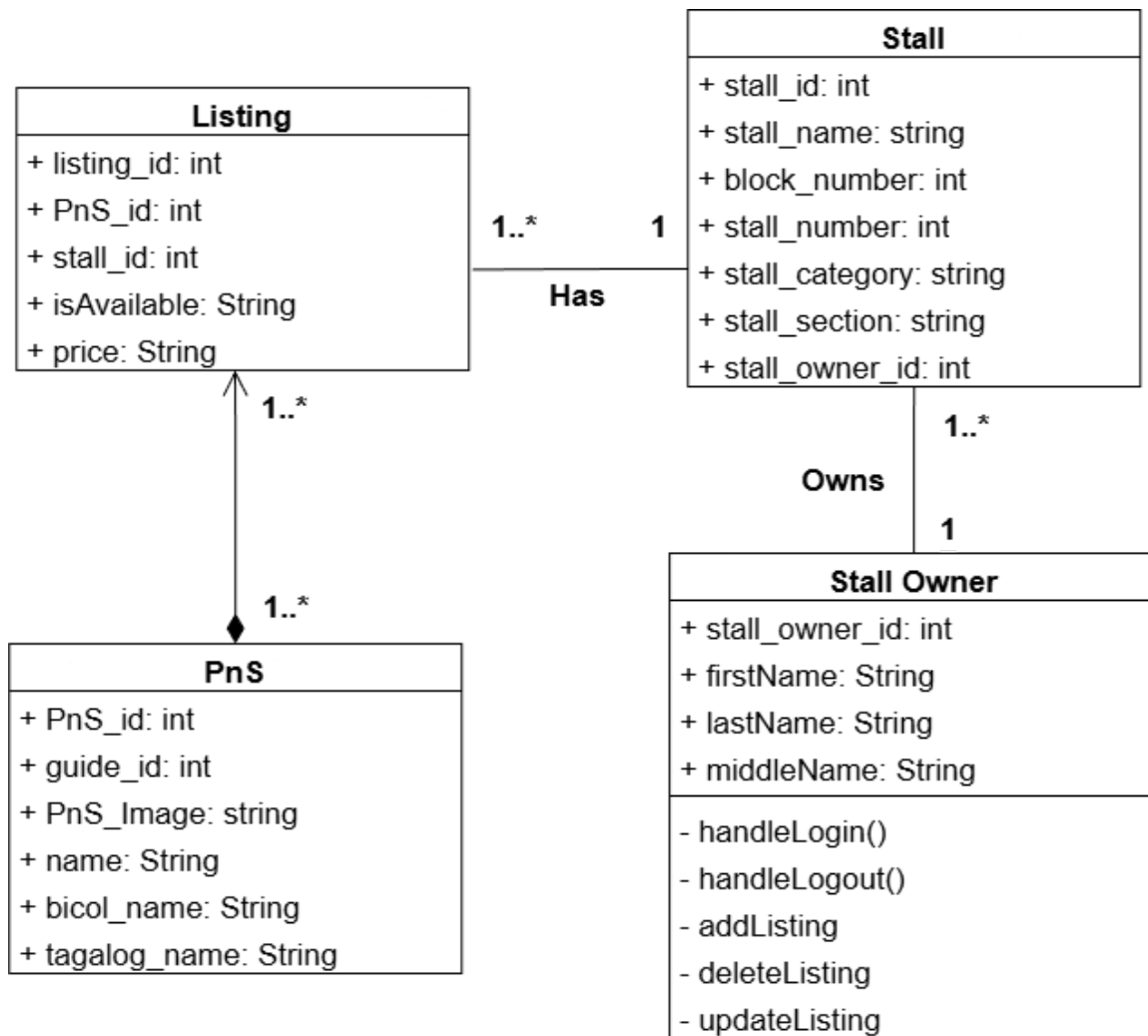


Figure 46: Class Diagram of Store Management Module (Unified Modeling Language)

The figure above represents the class diagram for the store management module, showing the structure of the system entities and their relationship to each other. In this module, It shows the main classes such as the Stall Owner, Stall, and so on—along with their attributes, operations, and relationships or connection.

Testing and Evaluation

This section outlines the testing and evaluation procedures employed in the PathSmart system to assess the system's functionality, performance, and usability.

Testing Methods

The following testing methods were used in assessing the PathSmart system functionality, performance, and usability:

- Unit Testing:
 - Unit testing was conducted to verify that each individual component of the PathSmart system worked correctly in isolation. Each module's functions — such as the Shopper's Guide, Stall Management, and Store management module — were tested independently using predefined test cases with known expected outputs. For each module, functions were executed with inputs to confirm that the functions responded with the correct output. For instance, the Shopper's Guide module was tested to evaluate the effectiveness of the A* algorithm when generating routes in terms of speed and accuracy.
- Integration Testing:
 - Integration testing was conducted to verify that the individual modules of the PathSmart system worked together correctly and

exchanged data as expected. For instance, the Stall Management and Shopper's Guide modules were tested together to ensure that all products and services in the Stall management module are reflected through the Shopper's Guide module accurately.

- User Acceptance Testing (UAT):
 - User Acceptance Testing was conducted to assess whether PathSmart meets the needs and expectations of its intended users. Algorithm performance was measured in terms of speed — the time taken to compute a route — and accuracy — the correctness of the generated path relative to the actual layout of NCPM.

Evaluation Criteria

The following criteria were used in evaluating the PathSmart system functionality, performance, and usability:

- **Functionality:** According to ISO/IEC it is “The degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions”
 - The stall management module supports all required MEPO employee functionalities (create stall owner accounts, assign stalls, manage listings, etc).
 - Each function performs as described in the system requirements.
 - Features are accessible and complete without missing functionality.

- **Reliability:** According to ISO/IEC it is “degree to which a system, product or component performs specified functions under specified conditions for a specified period of time”
 - Data is saved and persists across sessions.
 - Error messages are displayed when inputs fail validation.
 - The system can recover from minor errors without developer intervention.
- **Performance:** How quick and efficient the system responds to user actions and processes data.
 - Loading time for most actions is under 2 seconds (search results, submitting a form, account creation, etc.).
 - Pagination returns data without delay even with large datasets.
- **Usability:** The system is easy to understand, navigate, and use.
 - MEPO employee users can navigate the module with minimal guidance.
 - Main actions such as account creation or stall assignment are easy to navigate within 2–3 clicks.
 - Interface labels, buttons, and forms are clear and self-explanatory.

- **Security:** The system protects all user and market data from unauthorized access, misuse, loss, or corruption
 - All sensitive data (e.g., passwords) shall be stored using one-way encryption (e.g., bcrypt).
 - Users should be authenticated before they can access the system and only access the necessary resources in the system based on their user role.

CHAPTER 4

DISCUSSION OF FINDINGS

This chapter presents the discussion of findings based on the data gathered and analyzed in this study. It addresses the research objectives by interpreting the results obtained from interviews and document analysis.

Input requirements needed to develop the PathSmart system

Input Data	Input Source	Processing Logic	Output Data	Output Destination
Floor plan layout	City Planning Office, City Hall	Acts as the primary foundation for the map	Map Area and Partitions	Pathfinder Map
Manual area mapping	Manual area surveying	Used to define and label specific zones or partitions within the floor plan	Spatial Mapping Output	Floor Plan Layout
Stall Owner Masterlist	MEPO office	Document Analysis	Stall Owner Information (Name, Stall No., Stall name, etc.)	Supabase DB: Table: Stall Owner & Stalls

The findings revealed that several key data components are essential for building an effective indoor navigation system. These include the market floor plan layout, manual area mapping, and the stall owner masterlist.

The floor plan layout, obtained from the City Planning Office of City Hall, serves as the primary foundation of the Pathfinder Map, providing the base structure that represents the overall layout of the market. The manual area mapping, gathered through manual area surveying, was used to define and label specific zones and partitions within the floor plan, producing the spatial mapping output that is reflected in the final floor plan layout.

The stall owner masterlist, sourced from the MEPO office, underwent document analysis to extract important stall owner details such as the owner's name, stall number, and stall name, among others. This information was then stored in the Supabase database under the Stall Owner and Stalls tables, making it accessible for use within the PathSmart system.

Together, these data sources gave the proponents the accurate and complete information needed to develop a functional and reliable indoor navigation system.

Modifications needed to the A* algorithm to work the layout and navigation needs of the NCPM properly

Input Data	Input Source	Processing Logic	Output Data	Output Destination
Weighted Graph	connections.js on (node-to-node edges with distance weights)	The market layout is represented as a graph where each node has an ID with x and y coordinates, and each edge has a numeric weight (distance). The A* algorithm traverses this graph by tracking the cost from the start node to each neighbor using costSoFar[current] + distance	Navigable Node-Edge Map	A* findPath() function
Euclidean distance	f1nodes.json (x, y coordinates of each node)	Used as the heuristic function inside A*: heuristic = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. The algorithm calculates priority = costSoFar + heuristic to decide which node to explore next, guiding it	Heuristic Cost Value (f(n) = g(n) + h(n))	A* priority frontier (sorted list)

		toward the destination efficiently		
Start and End Points	User Input (selected stall)	The findPath(startId, endId) function runs A* from the user's current location to the target stall's stall_endNode . Once the destination is reached, the path is reconstructed by backtracking through cameFrom[]	Optimal Node Path (array of node IDs)	Pathfinder Map (rendered as a route)

The second objective focused on identifying the modifications needed to adapt the A* algorithm to fit the layout and navigation needs of the Naga City Public Market (NCPM). The findings revealed three key data components that were essential in making the algorithm work effectively within the market environment.

The first component is the **weighted graph**, which was built using a **connections.json** file that defines how each node in the market is connected to its neighbors, along with the distance between them. Instead of treating all paths as equal, each connection was assigned a numeric weight representing its actual distance, allowing the algorithm to distinguish shorter paths from longer ones.

The second component is the **Euclidean distance heuristic**. The A* algorithm requires a heuristic function to estimate how far a node is from the destination. In this system, each node was given x and y coordinates stored in **f1nodes.json**,

and the heuristic was calculated using the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. This estimate guides the algorithm toward the goal faster instead of blindly checking all possible paths.

These two components work together inside the **findPath()** function through the formula **f(n) = g(n) + h(n)**, where **g(n)** is the actual travel cost from the start node and **h(n)** is the Euclidean distance estimate to the destination. The algorithm always explores the node with the lowest combined score first, ensuring it finds the most efficient route.

The third component involves **stall end nodes**. Since stalls in the market occupy physical space and cannot be walked through, each stall was assigned one or more entry points called **stall_endNodes**. When a user selects a destination stall or product, the algorithm finds the path to the nearest accessible entry point of that stall rather than to the stall itself.

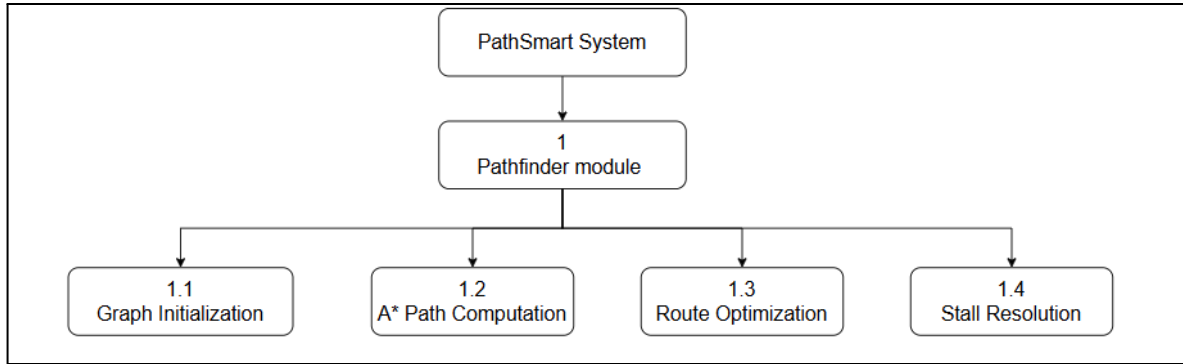


Figure #47: Function Decomposition Diagram of PathSmart System

The PathSmart System is composed of one main module for the algorithm: the Pathfinder Module. The Pathfinder Module is further decomposed into four sub-modules that collectively implement the indoor navigation and route planning functionality of the system. The first sub-module, Graph Initialization (1.1), is responsible for loading the map data — specifically the node coordinates and connection weights — and normalizing item identifiers before pathfinding begins. The second sub-module, A* Path Computation (1.2), implements the core A* algorithm, which calculates the shortest path between two nodes on the indoor map using a Euclidean distance heuristic, a priority queue for frontier traversal, and a path reconstruction mechanism. The third sub-module, Route Optimization (1.3), handles the intelligent reordering of the shopper's list by applying a greedy nearest-neighbor strategy to minimize the total travel distance across all desired stalls. The fourth sub-module, Stall Resolution (1.4), is responsible for resolving product and stall identifiers to their corresponding map entry points, determining which stalls carry a given product and selecting the closest accessible end node for navigation.

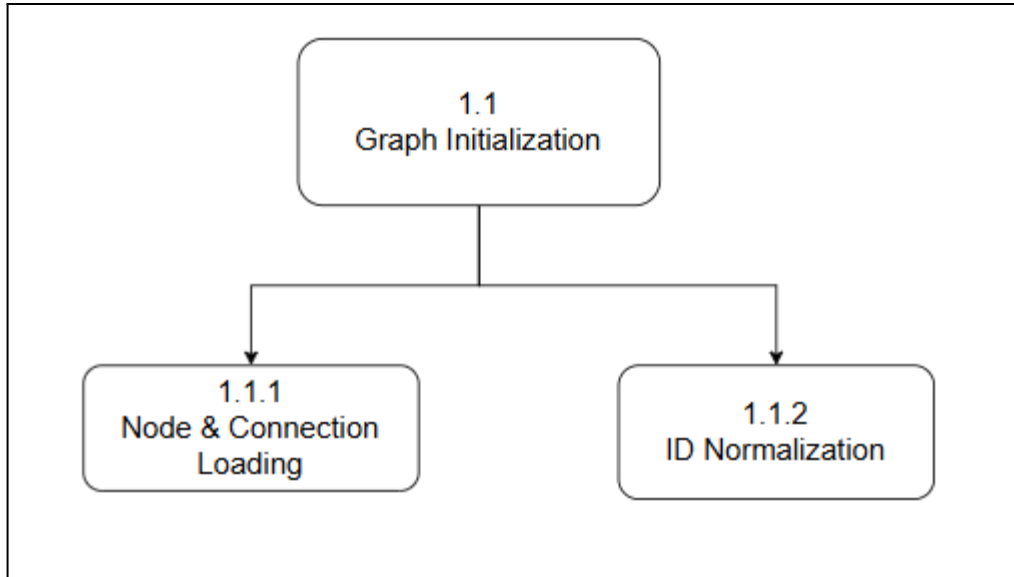


Figure #48: Function Decomposition Diagram of Graph Initialization

The Graph Initialization sub-module (1.1) is composed of two functions. The first, Node & Connection Loading (1.1.1), is responsible for loading the map's spatial data into memory — specifically, it reads **f1nodes.json**, which contains the coordinates of every navigable node on the indoor map, and **connections.json**, which defines the weighted edges between those nodes representing walkable paths. These are loaded once in the **PathfinderController** constructor and serve as the foundation for all subsequent pathfinding operations. The second, ID Normalization (1.1.2), is handled by the **normalizeId()** function in **pathfinder.jsx**, which standardizes product and stall identifiers into a consistent format before they are passed to the pathfinding logic — for example, stripping redundant leading

characters from product IDs such as **pp1** into **p1**, and converting stall IDs prefixed with **s** such as **s1** into their numeric string equivalent **1**.

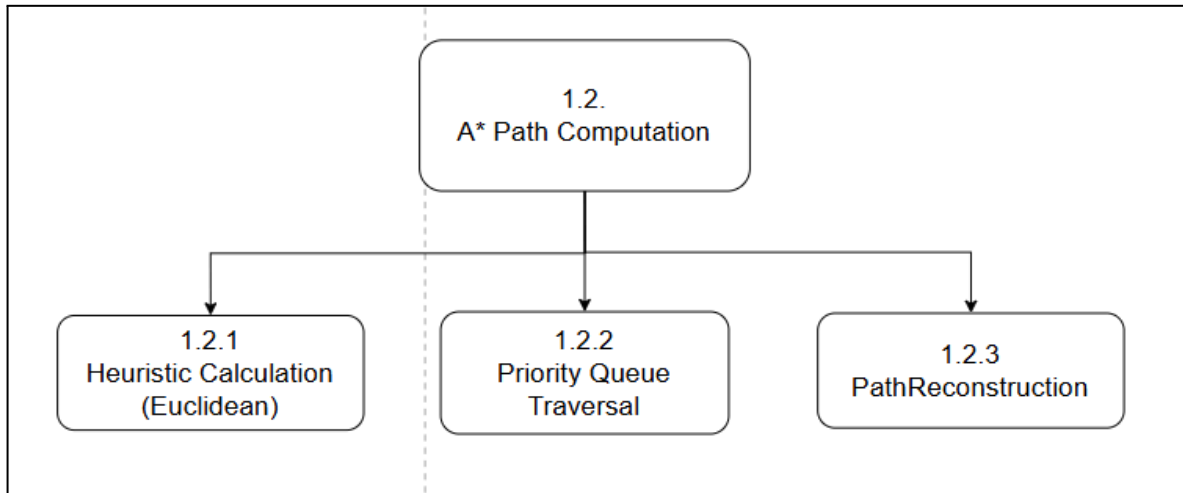


Figure #49: Function Decomposition Diagram of A* Path Computation

The A* Path Computation sub-module (1.2) is composed of three functions. The first, Heuristic Calculation (1.2.1), is implemented as the **euclidean()** function inside **findPath()**, which estimates the straight-line distance between two nodes using their x and y coordinates from **f1nodes.json**. This heuristic guides the algorithm toward the destination efficiently by prioritizing nodes that are geometrically closer to the goal. The second, Priority Queue Traversal (1.2.2), maintains a frontier list where each entry holds the total estimated cost, the cost so far, and the current node. At every iteration, the frontier is sorted so the lowest-cost node is processed first, and neighboring nodes are evaluated using **getNeighbors()** to retrieve their edge weights from **connections.json** — updating **costSoFar** whenever a cheaper path to a neighbor is found. The third,

Path Reconstruction (1.2.3), works by tracing back through the **cameFrom** dictionary once the destination node is reached — starting from the end node and following each parent pointer back to the start node, then reversing the resulting array to produce the correct ordered path from start to destination.

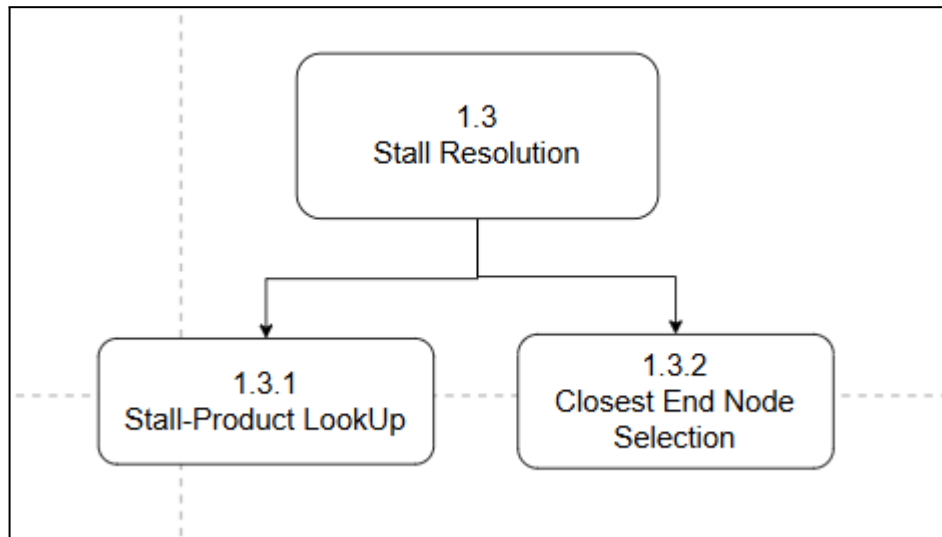


Figure #50: Function Decomposition Diagram of A* Stall Resolution

The Stall Resolution sub-module (1.3) is composed of two functions. The first, Stall-Product Lookup (1.3.1), is implemented through **getStallsForProduct()** in **PathfinderController.js**, which accepts a product ID and iterates over all stall entries in **saveData.json** to find and return every stall that carries that product — effectively mapping a shopper's desired item to its possible physical locations on the indoor map. This lookup also supports stall-direct navigation, where instead of a product ID, a stall ID is provided and the function returns that stall's entry directly. The second, Closest End Node Selection (1.3.2), is handled by **findClosestStall_endNode()** in

PathfinderController.js, which takes the shopper's current node and a list of possible end nodes for a stall and runs **findPath()** against each one, accumulating the total edge cost for each resulting path and returning the end node that produces the lowest cost — ensuring that when a stall has multiple entry points on the map, the shopper is always directed to the nearest and most efficient one.

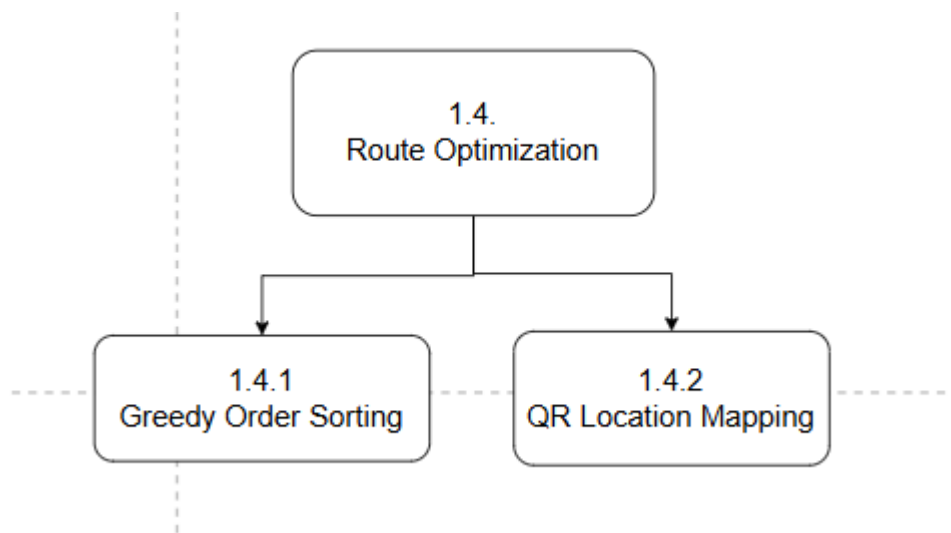


Figure #51: Function Decomposition Diagram of A* Route Optimization

The Route Optimization sub-module (1.3) is composed of two functions. The first, Greedy Order Sorting (1.3.1), is implemented through **findOptimizedShoppingOrder()** in **PathfinderController.js**, which reorders the shopper's list using a greedy nearest-neighbor strategy — starting from the shopper's current node, it repeatedly finds the closest unvisited stall or product among the remaining items by running **findPath()** for each candidate and selecting the one with the lowest cumulative edge cost, then updating the

current position to that stall's end node before repeating the process until all items in the shopping list have been accounted for. This ensures the shopper follows the most efficient sequence of stops rather than visiting stalls in an arbitrary order. The second, QR Location Mapping (1.3.2), is handled by the **findClosestNode()** function in **pathfinder.jsx**, which is triggered when the shopper scans a QR code placed at various points around the market — it parses the scanned data as either a direct node ID, a coordinate pair, or an integer node reference, and when coordinates are provided it iterates over all nodes in **f1nodes.json** to compute the Euclidean distance from the scanned position to every known node, selecting the nearest one as the shopper's current starting position. Once the starting node is set, the shopping list is immediately re-optimized from that new location.

Accuracy and speed of the A* algorithm when generating routes in the PathSmart system

To address the third objective of the study, the proponents evaluated the performance of the A* algorithm in PathSmart in terms of accuracy and speed. The evaluation involved route generation testing using pixel scaling to convert map distances into real-world measurements in meters.

The test was conducted from Entrance 6 to Stall 166, which is located at the center of Naga City People's Mall (NCPM). Three routes were compared: the route generated by the PathSmart pathfinder (Test A) and two alternative routes (Test B and Test C).

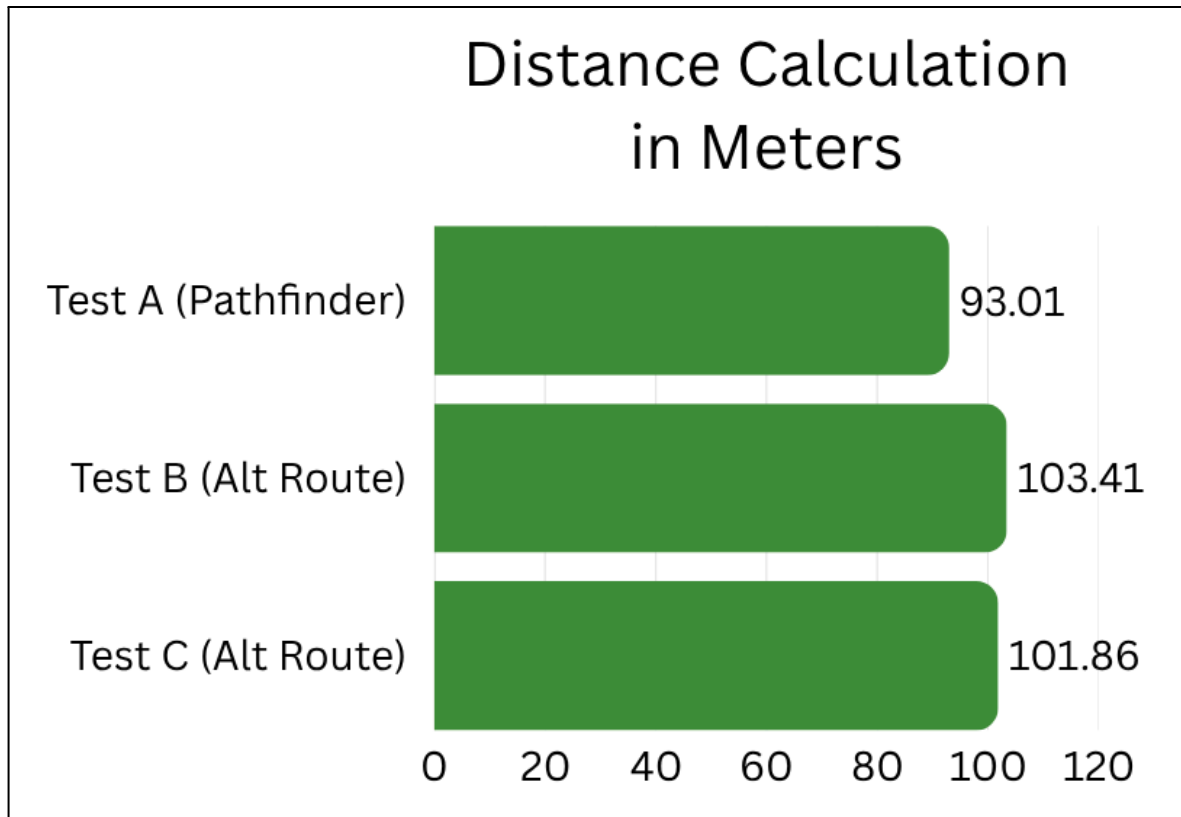


Figure #52: Testing Distance Calculation in Meters

In terms of accuracy, the results show that the A* algorithm successfully generated the shortest route with a distance of 93.01 meters (Test A). This outperformed the alternative routes, which measured 103.41 meters (Test B) and 101.86 meters (Test C). These results confirm that the algorithm correctly identifies and produces the optimal path within the market layout.

In terms of speed, the algorithm demonstrated efficient performance across different path sizes. The computation time recorded was 0.11 seconds for a 10-node path, 0.20 seconds for a 20-node path, and 0.32 seconds for a 30-node path. These results indicate that the algorithm performs efficiently and that its

computation time increases predictably with path complexity, while still remaining under one second.

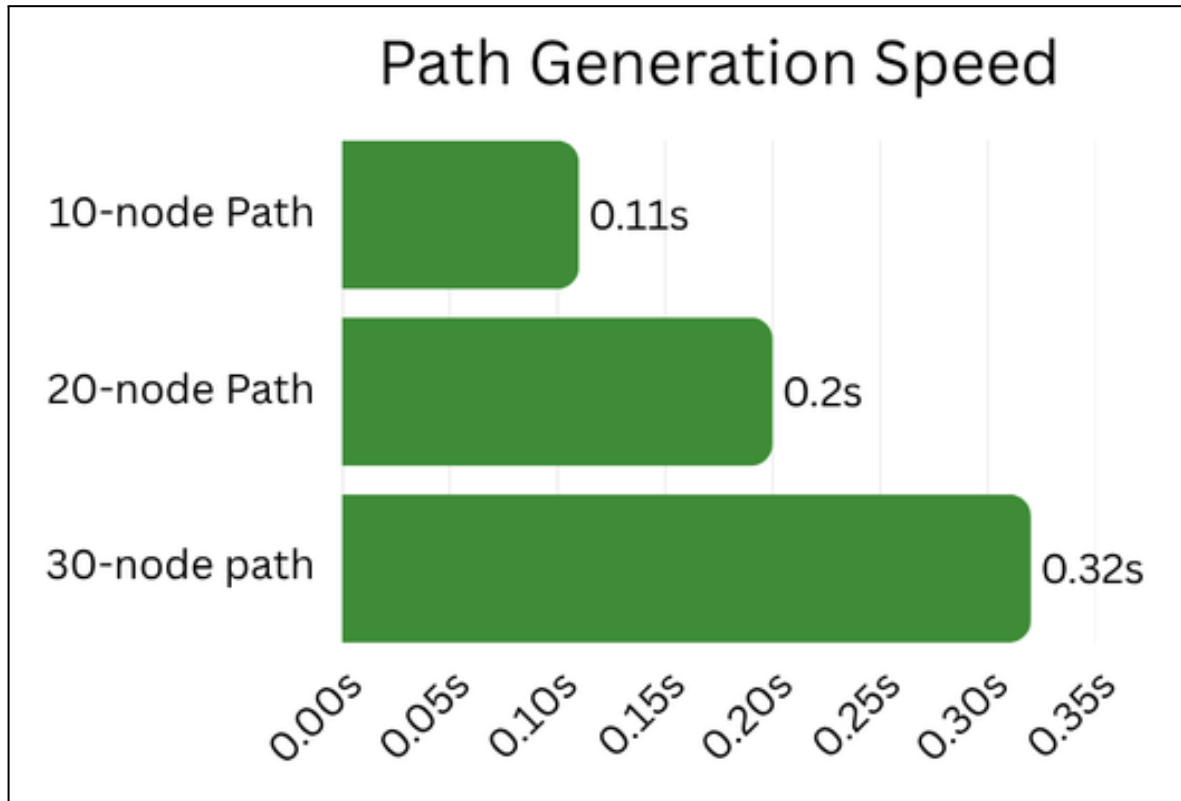


Figure #53: Testing Path Generation Speed

Overall, the findings confirm that the adapted A* algorithm is both accurate and efficient in generating optimal navigation routes within NCPM. This fulfills the third objective of the study and demonstrates the suitability of the algorithm for indoor navigation applications.

CONCLUSIONS AND RECOMMENDATIONS

This section presents the synthesis of findings based on the three statements of the problem in this study. It summarizes the key outcomes derived from the development and evaluation of the PathSmart system.

Conclusions

This section presents the synthesis of findings based on the three statements of the problem in this study. It summarizes the key outcomes derived from the development and evaluation of the PathSmart system.

(1) Synthesis of SOP 1 Findings. The findings indicate that the development of the PathSmart system requires several essential input requirements to ensure the accuracy and reliability of the indoor navigation system. These include the market floor plan layout, manual area mapping, and the stall owner masterlist. The floor plan layout served as the primary foundation of the Pathfinder Map by representing the overall structure of the market, while the manual area mapping helped identify and label specific zones, partitions, and stall locations to produce a more accurate spatial mapping output. Additionally, the stall owner masterlist provided important stall-related information, such as stall numbers, stall names, and owner details, which were organized and stored in the Supabase database for system integration.

(2) Synthesis of SOP 2 Findings. The findings revealed that several important modifications were necessary to adapt the A* algorithm to the complex layout and navigation requirements of the Naga City Public Market (NCPM). The use of a weighted graph through **connections.json** ensured that path costs reflected real distances rather than arbitrary steps, allowing the algorithm to meaningfully distinguish between routes. The Euclidean distance heuristic, derived from the x and y coordinates stored in **f1nodes.json**, provided the algorithm with directional guidance through the formula $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$, reducing unnecessary exploration and improving efficiency. Finally, the introduction of **stall end nodes** addressed the physical reality of the market layout, where stalls occupy space and cannot be traversed, by giving each stall one or more accessible entry points that serve as navigation targets. Together, these three components — weighted connections, a coordinate-based heuristic, and stall end nodes — form the structural foundation that makes the A* algorithm functional and practical within the complex environment of the NCPM. Without these adaptations, a standard implementation of A* would have been insufficient to meet the spatial and navigational demands of the market setting.

(3) Synthesis of SOP 3 Findings. The study concludes that the adapted A* algorithm performs efficiently in terms of both accuracy and speed. In terms of accuracy, the algorithm successfully generated the shortest route from Entrance 6 to Stall 166 with a distance of 93.01 meters, outperforming alternative routes of 103.41 meters and 101.86 meters. In terms of speed, the algorithm demonstrated consistent performance, completing route computations in 0.11 seconds for 10

nodes, 0.20 seconds for 20 nodes, and 0.32 seconds for 30 nodes. These results confirm that the A* The algorithm is both accurate and efficient, making it suitable for indoor navigation within NCPM.

Recommendations

This section presents the recommendations based on the findings and conclusions of the study. These aim to improve the PathSmart system and guide future developers and researchers in enhancing indoor navigation systems in similar environments.

1. For future developers and administrators of PathSmart, it is suggested to create a data maintenance protocol to keep the system's required data, such as the market floor plan and product listings, accurate and updated. Assigning a responsible person, such as a MEPO employee, to manage and update the data may help ensure that the system stays reliable and useful over time.
2. Future versions of the system may also consider adding positioning technologies like Bluetooth Low Energy (BLE) or Wi-Fi-based positioning. This can help improve the current ISP, which uses a user-defined starting point, and make the navigation more accurate.
3. In terms of algorithm improvement, future developers may further enhance the adapted A* algorithm to handle more complex navigation cases within Naga City People's Mall. They may also add features such as suggesting alternative routes based on crowd levels or user behavior to improve efficiency and user experience.

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Web Development

- Frontend: Vue.js, React, HTML, CSS
- Backend: Node.js, Express.js, Supabase
- Database: MySQL, MongoDB, Supabase

Specialized Solutions

- Able to create systems that use A* algorithm and Dijkstra algorithm.
- Able to create real-time updating systems.

Create Systems

- Thinkers' Online Classes Prototype: The website is able to accept student applications, has real-time tracking of teacher schedules, and most importantly, stores data securely.
- PathSmart Application: The application is a pathfinder that uses A* algorithm to locate a product or store.

CERTIFICATIONS

- AWS Cloud Fundamentals
- Sololearn: Intermediate Java
- Sololearn: Intermediate SQL
- Cisco – Networking Essentials
- LinkedIn – React Building Progressive Web Apps PWAs
- LinkedIn – Git Essential Training 2023
- LinkedIn: Systems Thinking
- LinkedIn – Software Design Modeling with UML
- LinkedIn: Software Development Life Cycles
- LinkedIn: Software Architecture Foundations
- LinkedIn: Software Architecture Patterns for Developers

WORK EXPERIENCE

OJT/Internship

AWS Cloud Foundations Training

- Learned basic management of AWS Cloud Services

January 2026 – March 2026

REFERENCES

Hilary Joan Prilles

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I certify that the above information is true and correct

JOVY JAY PALMONES DELA CRUZ

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SUMMARY

A detail-oriented developer with knowledge of Java and JavaScript programming, complemented by hands-on experience in front-end development using HTML and CSS. Skilled in designing structured solutions, developing responsive interfaces, and managing databases using MySQL and Supabase. Demonstrates logical problem-solving and debugging abilities, along with adaptability in learning new technologies and development tools such as Git and Visual Studio Code. Known for a proactive mindset, analytical thinking, and a commitment to continuous learning, contributing to the development of organized, functional, and user-focused software solutions.

EDUCATION

Baccalaureate Degree: Bachelor of Science in Computer Science
University of Nueva Caceres
Naga City
2022-2026

WORK EXPERIENCE

Special Program for Employment of Students (SPES)

LGU Naga
2019

- Provided front desk assistance to clients by answering inquiries and guiding them through government services and requirements.
- Assisted in processing documents and ensuring accurate completion of client forms.
- Maintained a professional and approachable demeanor while supporting daily public service SSoperations.

SKILLS AND ABILITIES

Personal Skills

- **Adaptability and Continuous Learning**
 - Quickly learned new programming languages and development tools.
 - Adapted to different project requirements and technologies.
 - Actively improved technical skills through practice and research.
- **Teamwork and Collaboration**
 - Worked in group projects following structured development processes.
 - Shared responsibilities and contributed ideas to improve system outputs.
 - Supported team members in completing technical tasks.
- **Good work ethic**
 - Dedicated to completing tasks efficiently and meeting deadlines.
 - Committed to producing quality outputs and continuously improving skills.
 - Willing to learn new technologies independently.

Technical Skills

- **Programming languages**
 - Java
 - JavaScript
- **Web development**
 - Html
 - Css
 - Basic Front end development
- **Database Management**
 - Basic Database Design and Querying
 - MySQL
 - SupaBase
- **Software & Tools/Platforms**
 - Visual Studio Code
 - Git (Basic Version Control)
 - Microsoft Office (Word, Excel, PowerPoint)
 - Zoom
 - Automation tools (Chatgpt, Gemini, Claude, Copilot)

TRAINING/CERTIFICATION/LICENSE

Software Development Life Cycle (SDLC)
 LinkDin learning
 January 29, 2025

Software Architecture: Patterns for Developers
 LinkDin learning
 March 5, 2025

Software Architecture Foundations
 LinkDin learning
 April 01, 2025

REFERENCES

Christine Mae D.C Bisana
Q/A Tester
 ClinkIT
 0916 802 6776

Christian Jay P. Dela Cruz
Debugger
 Accenture
 0949 782 1254

Jacob B. Bisana
Teach lead
 ClinkIT
 0916 802 6776

I certify that the above information is true and correct.

Jovy Jay P. Dela Cruz

JOSEPH FRONDOZO MARTIN JR.

[Portfolio](#) • [LinkedIn](#) • [GitHub](#)

Naga City, Philippines • se.josephmartin@gmail.com • 09476373794

SUMMARY

Fourth-year Computer Science student at the University of Nueva Caceres seeking a Software Developer role. Proficient in HTML, CSS, JavaScript, ReactJS, Git, and GitHub, with experience contributing to open-source programs such as GirlScript Summer of Code (GSSoC), Hacktoberfest, and Social Winter of Code (SWOC). Currently completing my internship at SmartBridge, expanding my skills in MERN stack development. Passionate and eager to apply my technical knowledge to build impactful solutions, contribute to innovative projects, collaborate with others, and grow as a developer.

INTERNSHIP

SmartBridge: *Full-Stack MERN Stack Intern*
Present

March 03, 2026 –

- Currently completing in SmartBridge's LastMile Program, an industry-supported, project-based internship designed to develop job-ready full-stack development skills

ACHIEVEMENTS

Open-Source Contributor

Girlscript Summer of Code (GSSoC) | May 2023

- Selected as a contributor among thousands of applicants, collaborating with mentors and contributors globally over a 3-month program.
- Contributed to **10+ repositories**, focused on UX and accessibility improvements.
- Ranked **#183 out of 1,559 active contributors** on the GSSoC leaderboard.

Open-Source Admin & Maintainer

Girlscript Summer of Code (GSSoC) | May 2024

- Led **ResourceHub** as a Selected as a Project Admin in GSSoC 2024, overseeing contributions from a global pool of open source contributors over 3 months.
- Reviewed and merged pull requests, ensuring PR quality in the project.
- Guided and mentored contributors — including first-time open source contributors — through the contribution process.
- Provided ongoing support and feedback for contributors

EDUCATION

Baccalaureate Degree:

Bachelor of Science in Computer Science
University of Nueva Caceres
Naga City
2022-2026

SKILLS AND ABILITIES

Personal Skills

- **Communication skills (written and verbal)**
 - Developed strong communication skills through participation in global open source programs — [GSSoC](#), [Hacktoberfest](#), and [SWOC](#) — engaged with mentors and contributors from different backgrounds and levels to discuss tasks, provide updates, and resolve issues.
 - Collaborated with 2 developers in the MLH [Hack Your Portfolio](#) hackathon, where clear and constant communication was key to successfully building a mental health chatbot app within a limited timeframe.
 - Served as **Project Admin/Maintainer** for **ResourceHub** in GSSoC 2024, communicating consistently with contributors over 3 months — providing guidance, reviewing work, and offering feedback to support contributors.
- **Teamwork and Collaboration**
 - Having worked with contributors, mentors, developers across multiple international open source programs and hackathons online (e.g. MLH, SWOC)
- **Time management**
 - Applies time management techniques such as **Time Blocking**, **Deep Work**, **Pomodoro**, and other techniques to effectively balance academic responsibilities, self-taught learning, and personal life.

Technical Skills

- **Web Development**
 - Proficient in HTML, CSS, JavaScript, ReactJS, TailwindCSS, TypeScript
 - Supabase, MySQL, NodeJS, ExpressJS, MongoDB, Git, GitHub, Figma
- **Technical writing/Blogging**
 - Writes tech blogs about my experiences/learnings, open source, and other technical stuff in [dev.to](#)
 - Accumulated over **15,000 post views**, **1170 followers**, and **94 reactions** on the platform
- **Others**
 - Git & GitHub, Figma, Data Structures & Algorithms, AWS cloud

TRAINING/CERTIFICATION/LICENSE

AWS Academy Cloud Foundations

March 2026

Amazon Web Services Training and Certification

Master the Coding Interview: Data Structures + Algorithm

December 2024

Zero to Mastery Academy

JavaScript Algorithms and Data Structures

December 2023

freeCodeCamp

Communication Foundations

December 2023

LinkedIn

REFERENCES

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Joseph F. Martin Jr.